Lab 1

a)- Introduction and gathering addresses (IP/MAC) information

In this lab we will learn about

1. Introduction to IP/Mac address
2. Gather information, including the connection, host name, MAC(Layer2) address, and TCP/IP Network(Layer 3)
3. Learn to use the TCP/IP packet Internet groper (ping) command.
4. Learn to ping Loopback address
5. Learnt to use the trace route (tracert) command.

Introduction to IP and Mac addressing

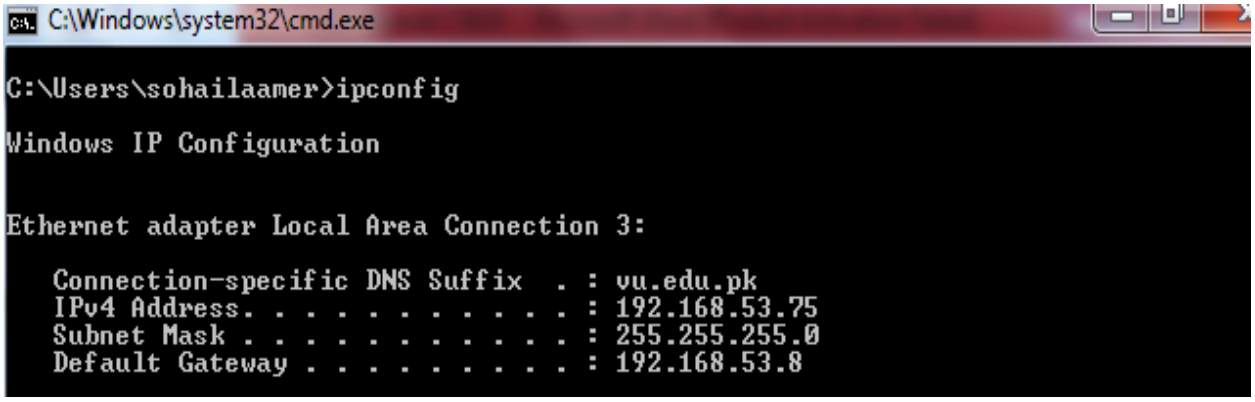
Every machine on a network has a unique identifier. An IP address provides an identity to a networked device. Similar to a home or business address supplying that specific physical location with an identifiable address; devices on a network are differentiated from one another through IP addresses.

IPv4 uses 32 binary bits to create a single unique address on the network. An IPv4 address is expressed by four numbers separated by dots. Each number is the decimal (base-10) representation for an eight-digit binary (base-2) number, also called an octet. For example: 216.27.61.137

A MAC address, or Media Access Control address, is a 48 bit address associated with a network adapter. While IP addresses are associated with software, MAC addresses are linked to the hardware of network adapters. The MAC address is a unique value associated with a network adapter. MAC addresses are also known as hardware addresses or physical addresses. They uniquely identify an adapter on a LAN.

Gather TCP/IP configuration information.

1. Use the Start menu to open the command prompt (Start>Programs>Accessories>Command Prompt or Start>Programs>Command Prompt).
2. Type ipconfig and press Enter key. The spelling of the ipconfig is critical, but the case is not.
3. The screen shows the IP address, subnet mask and the default gateway. The IP address and the default gateway should be in the same network or subnet; otherwise this host wouldn't be able to communicate outside the network.



- **Write down the TCP/IP information for your computer.**

a. IP address: _____

b. Subnet mask: _____

c. Default gateway: _____

- **Compare your computer's TCP/IP configuration to that of others on the LAN. If this computer is on a LAN, compare the information of several machines (Hosts).**

a. Are there any similarities? _____

b. What is similar about the IP addresses? _____

c. What is similar about the default gateway? _____

d. Record a couple of the IP addresses (of your nearby hosts)

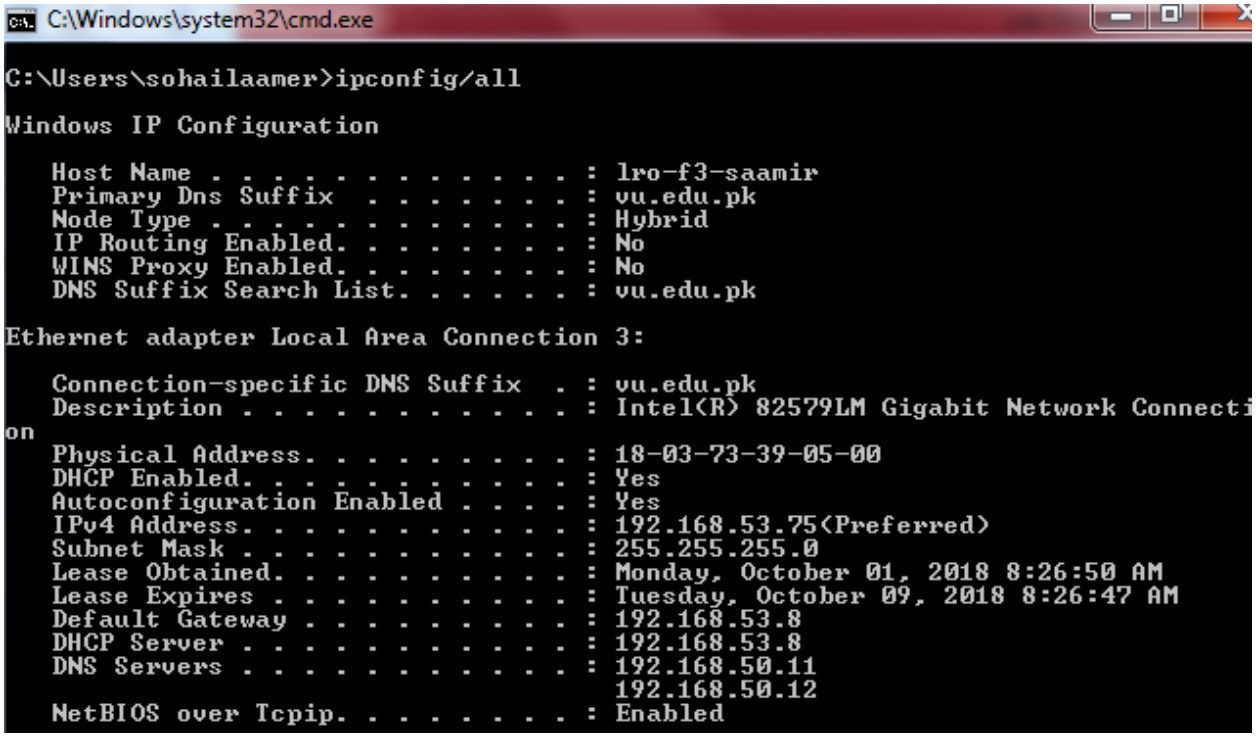
1._____

2._____

3._____

Check additional TCP/IP configuration information.

1. To see more information, type ipconfig/all and press Enter key. The figure shows the detailed IP configuration of the computer on the screen.



2. You should see the following information:

   - The host name (computer name), the Physical address of your machine, IP address, subnet Mask, Default Gateway and DNS Servers.

3. In the LAN, compare your result with a few nearby computers. What similarities do you see in the physical (MAC) address?

_____

d. Write down the computer's host name: _____

e. Write down the host names of a couple of other computer:

a. _____

b. _____

c. _____

d. _____

b) Verification of network connectivity using Ping and trace route

Using PING and TRACERT from a Workstation

Open the Command prompt (MS-DOS). Ping the IP address/domain name of another computer.

In the window, type ping, a space, and the IP address/domain name.

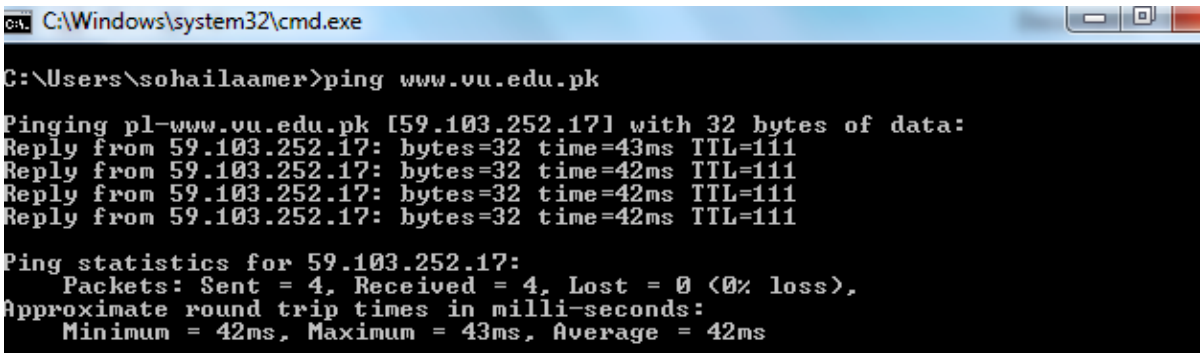
Steps to pinging vu website

1-your computer should be connected to Internet

2-GO to Dos Prompt

3-On Dos Prompt write ping www.vu.edu.pk

4-you will see the Pinging statistics



Ping uses the Internet Control Message Protocol (ICMP) echo-request and echo-reply feature to test physical connectivity. If there will some Dropped or lost packets then a ping/trace route will usually show as asterisks (*).

Ping the following addresses and verify that result was successful.

1. www.pucit.edu.pk

2. www.aiou.edu.pk

3. www.pu.edu.pk

4. www.microsoft.com

5. www.mit.edu.

Understanding OUTPUT:

Interesting result from ping is round-trip time calculation. Measured in milliseconds, round-trip time indicates the delay between the sending of a ping request packet and the receipt of the corresponding response packet. The network delay or latency indicated by ping offers a good indicator of the responsiveness of network services on that remote host.

By default, ping waits Approximately **4,000 milliseconds (4 seconds)** for each response to be returned before displaying the "Request Timed Out" message.

In computer networks, **hop refers** to the number of routers that a packet (a portion of data) passes through from its source to its destination. The hop count represents the total number of routing devices **Time-to-live (TTL)** fields. The TTL value specifies approximately how many router (i.e. hops) the packet has gone through.

Adding to other answers that TTL value at max can be 255 because of its size of 1 byte (8 bits -> max. value = 255). On the internet, it usually takes very less Round Trips (RT)

to finish a request. And within each trip, with each hop TTL decreases. Most of the requests get finished without even reaching TTL value zero

You sent a request to some non-existent server/URL. At first its TTL was 255. It keeps on hopping from 1 router to other, until TTL becomes zero, which is when a packet is dropped.

Why we do not need a bigger TTL value.

Because if you set TTL of a packet to 10000, it keeps on hopping from one device to another (eventually TTL reaches zero and packet gets dropped) if the server/URL is not found. But think if there are lots of packets circulating, without a destination, slowing the whole network.

Error messages:

- **Destination Host Unreachable**

This message indicates one of two problems: either the local system has no route to the desired destination, or a remote router reports that it has no route to the destination. When you ping to an IP address in a different network, ping packets reach default gateway. My default gateway sends the packet to remote gateway. So ping packet finally reaches the remote network. However, if the remote gateway failed to find the remote host, it will send an echo Destination host unreachable.

The error message "Destination Host Unreachable" tells that the ping request from our computer cannot find the route to the destination IP address (destination host). It means, the Packet send from your computer reached the destination network successfully but the remote gateway failed to find the destination host. So the remote gateway sends an Echo message Destination host unreachable. One major possibility for this error; there is no route listed in the remote gateway, for the packet send from your computer, to the destination host. If there is no route available, it is impossible to find the destination host and your computer will receive an error message "Destination host unreachable" from remote gateway. So one of the reason for this can be faulty routing table. If the

destination host is down at the time the packet send, it may result destination host unreachable error message.

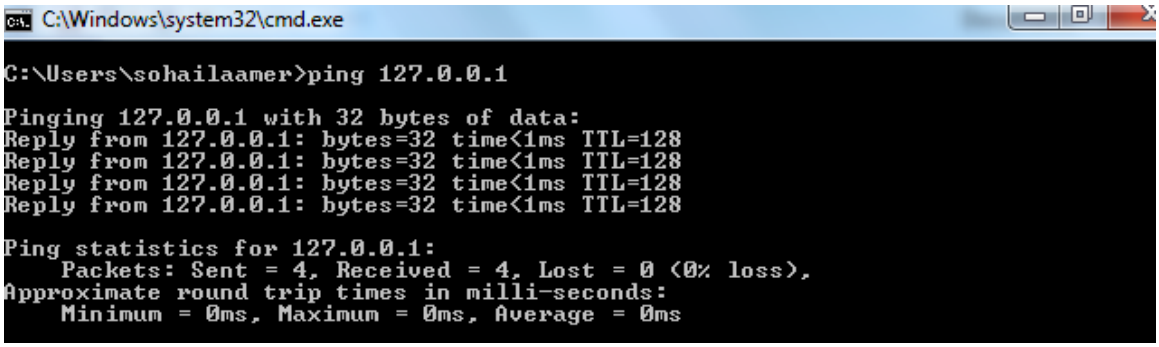
- **Request Timed Out**

By default, ping/trace route waits Approximately 4,000 milliseconds (4 seconds) for each response to be returned before displaying the "Request Timed Out" message. This error message indicates that your host did not receive the ping message back from the destination device within the designated time period. This is an indicator that the destination device is not connected to the network, is powered off, or is not configured correctly.

Step-5: Ping the computer's loopback IP address.

Type the following command:

ping 127.0.0.1



The address 127.0.0.1 is reserved for loopback testing. If the ping is successful, then TCP/IP is properly installed and functioning on this computer.

Step-6: Using trace route command

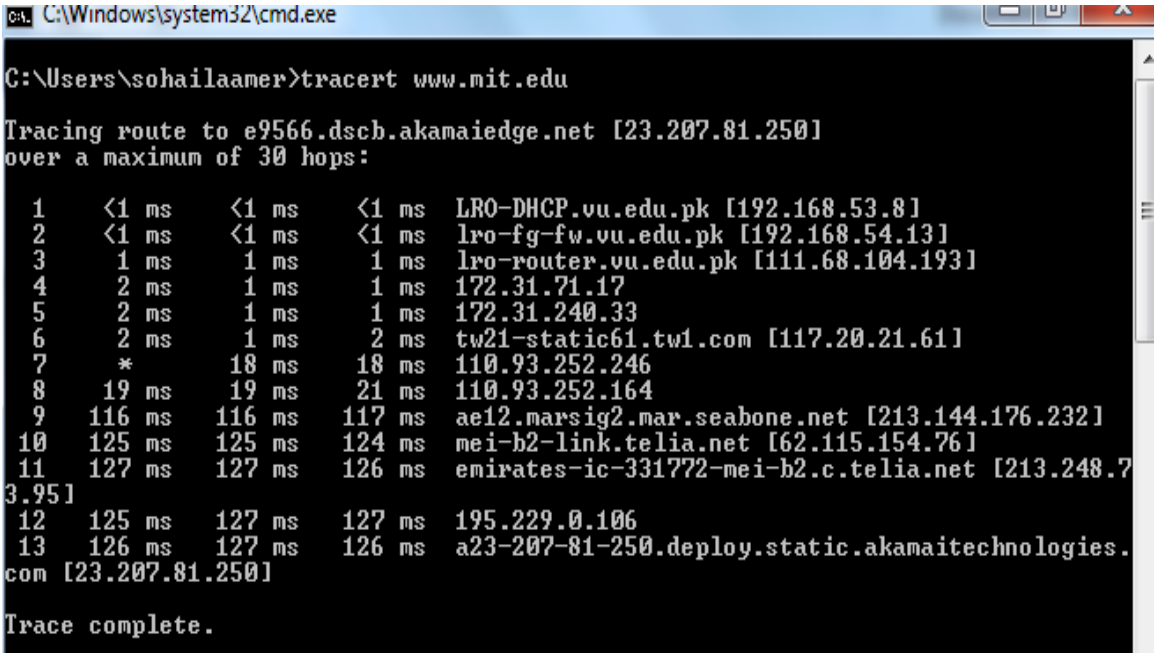
Steps to tracing

1-your computer should be connected to Internet

2-GO to Dos Prompt

3-On Dos Prompt write tracert www.mit.edu

4-you will see the trace route statistics



The result shows the complete route to the site and the number of hops in path. Traces out the route and permits you to access a particular server. The tracert command gives you a long list of all servers you pass through on your way to your final destination. (Maximum up to 30hops- if your route is longer than that, you are expected to give up waiting for the connection).

Tracert uses the same echo requests and replies as the ping command but in a slightly different way. Observe that tracert actually contacted each router three times. Compare the results to determine the consistency of the route. Each router represents a point where one network connects to another network and the packet was forwarded through.

- Trace a local host name or IP address in your local area network (LAN).

Mechanism to Conduct Lab:

Students and teacher communicate through Adobe Connect.

Lab 2

Before starting this lab you are required to follow the below procedures:

1. Download Packet Tracer Simulation Tool on your PC.

2. To get familiar with the Packet Tracer environment, watch this video named "Interface Overview" from the Help Tutorials.

In this Lab, we will design a simple network topology by selecting some devices and suitable media. In this lab we will keep it simple by using End Devices, Switches, Hubs, and Connections.

Packet Tracer Workspaces

Two workspaces are supported; logical and physical. In the logical workspace, we can build logical network topologies by placing, connecting, and clustering virtual network devices. The physical workspace provides a sense of scale and placement in how network devices such as routers, switches, and hosts would look in a real environment

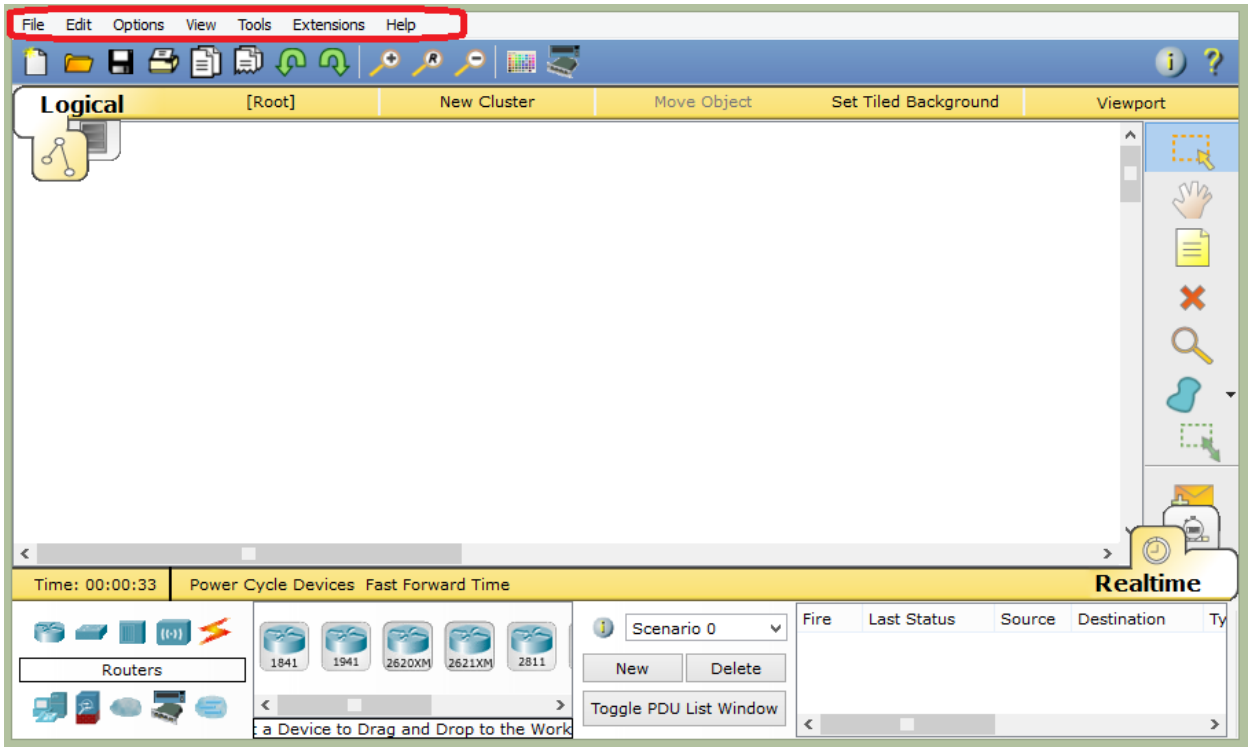
Packet Tracer Modes

It supports two operating modes—real-time mode and simulation mode. In real-time mode, all network activities take place with immediate real-time response. The simulation mode allows a user to control time intervals, and the propagation of data across a network.

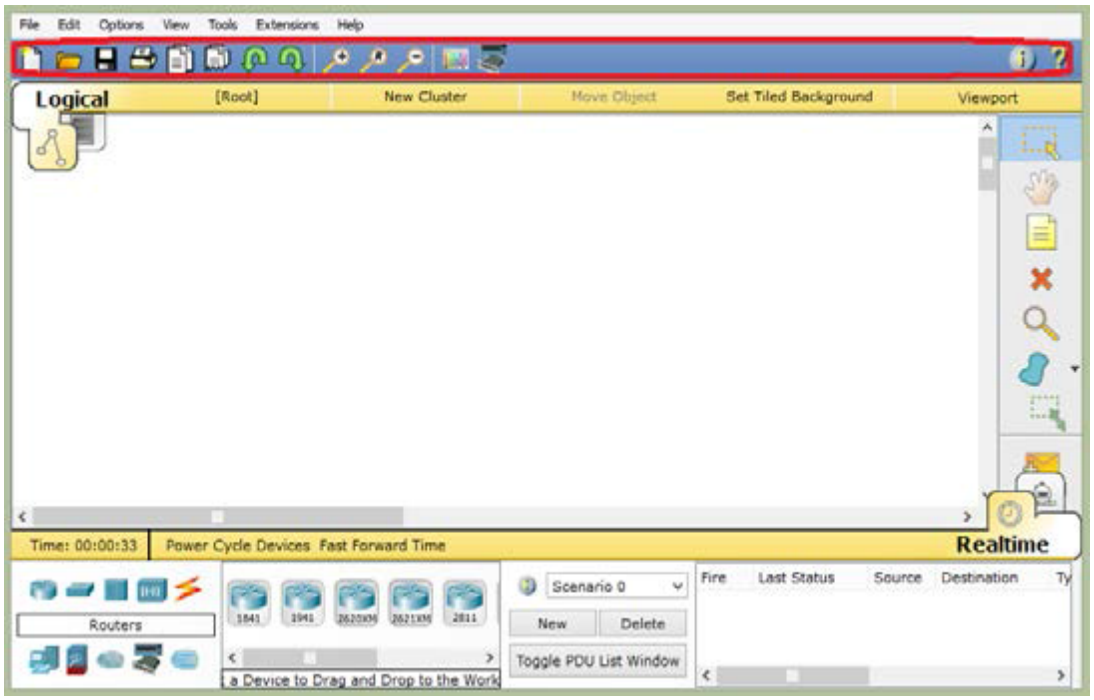
Installation of Packet Tracer

You can download the latest version of Packet Tracer for free from www.netacad.com/about-networking-academy/packet-tracer.

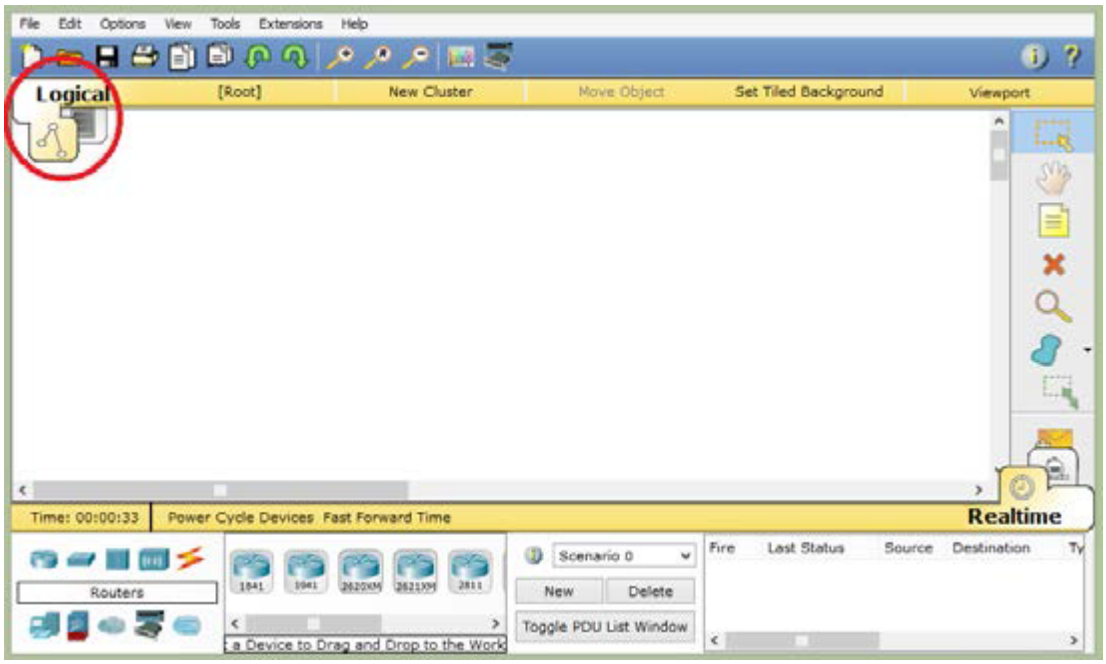
Menu bar: This is a commonly found menu. It is used to open, close, print, save, change preferences, and so on.



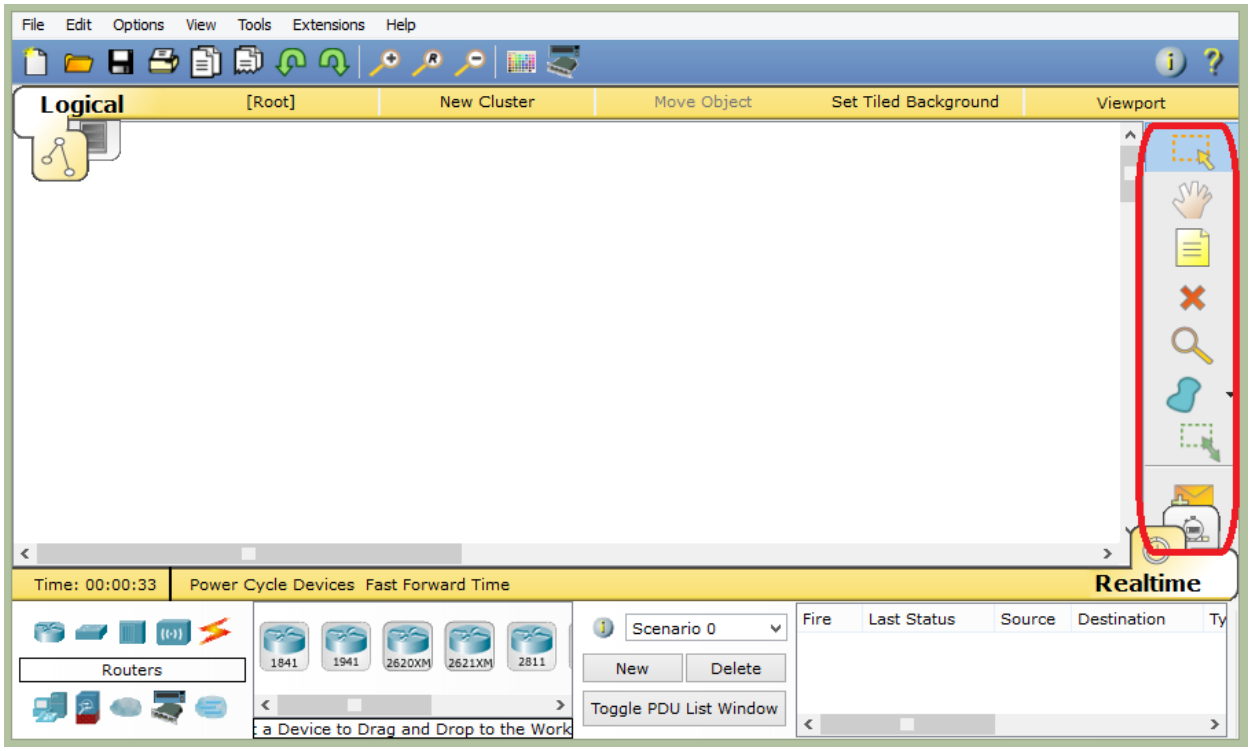
Main toolbar: This bar contains shortcut icons to menu options that are frequently accessed. For example, open, save, zoom, undo, and redo.



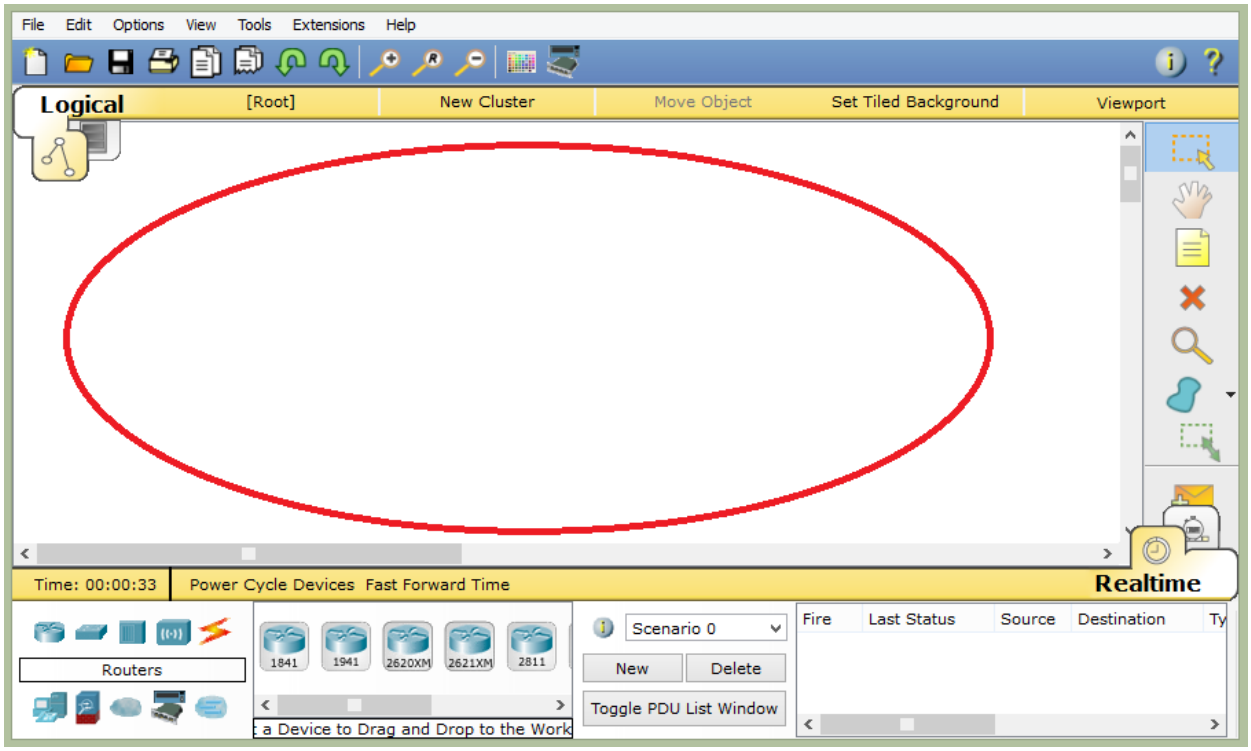
Physical/Logical workspace tabs: These tabs allow you to choose between the **Logical** and **Physical** work areas.



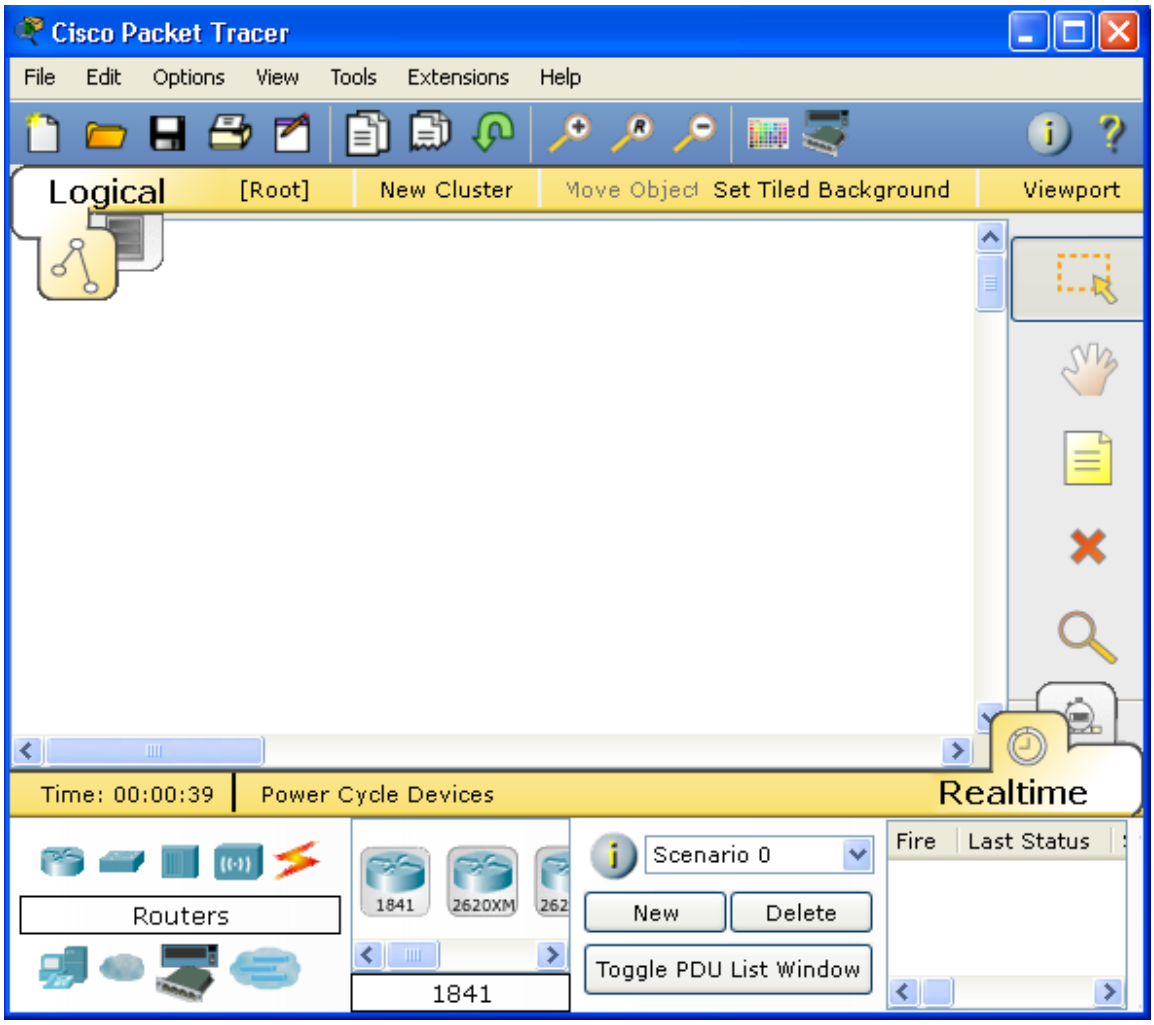
Common tools bar: With help of this toolbar, we can manipulate topologies. For example, select, move layout, place note, delete, resize shape, add simple/complex Protocol data unit (PDU).



Workspace: Here, we create topologies and display simulations.

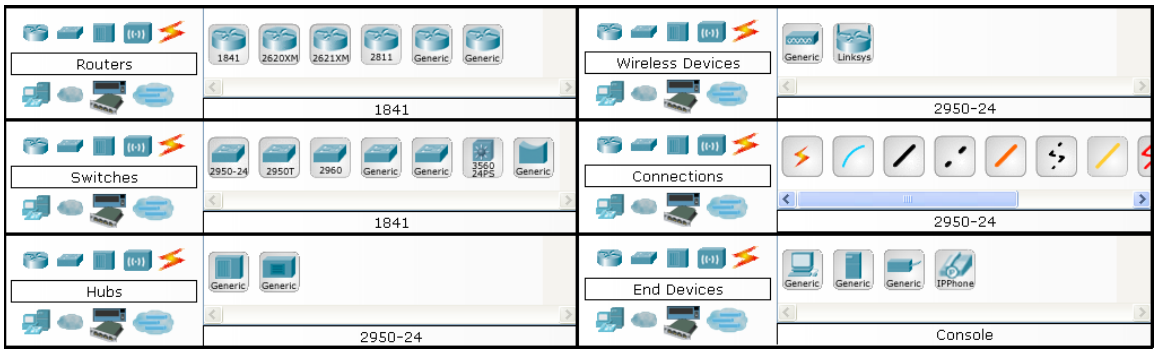


Step 1: Start Packet Tracer and enter in logical work space



Step 2: Choosing Devices and Connections

Single click on each group of devices and connections to display the various choices

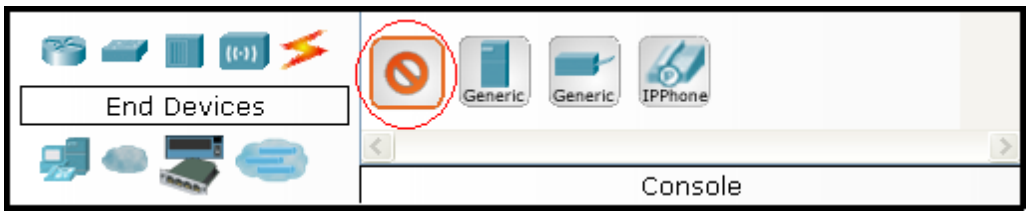


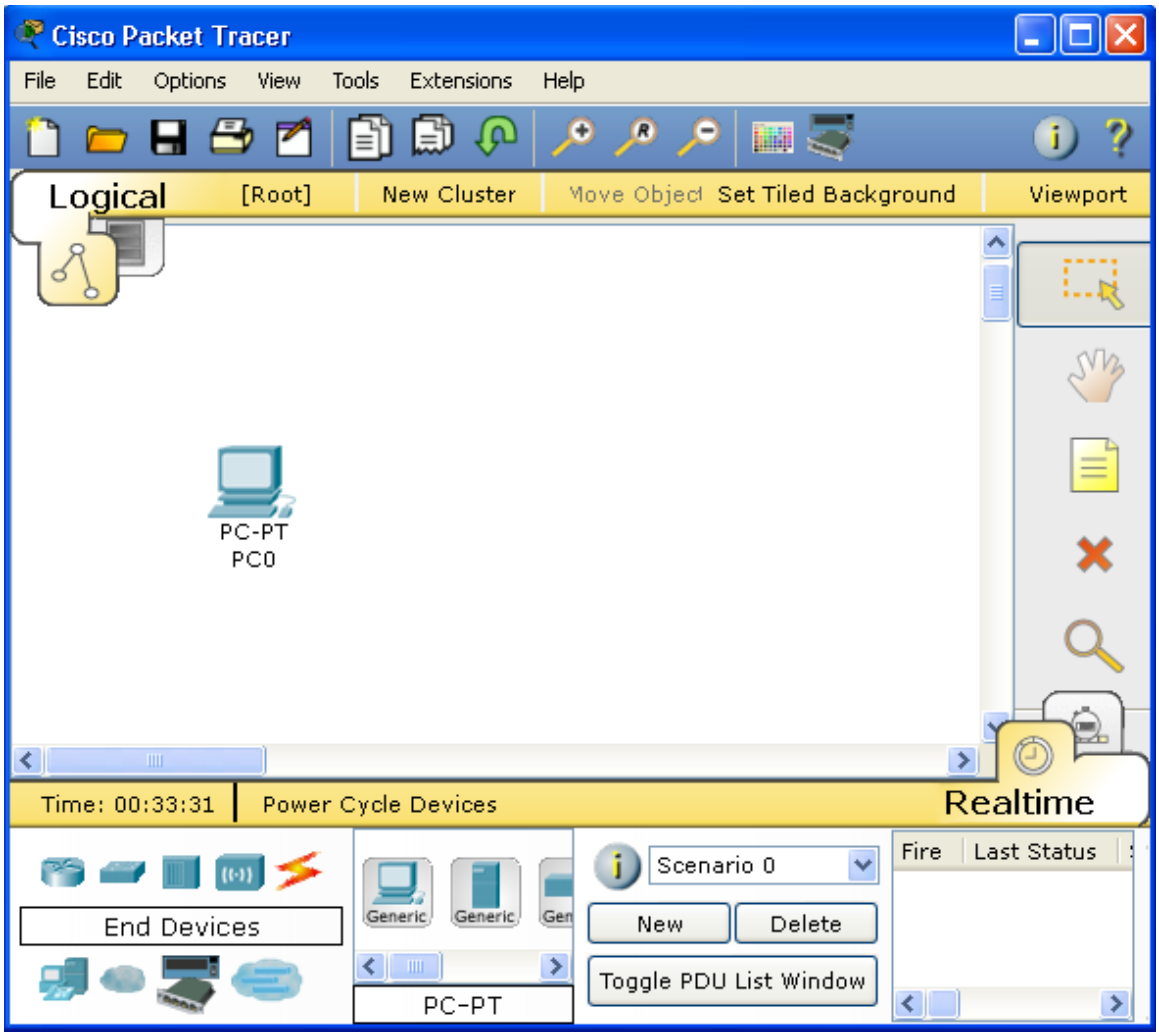
Step 3: Adding Hosts using device selection option

- Single click on the End Devices icon.

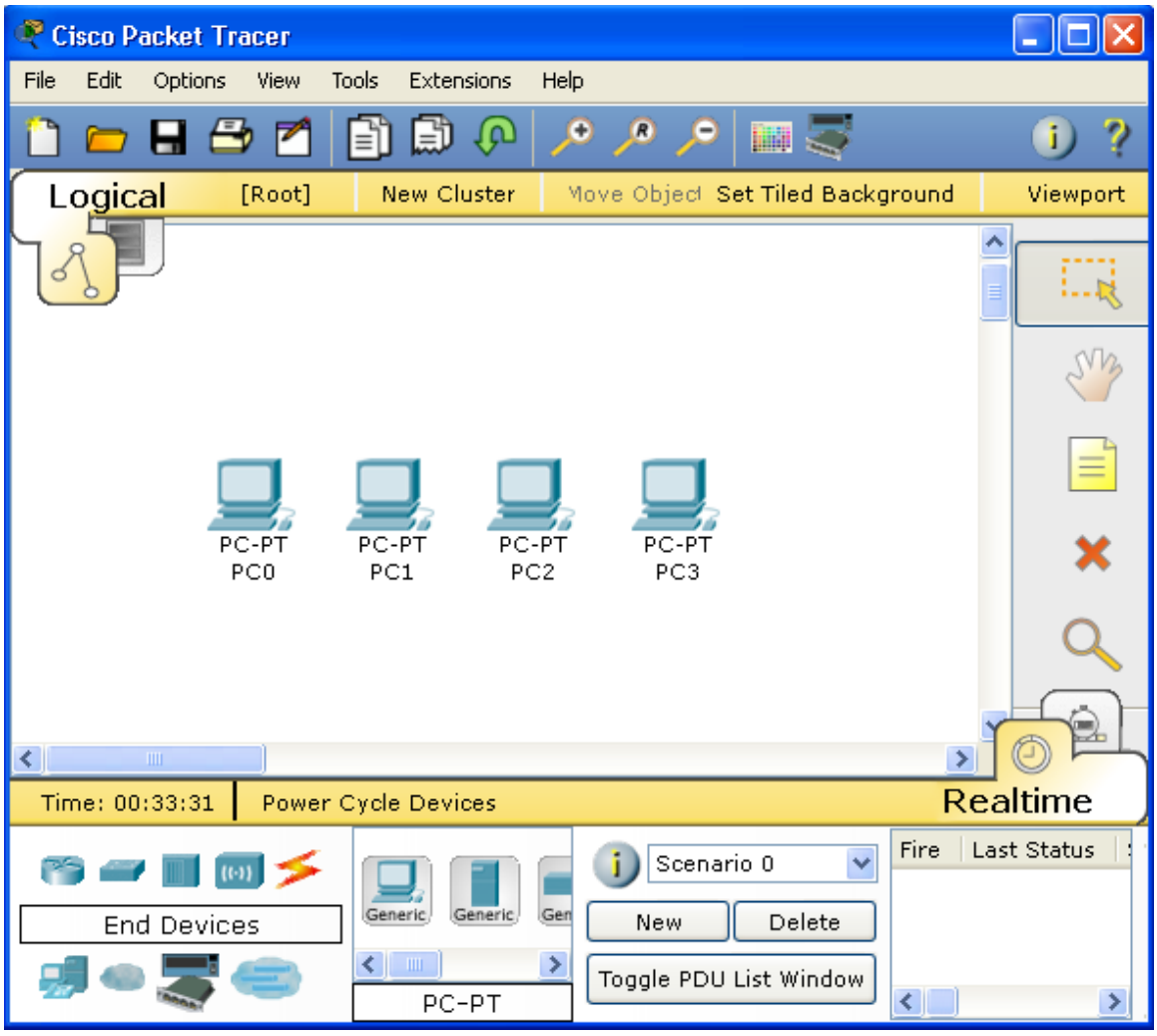


- Single click and drag the Generic host.



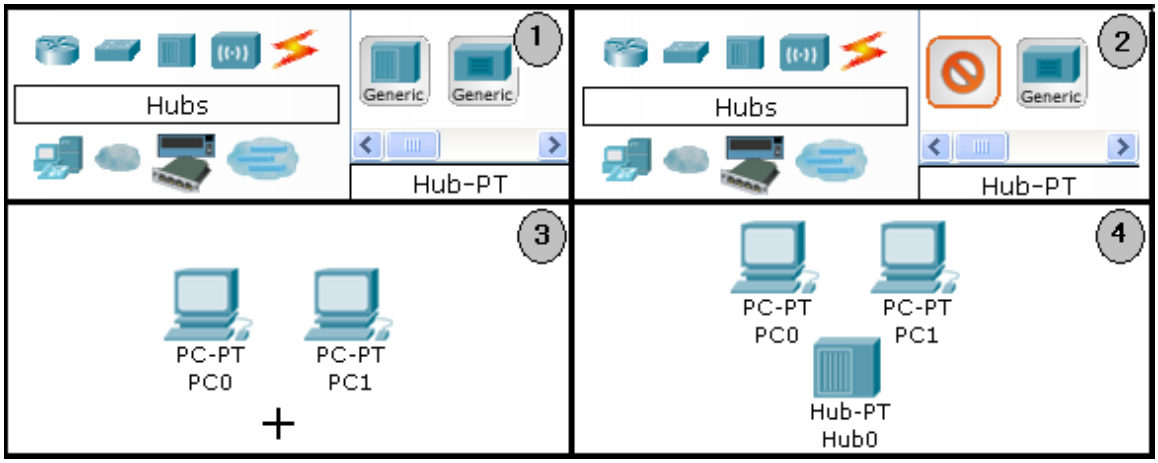


- Add three more hosts.

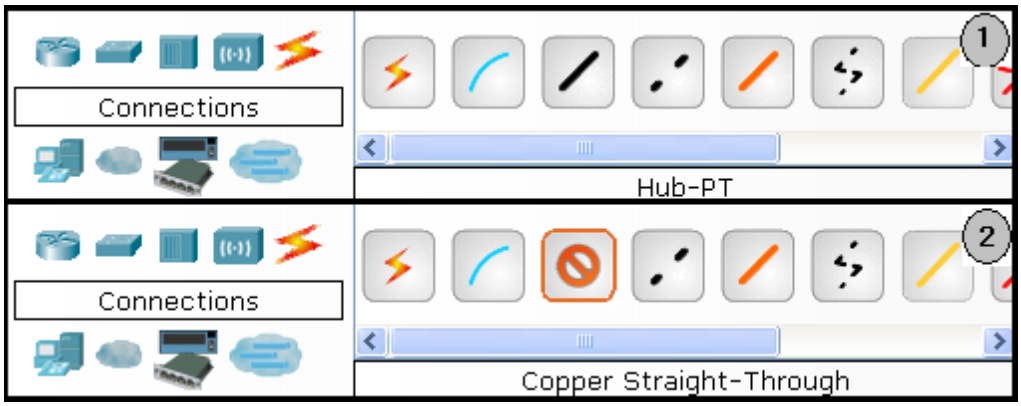


Step 4: Connect the Hosts to Hub and Switch

Adding a Hub: Select a hub, by clicking once on Hubs and once on a Generic hub.

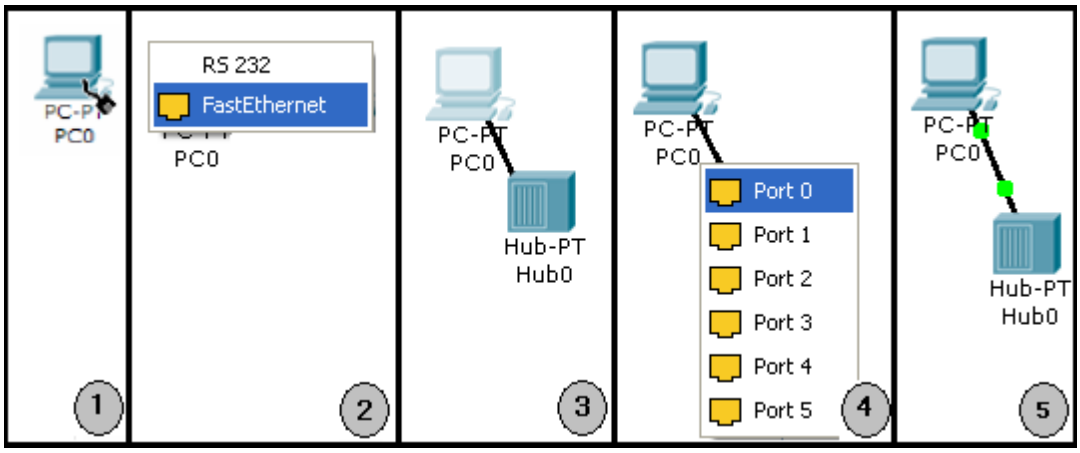


- Connect PC0 to Hub0 by a suitable Connection.
- Click once on the Copper Straight-through cable.

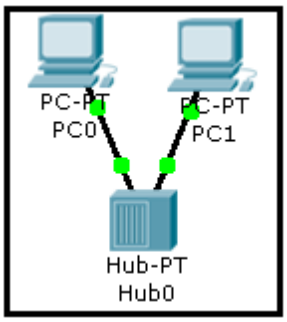


Perform the following steps to connect PC0 to Hub0:

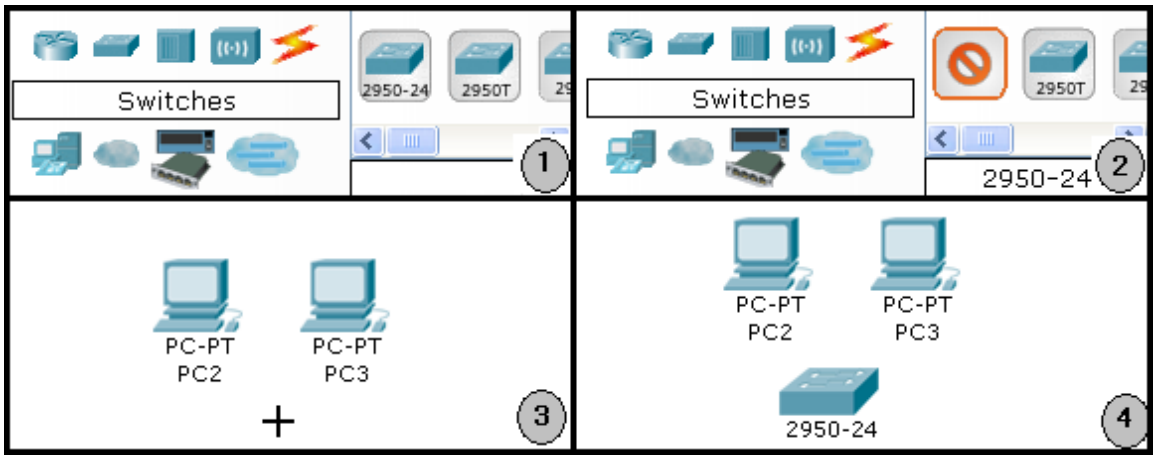
1. Click once on PC0
2. Choose Fast Ethernet
3. Drag the cursor to Hub0
4. Click once on Hub0 and choose Port0
5. Notice the green link lights on both the PC0 Ethernet NIC and the Hub0 Port0 showing that the link is active



- Repeat the steps above for PC1 connecting it to Port1 on Hub0.



Adding a Switch: Select a switch, by clicking once on Switches and once on a 2950-24 switch.



- Connect PC2 to Switch0 by first choosing Connections.
- Click once on the Copper Straight-through cable.
- Perform the following steps to connect PC2 to Switch0:

1. Click once on PC2

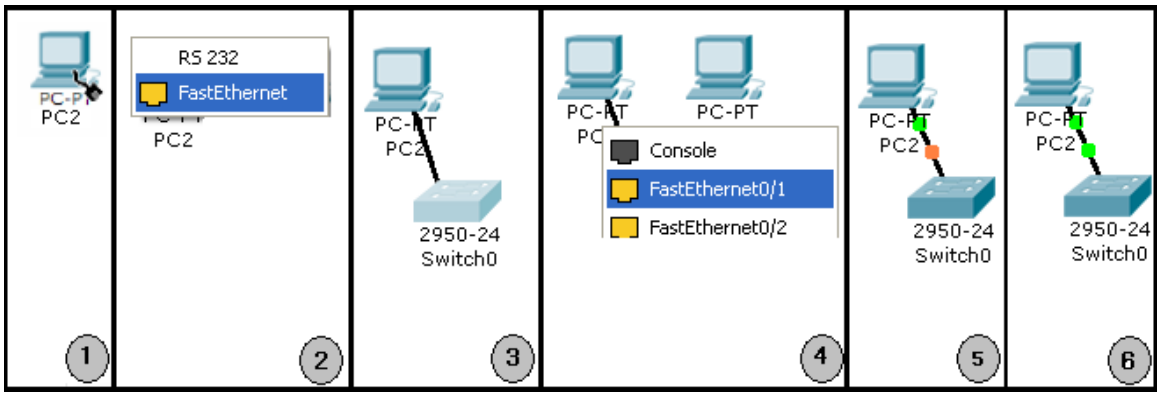
2. Choose Fast Ethernet

3. Drag the cursor to Switch0

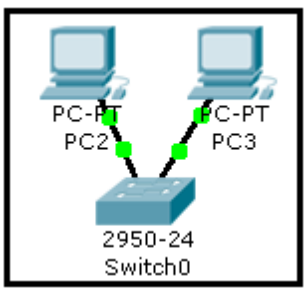
4. Click once on Switch0 and choose FastEthernet0/1

5. Notice the green link lights on PC2 Ethernet NIC and amber light Switch0 FastEthernet0/1 port. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.

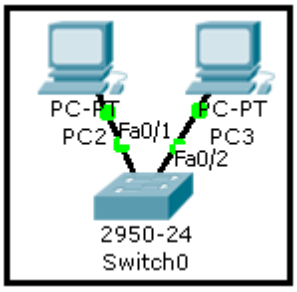
6. After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now be forwarded out the switch port.



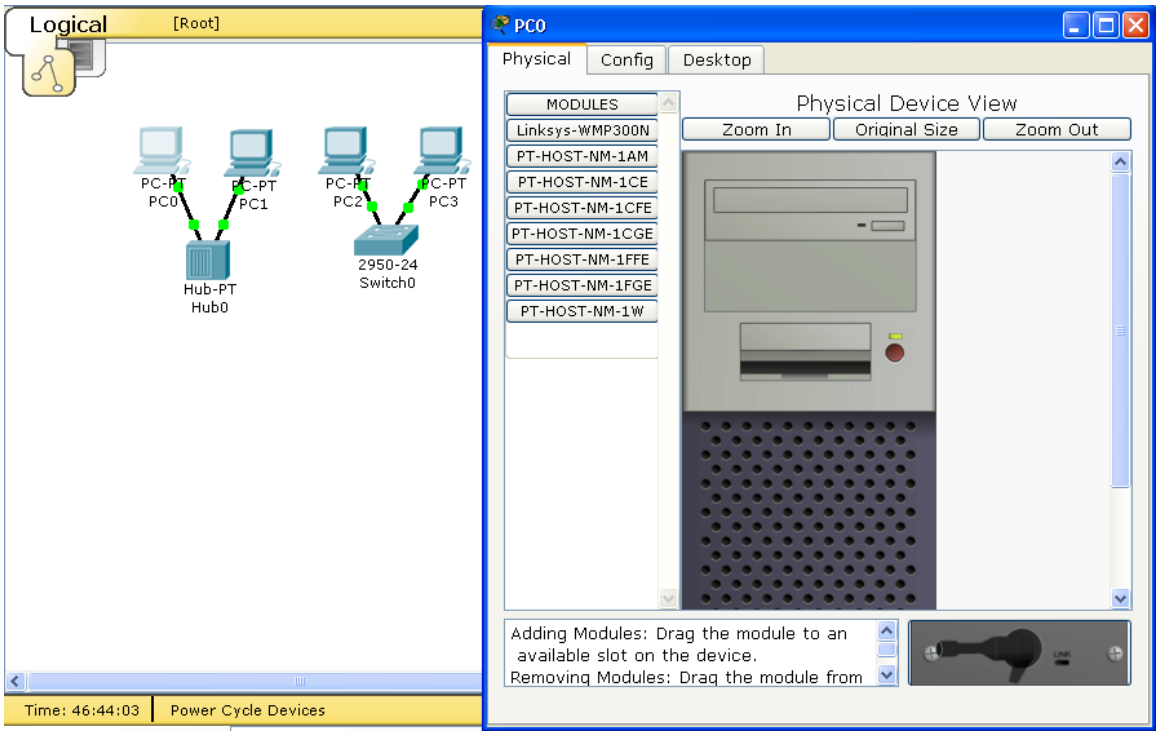
- Repeat the steps above for PC3 connecting it to Port3 on switch0 on port FastEthernet0/2.



- Move the cursor over the link light to view the port. Fa means FastEthernet, 100 Mbps Ethernet.

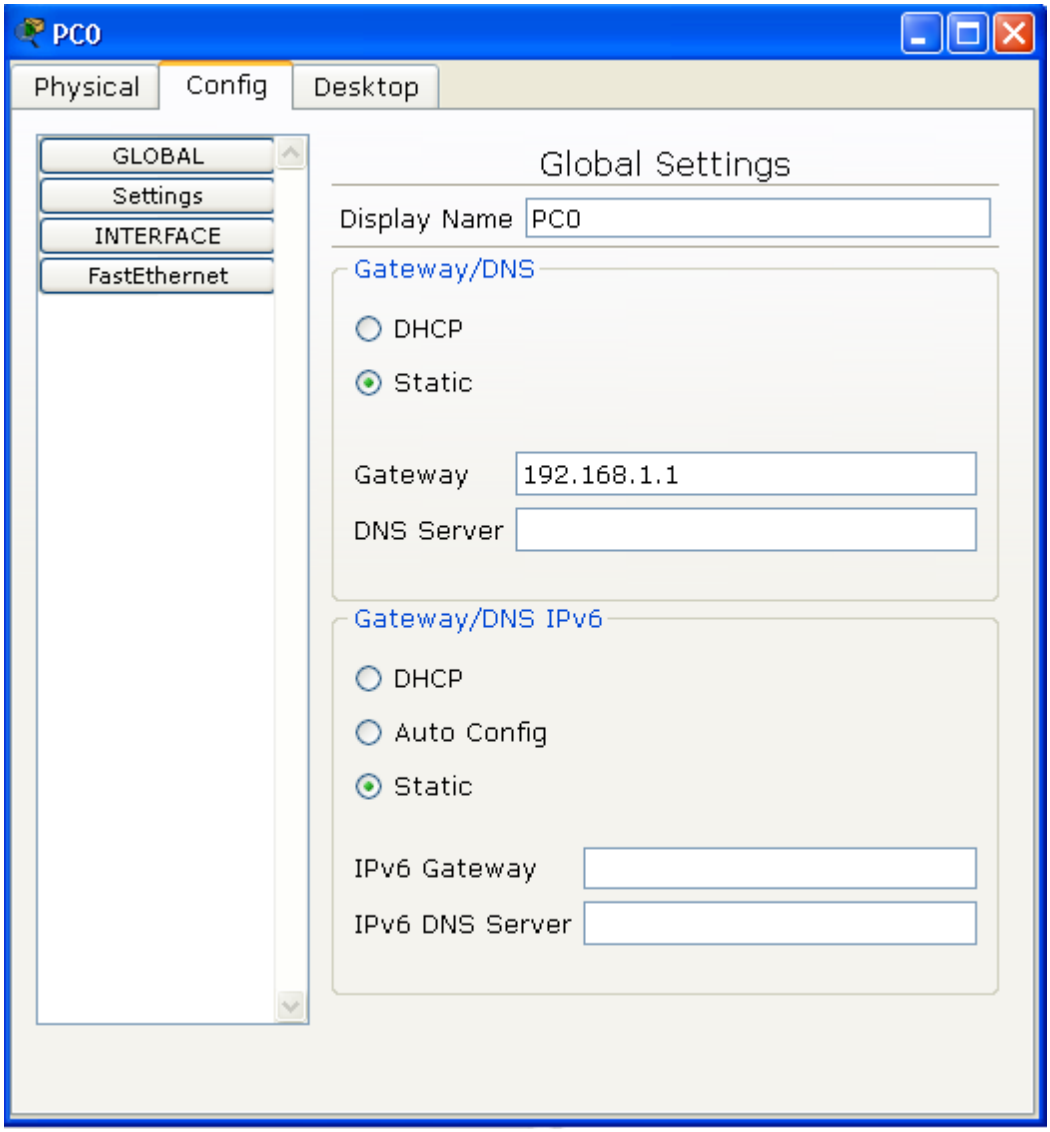


Step 5: Configuring IP Addresses and Subnet Masks on the Hosts Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

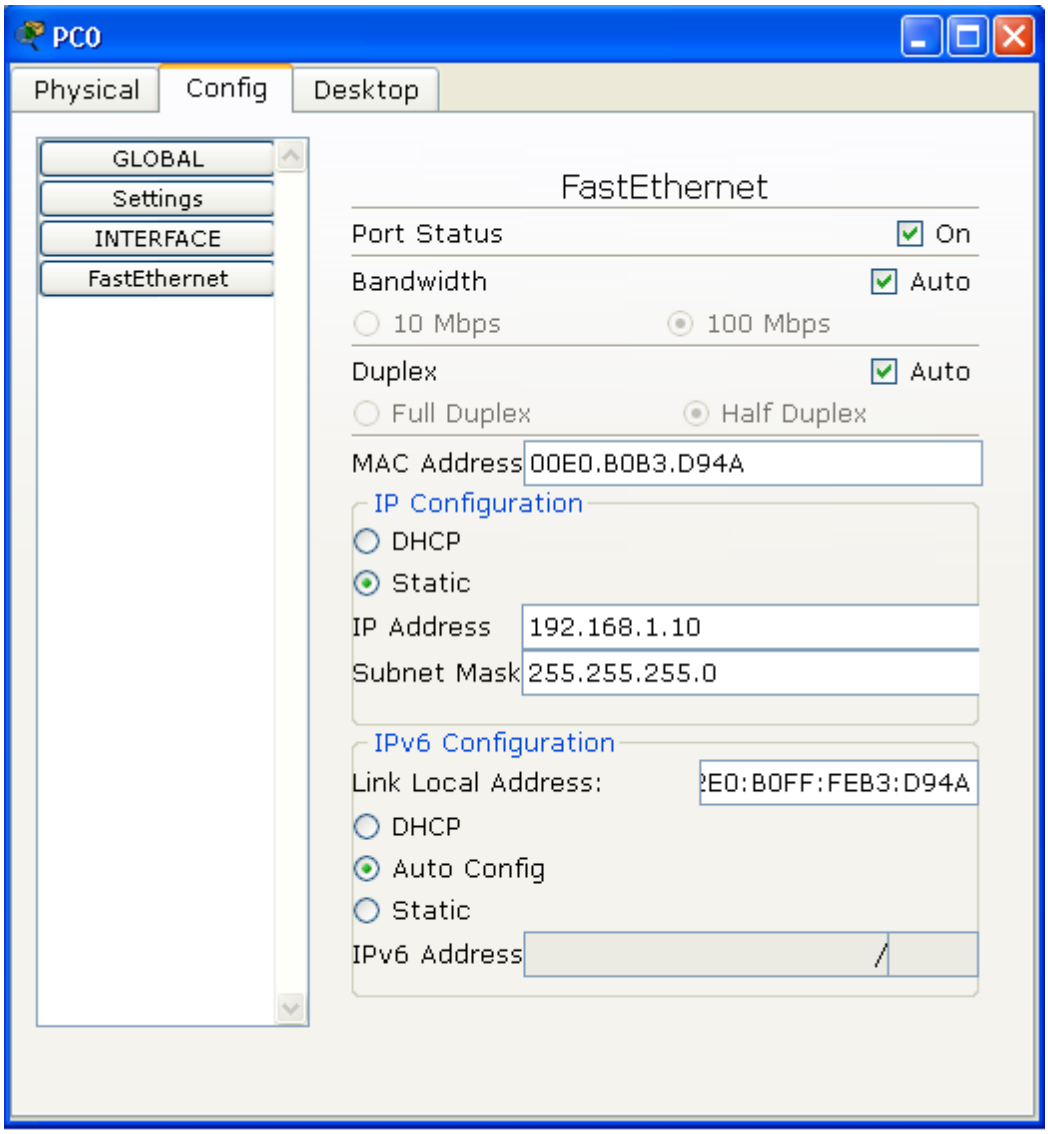


- Click once on PC0.

- Choose the Config tab. It is here that you can change the name of PC0. It is also here where you would enter a Gateway IP Address, also known as the default gateway. If you want, you can enter the IP Address 192.168.1.1.



- Click on FastEthernet. Add the IP Address to 192.168.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.255.0.

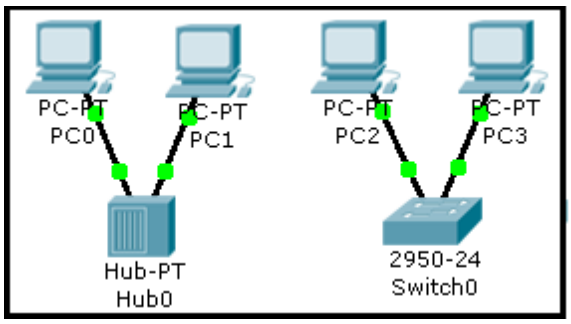


Connecting Hub0 to Switch0

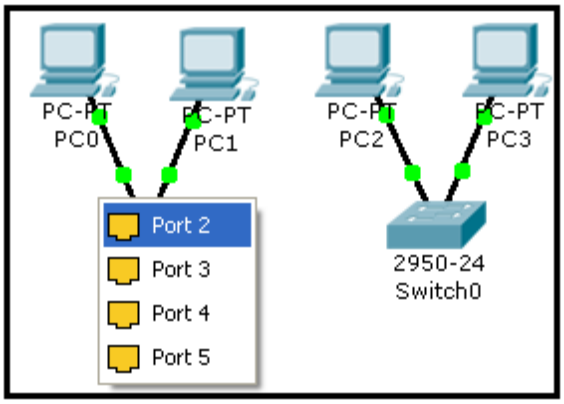
- To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable. Click once the Cross-over Cable from the Connections options.



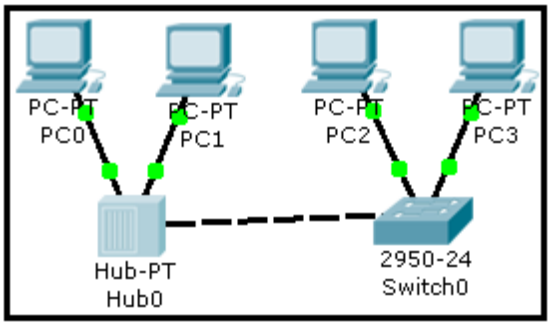
- Move the Connections cursor over Hub0 and click once.



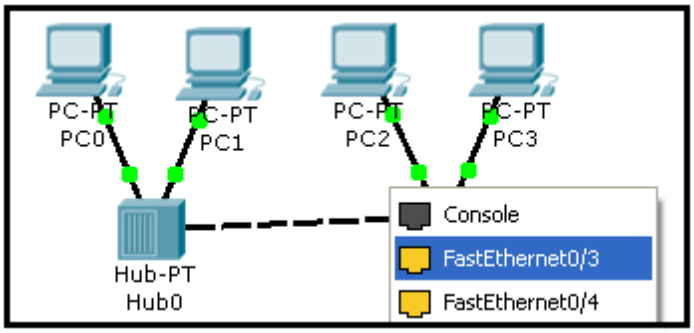
- Select Port2



- Move the Connections cursor to Switch0.



- Click once on Switch0 and choose FastEthernet0/3 (actual port does not matter).



- The link light for switch port FastEthernet0/3 will begin as amber and eventually change to green

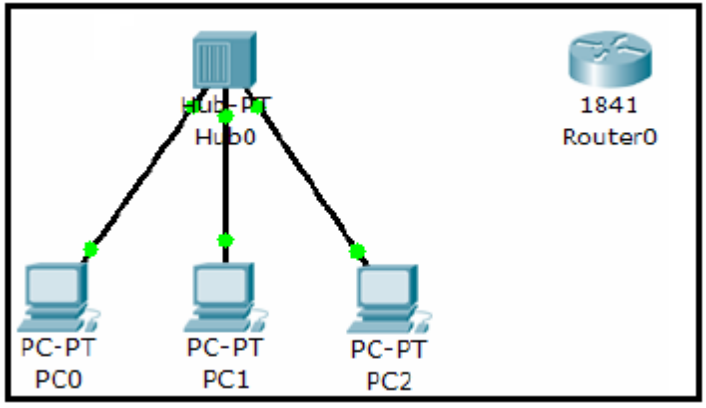
Adding routers

- In the Network Component Box, click on the router



- Select an 1841 router.

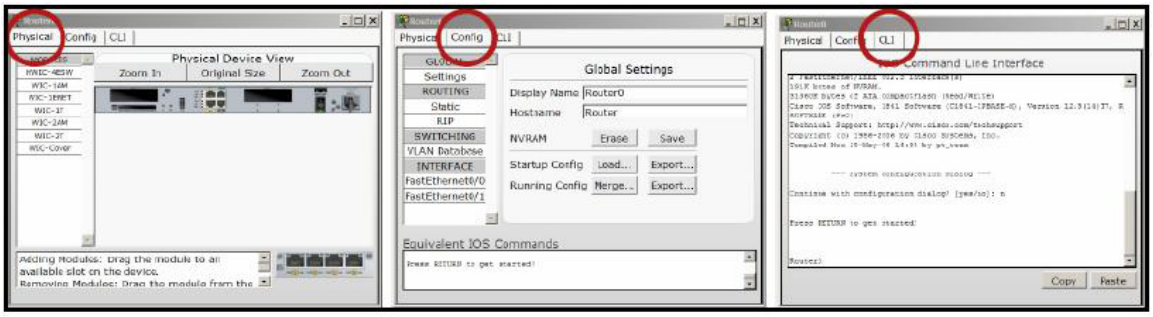




- Move the cursor to the Logical Workspace and click on the desired location.

NOTE: If multiple instances of the same device are needed press and hold the **Ctrl** button, click on the desired device, and then release the **Ctrl** button. A copy of the device will be created and can now be move to the desired location.

- Click on the router to bring up the Configuration Window. This window has three modes: Physical, Config, and CLI (Physical is the default mode).



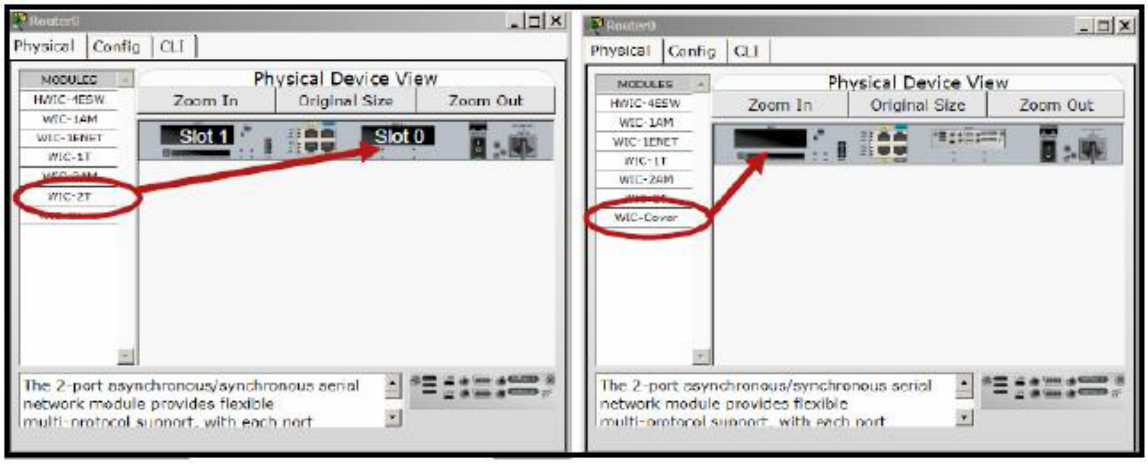
The Physical mode is used to add modules to a device, such as a WAN Interface Card (WIC). The Config mode is used for basic configuration. Commands are entered in a simple GUI format, with actual equivalent IOS commands shown in the lower part of the

window. The CLI mode allows for advanced configuration of the device. This mode requires the user to enter the actual IOS commands just as they would on a live device.

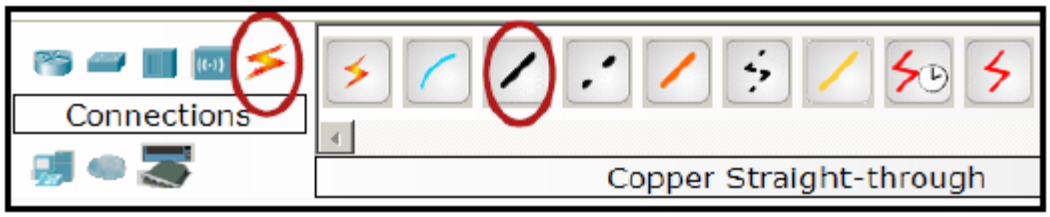
- In the Physical mode, click on the router power switch to turn the device off.



- Select the WIC-2T module and drag it to Slot 0 on the router. Then drag a WIC Cover to Slot1.



- Power the device back on.
- Click on the Network Component Box and select Connections. Then select a Copper Straight-through connection to connect the router to the hub.



- Click on the hub and choose Port 3. Then click on the router and choose interface FastEthernet 0/0.

Mechanism to Conduct Lab:

Students and teacher communicate through Adobe Connect. Students perform the task using the following simulator:

[https://www.netacad.com/courses/packet-tracer]

Packet Tracer Basics-1

https://vulms.vu.edu.pk/Courses/CS610/Downloads/Packet%20Tracer%20Basics-1.mp4

Creating simple topology in Packet tracer

https://vulms.vu.edu.pk/Courses/CS610/Downloads/Creating%20simple%20topolgy%20in%20Packet%20tracer.mp4

Lab 3

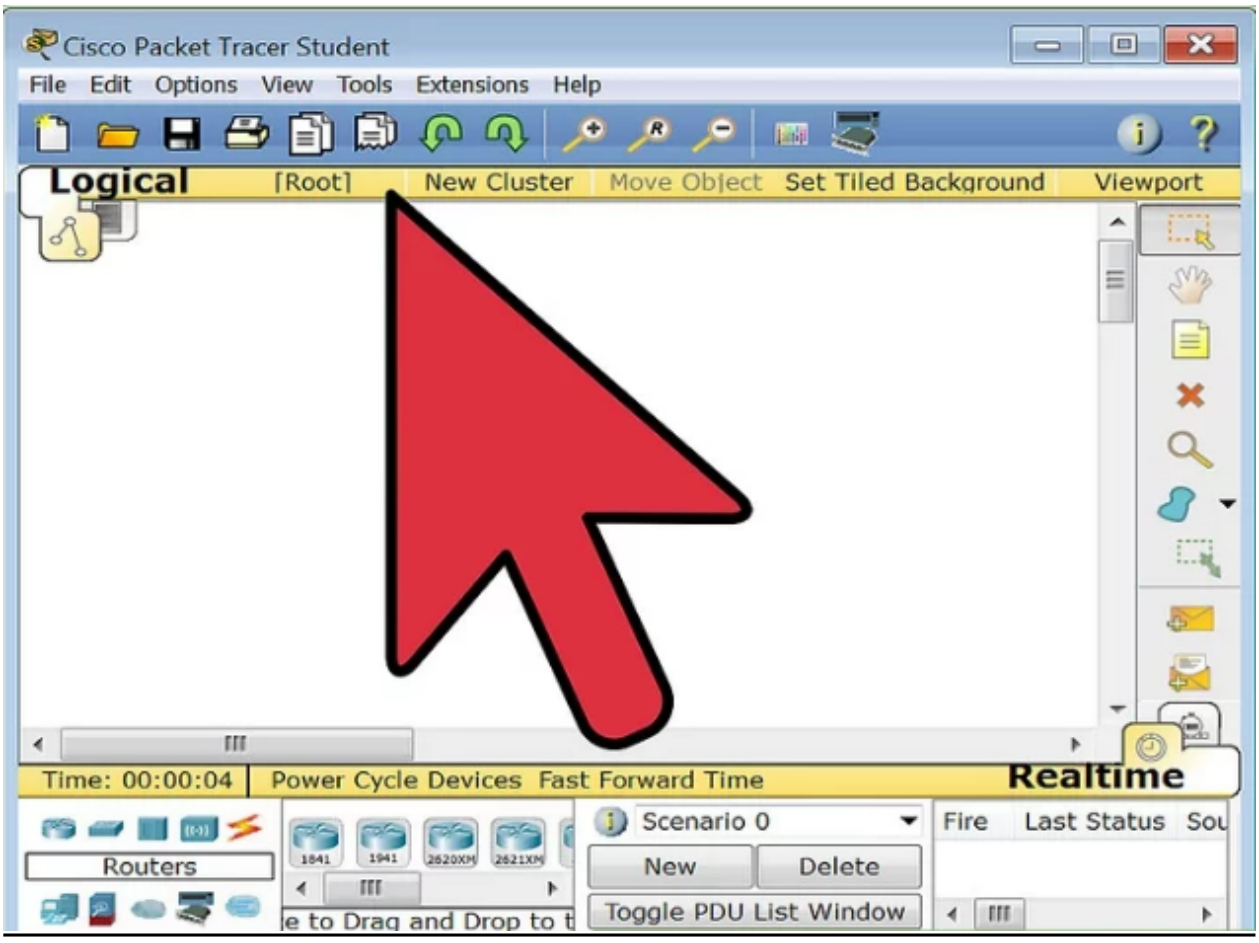
Task: How to configure a network on Packet Tracer.

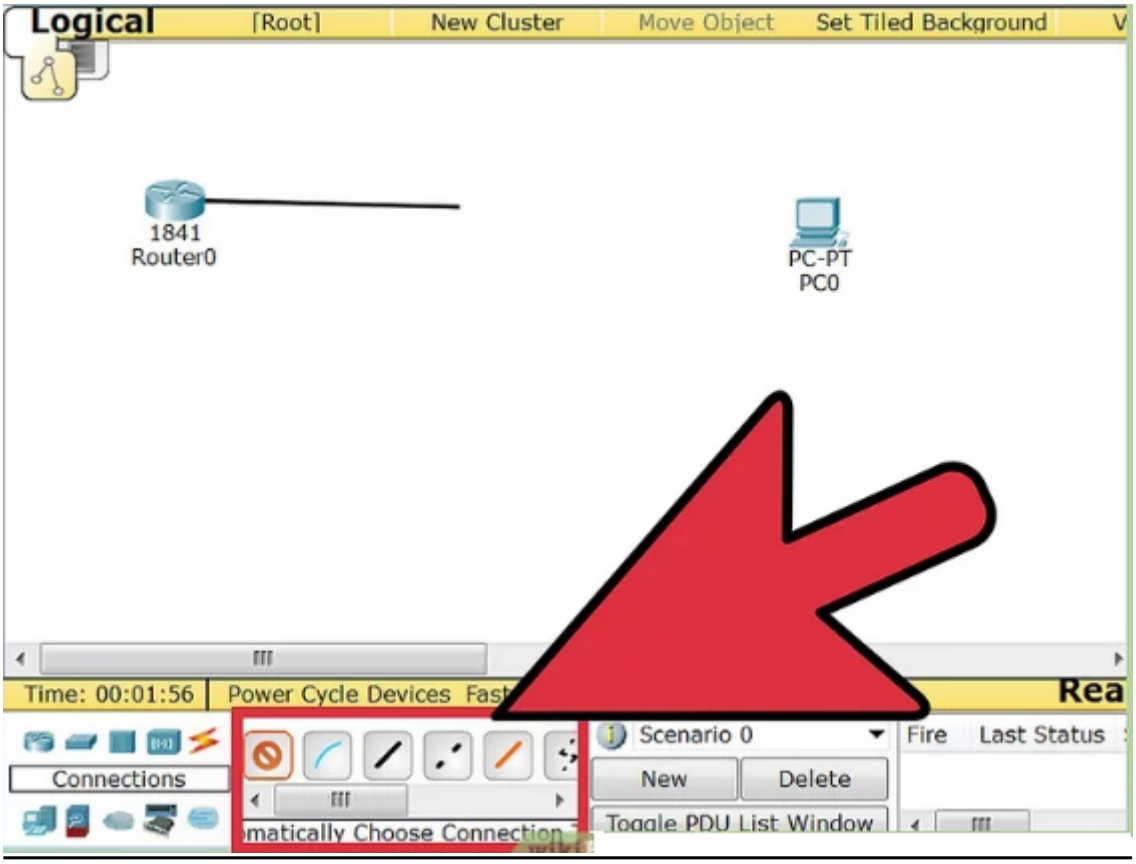
Software: Packet tracer Software

Configure a Network on Cisco Packet Tracer

Step1: Open Network Topology.

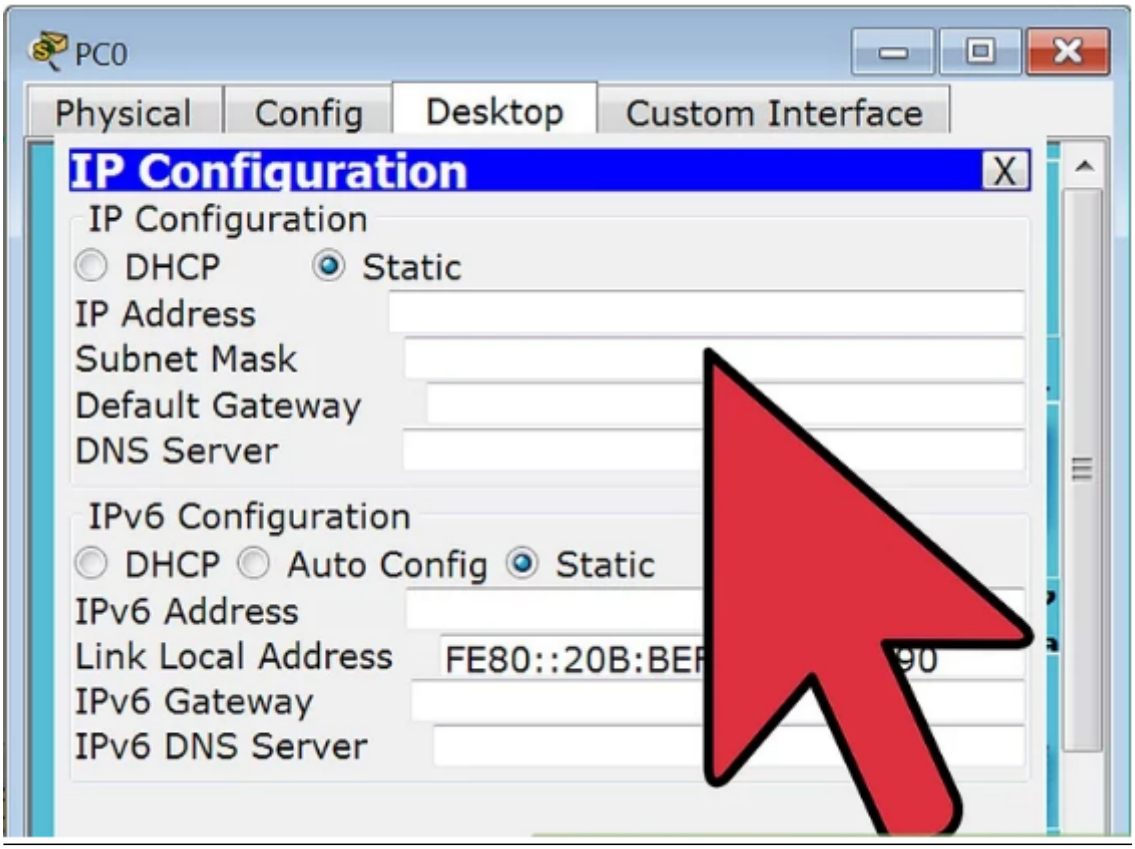
Access your network and identify the components of your network, for example; Servers, Routers, End Devices, etc.





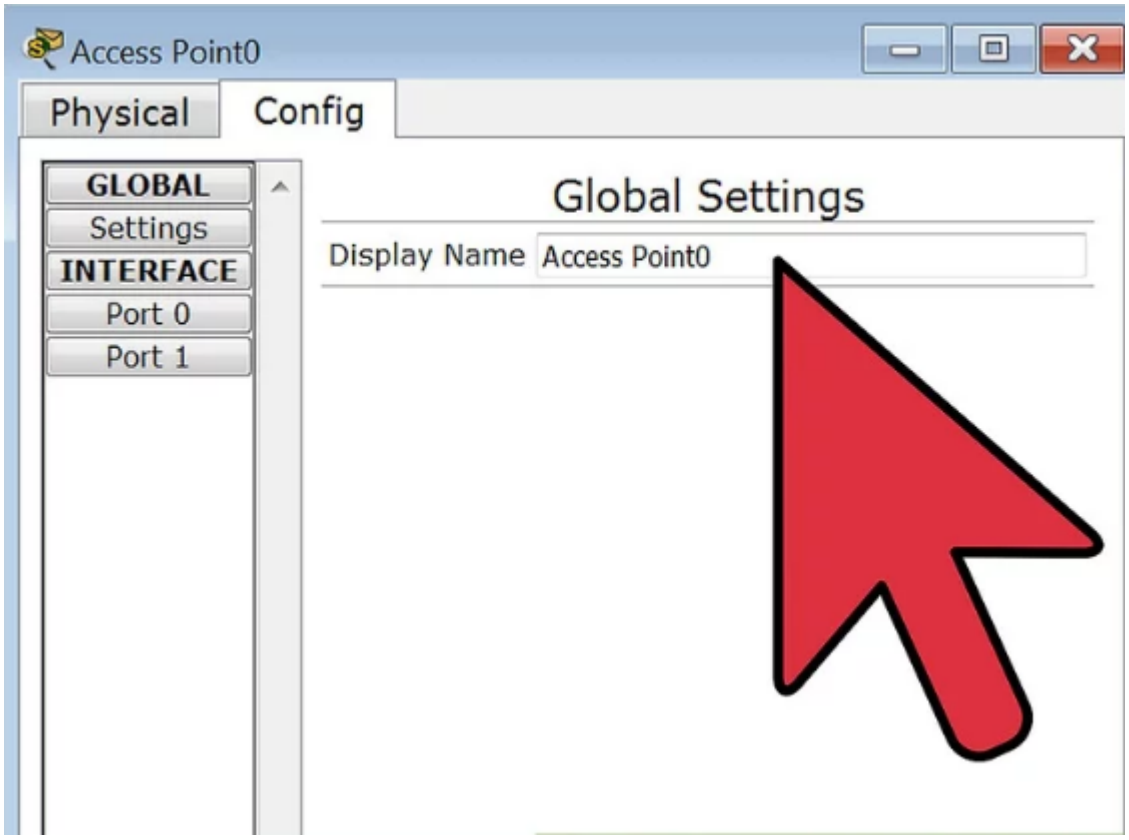
Step2: Complete the cabling.

Access the cables section and connect completely and correctly the cables between the networks in order to ensure connectivity between the devices in the network using the connections table given.



Step 3: Configure the IP addresses on the end devices.

Configure the IP addresses on the end devices. Using the address table still, correctly and completely configure the IP addresses on all end devices. This can be done by accessing the desktop platform on each device and locating the IP configuration section. The reason for doing this is to enable the devices be on the right network.

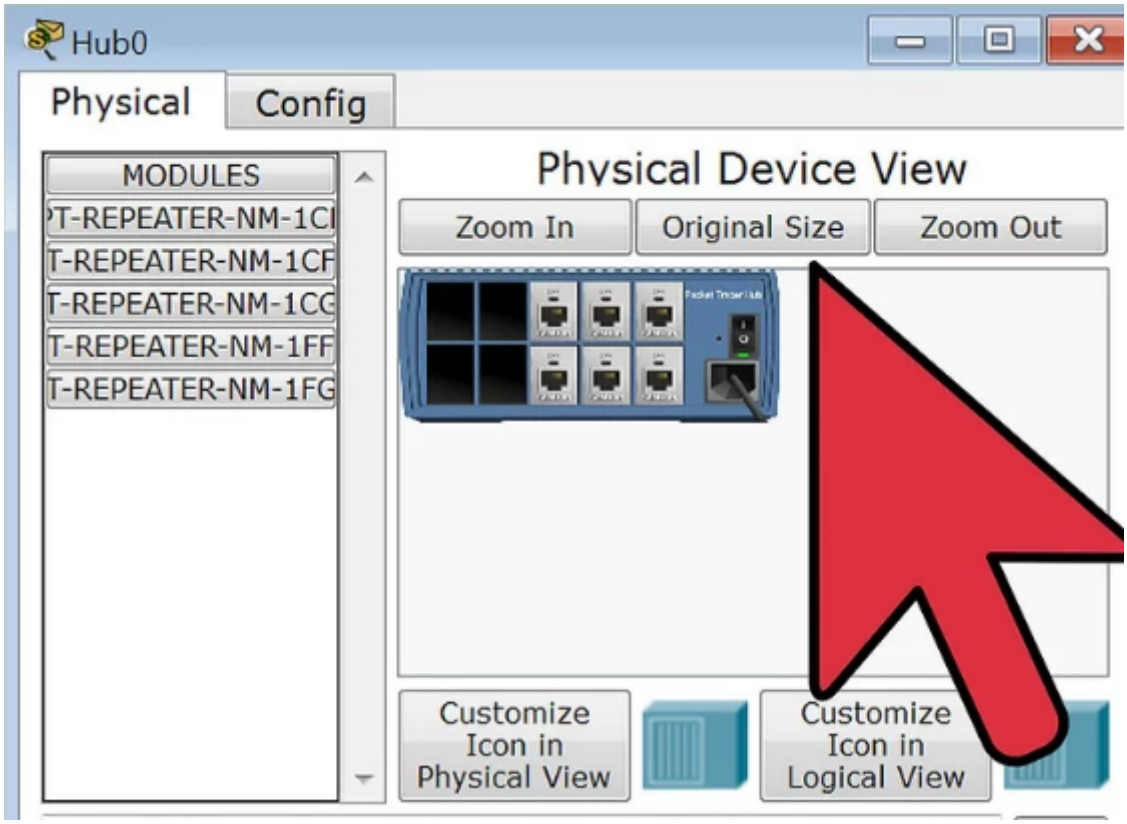


Step 4: Configure the IP addresses on your routers and switches.

After configuring the right IP addresses on the end devices, you will have to do the same on the routers and switches also, using the address table. But this time in a different way because there's no desktop platform on the routers and switches. You will have to access the configuration panel on both devices and this can be done in two ways:

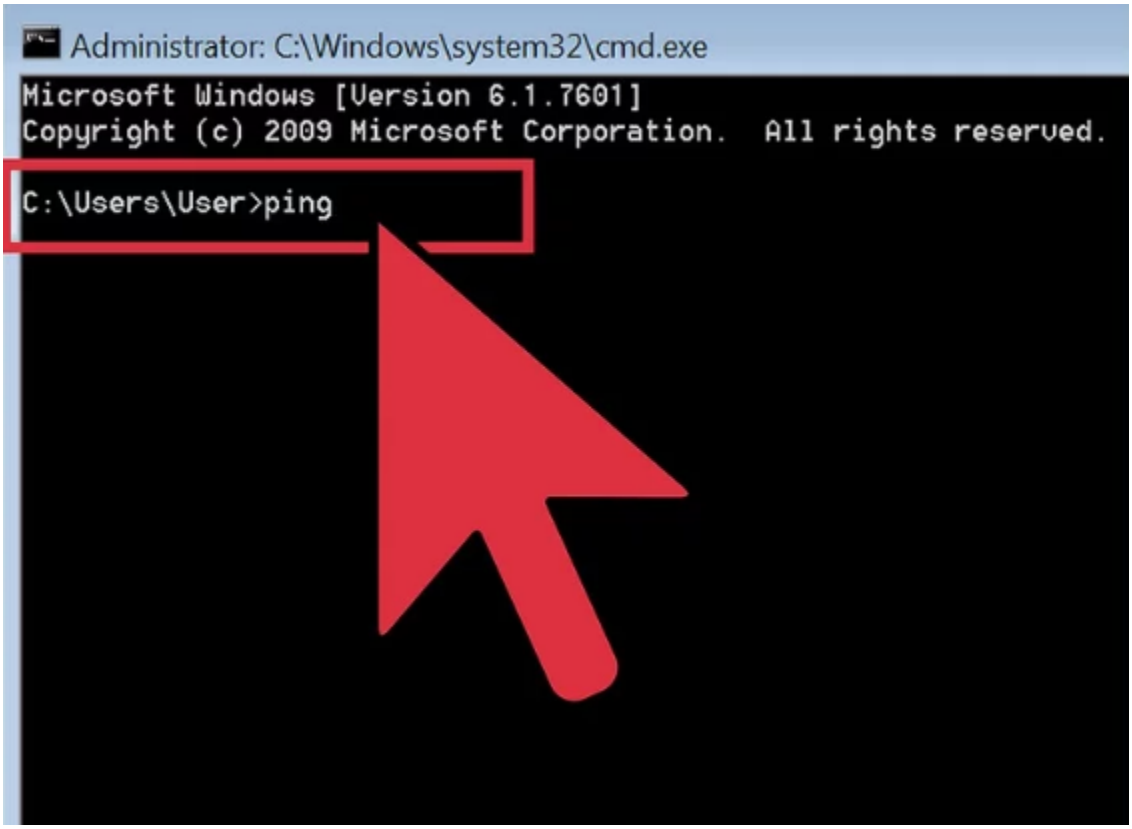
- Click on the device and open the Command Line Interface (CLI) and then type in the right commands to configure the right addresses for the router using the addressing table.

- Use a console cable from an end device and connect it to the device you wish to configure and access the terminal platform on the end device and it will take you to the device's Command Line Interface and then type in the commands in other to configure the right addresses.



Step5: Configure the default gateway.

After configuring the IP addresses, you will need to configure the default gateway also. The reason for this is so the end devices would know what network they are operating on. You can find the default gateway either in the addressing table (if given) or in the network topology.



Step 6: Test connectivity

After configuring the addresses, you will have to test connectivity by opening a command prompt window on the end devices and try pinging the address which the network operates on. If it gives you a reply, it means your network was configured correctly.

Mechanism to Conduct Lab:

Students and teacher communicate through Adobe Connect. Students perform the task using the following simulator:

[https://www.netacad.com/courses/packet-tracer]

Lab 4

To study about different physical equipment used for networking Cables

To transmit the data, medium must exist usually in the form of cables or wireless media. Here are some most commonly used cable types.

Thick Coaxial Cables (thick net) (RG-11)

Thick coaxial cables or thick wire is known as the Ethernet standard RG-11. This cable is mostly used as backbone cable, distributing Ethernet signal throughout a building, an office complex or other large installation. It is used in 10base5 Ethernet standard. The length may be up to 500 meters with a max of five segments connected by repeaters. This gives a total distance of 2500 meters. This is called a network diameter. RG-11 cable is typically orange; with black rings around the cable every 2.5-meter to allow taps into the cable.

Thin coaxial cables (thin net) (RG-58)

RG-58 is typically used for wiring laboratories and offices, or another small group of computers. The maximum length of thin wire Ethernet segment is 185 meters, which is due to the nature of the CSMA/CD method of operation, the cable attenuation, and the speed at which signals propagate inside the coax.

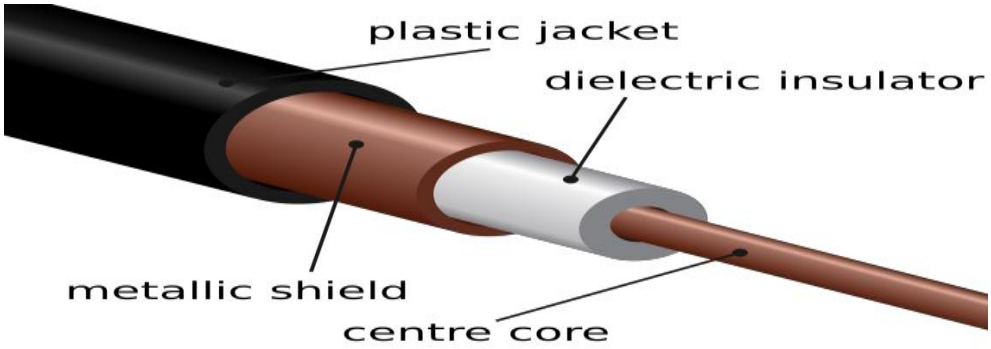


Fig 1: Thin coaxial cables (thin net) (RG-58)

The length is limited to guarantee that collision is detected when machines that are apart transmit at the same time. BNC connectors are used to terminate each end of the cable. When many machines are connected to the same Ethernet segment, a daisy chain approach is used. The BNC connectors allow the network interface card to the next machine. The machine each

`

end of the cable must use a terminating resistor to eliminate collision-causing reflection in the cable.

Coaxial Cable Connectors

Coaxial connectors are needed to connect coaxial cable to devices. The most common type of connector used today is the Bayone-Neil-Concelman, in short, BNC connector.



Fig 2: Coaxial Cable Connector

The three popular types of connectors are: the BNC connector, the BNC T connector, and the BNC terminator. The BNC connector is used to connect the end of the cable to a device, such as a TV set. The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.

The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

Applications

1. Coaxial cable was widely used in analog telephone networks, and later with digital telephone networks.
2. Cable TV networks use coaxial cables (RG-59) at the network boundaries. However, coaxial cable has largely been replaced today with fiber-optic cable due to its higher attenuation.
3. Traditional Ethernet LAN

- 10Base-2, or thin Ethernet, uses RG-58 coax cable with BNC connectors.
- 10Base-5, or thick Ethernet, uses RG-11 coax cable with specialized connectors.

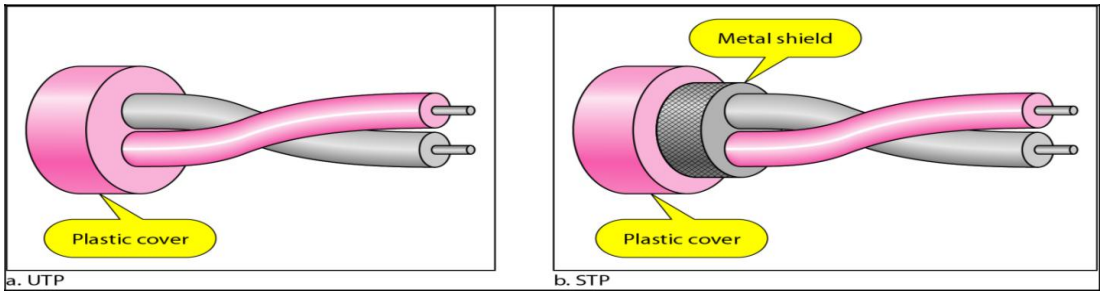
`

Twisted pair cables

Twisted pair is probably the most widely used cabling system in Ethernet in networks. Two copper wires twist around each other to form the twisted pair cable. Depending on category several insulated wire strands can reside in the cable.

Twisted pair is available in two basic types

a) Unshielded Twisted Pair (UTP)
b) Shielded Twisted Pair (STP)



Twisted pair cables

Unshielded Twisted Pair

Mostly the UTP is used. A twisted pair segment can't exceed 100 meters. This limitation is the only drawback to twisted pair. Twisted pair is used for 10/100 based Ethernet networks. UTP cables are wired as straight through or crossover cables. Straight through cables typically connect the computer's networks interface can't to be a port on the hub. Crossover cables are used for NIC to communication and for hub-to-hub connections when no crossover port is available.

Characteristics of twisted pair cable

1. Requires amplifiers every 5-6 km for analog signals
2. Requires repeaters every 2-3 km for digital signals
3. Attenuation is a strong function of frequency
4. Susceptible to interference and noise

`



Fig 3: RJ45 Cables

Applications

1. Used in telephone lines to provide voice and data channels.
2. The local loop –the line connecting the subscriber to the central telephone office- commonly consists of UTP cables.
3. DSL lines are also UTP cables.
4. LANs such as, 10Base-T and 100Base-T use UTP cables.

Fiber Optic

Fiber Optic relies on pulsed as light to carry information. Two types of plastic or glass with different physical properties are used (the inner core and the outer cladding) to allow a beam of light to reflect off the boundary between the core and cladding. Some fiber optic cables allow many different paths other allow one single mode. They are called multimode and single mode fibers. A popular multimode fiber has core/cladding dimensions of 62.5/125 nanometers.



Fig 4: Fiber Optic cable connector

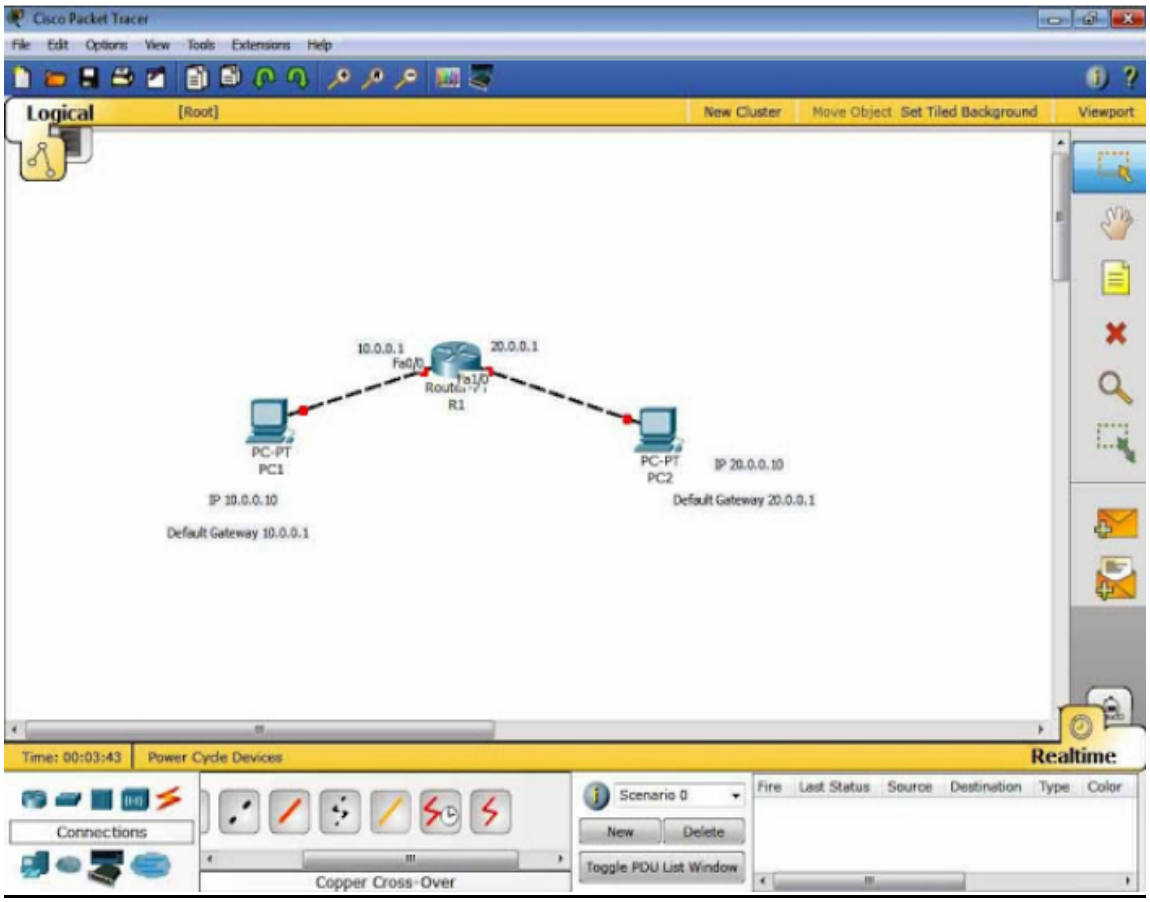
`

Task: How to Configure Ip address to PC and Routers in Packet Tracer

Software: Packet Tracer

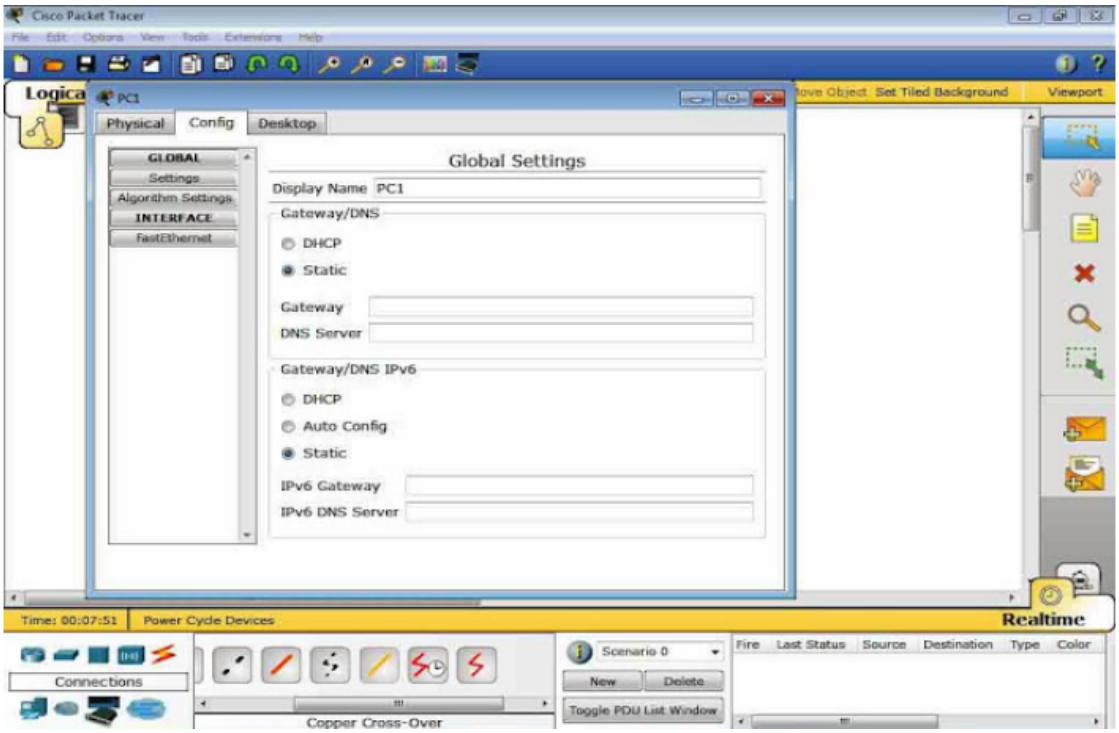
Configure Ip address to PC and Routers

Step 1: Create a simple topology like this,



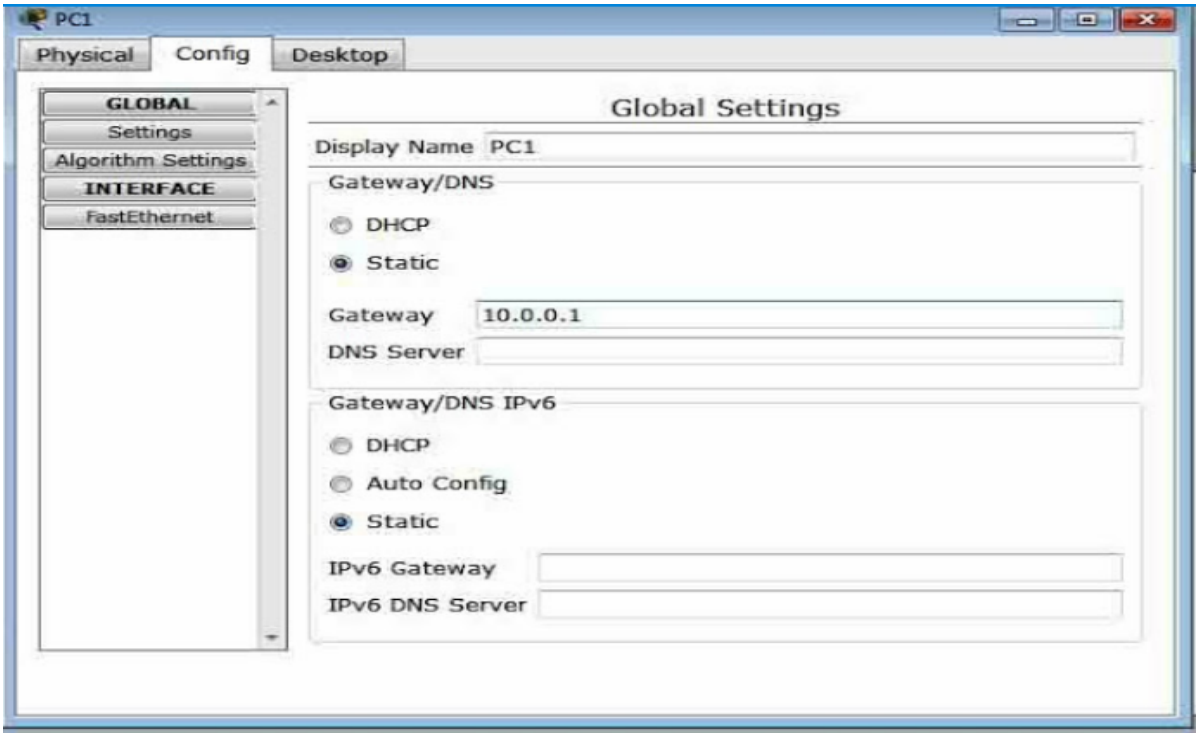
Step 2: Click on PC 1->Config, assign Gateway in this case, 10.0.0.1

`

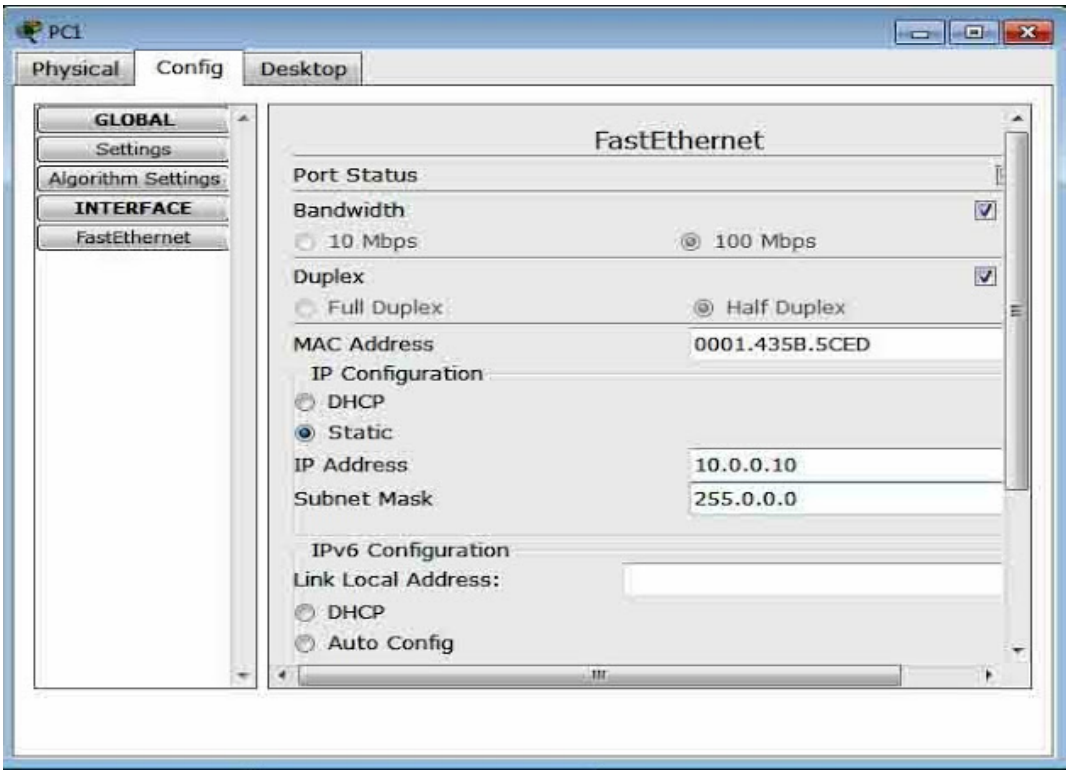


Step 3:

Click on Fastethernet and assign ip address and subnet mask, In this case ip 10.0.0.10 and subnet mask 255.0.0.0,Now close PC 1 window. Do like this for PC2 with appropriate ip and subnet mask.







Step 4: Now Click on Router R1,then click on

CLI(Command Line Interface).

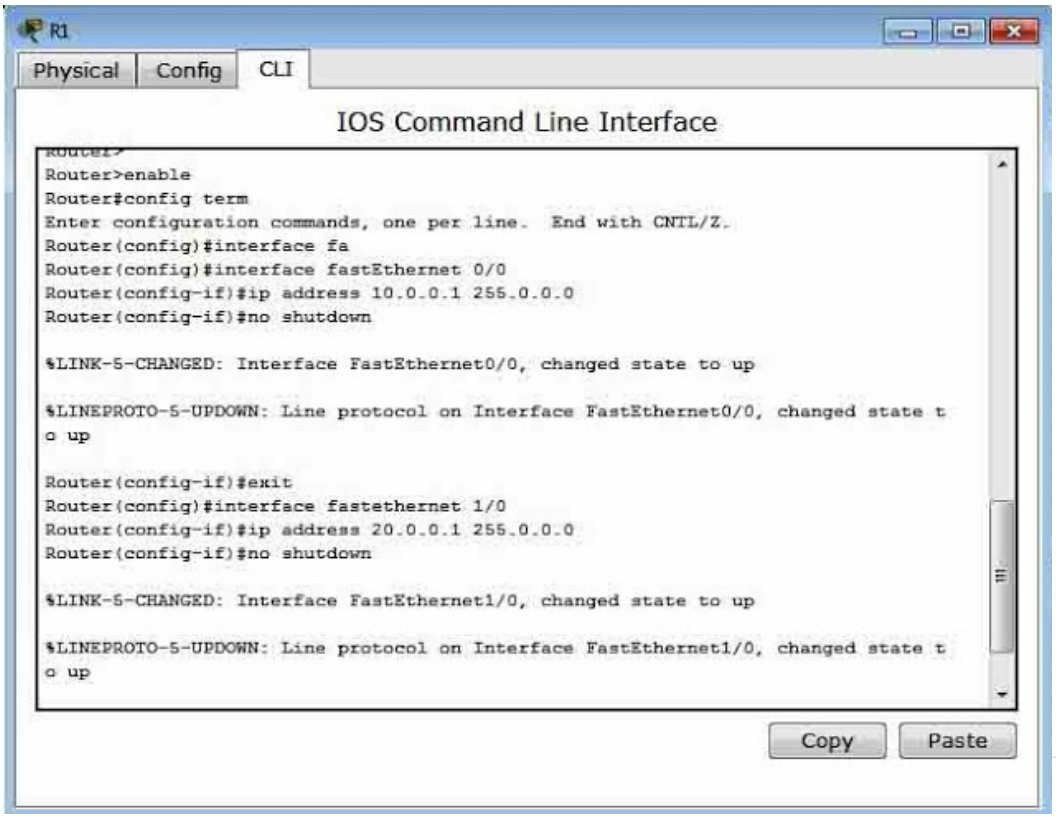
You will see like this, "Continue with configuration dialog? [yes/no]:".Give "no" and Press enter. Now you will go to user mode.

`

Step 5: Now give "enable" and press enter. Now you get into the Privileged Mode, now type "configure terminal" and press enter to get into global configuration mode.

Step 6: Now configure router interface with ip address and subnet mask then give no shutdown to make this interface and line protocol up(i.e. Carefully configure ip address with proper interfaces in this case f0/0 and f1/0,f is short form of fast ethernet.

Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Interface Line protocol on FastEthernet0/0, changed state to up
Router(config)#interface fastethernet 1/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Interface Line protocol on FastEthernet1/0, changed state to up

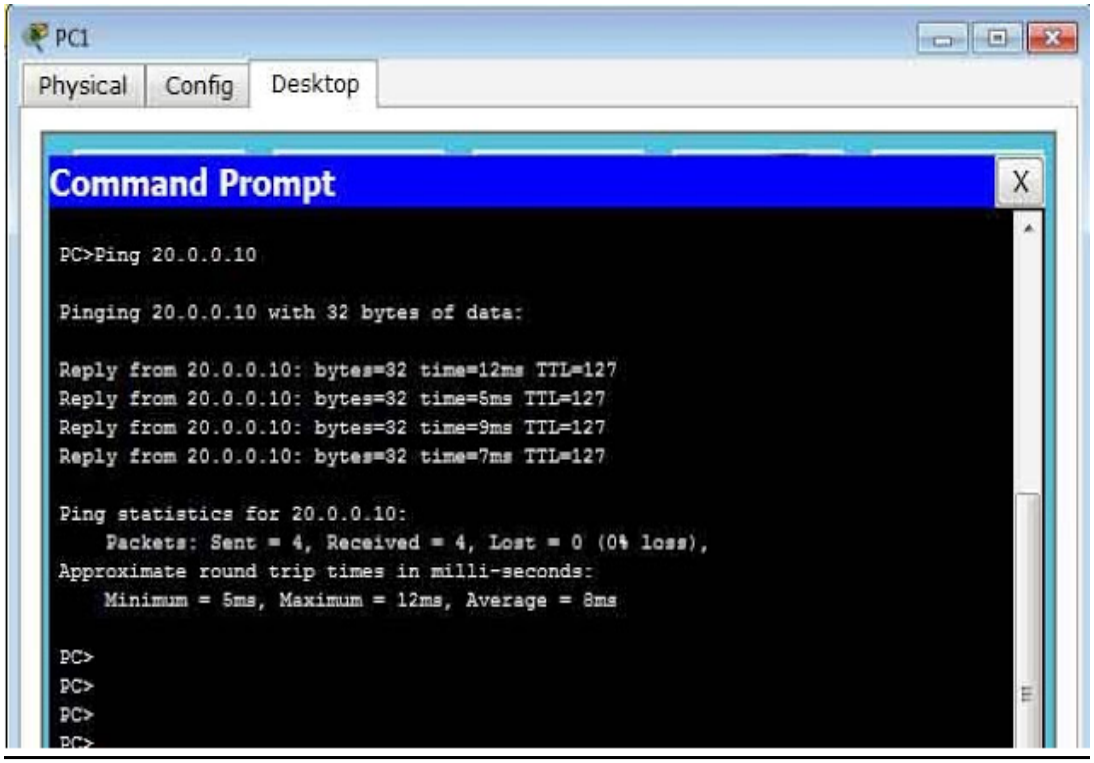


`

Step 7: Now lights on all ports become green from red.

Now click on PC1->Desktop->Command Prompt.

Step 8: Now give this command "ping 20.0.0.10" and press enter. You will get, connectivity between 10.0.0.10 and 20.0.0.10 is ok. Now PC1 communicates with PC2



Mechanism to Conduct Lab:

Students and teacher communicate through Adobe Connect. Students perform the task using the following simulator:

[https://www.netacad.com/courses/packet-tracer]

`

Lab 5

Objective: Study of following Network Devices in Detail

- Repeater
- Hub
- Switch
- Bridge
- Router
- Gate Way

Tool (Software): No software or hardware required.

Method: Following should be done to understand this practical.



1. Repeater: Functioning at Physical Layer. A **repeater** is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports ,so cannot be used to connect for more than two devices



2. Hub: An Ethernet hub, active hub, network hub, repeater hub, hub or concentrator

is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

`

3. Switch: A **network switch** is a computer networking device that connects network segments. The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.



4. Bridge: A **network bridge** connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term bridge formally means a device that behaves according to the IEEE 802.1 D standards. A bridge and switch are very much alike; a switch being a bridge with numerous ports. Switch or Layer 2 switch is often used interchangeably with bridge. Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.



5. Router: A **router** is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large collection of interconnected networks, the routers exchange information about target system addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected



`

networks.

6. **GateWay:** In a communications network, a network node prepared for interfacing with another network that uses different protocols.



- A gateway may contain devices such as protocol translators, impedance matching devices, rate converters, fault isolators, or signal translators as necessary to provide system interoperability. It also requires the establishment of mutually acceptable administrative procedures between both networks.
- A protocol translation/mapping gateway interconnects networks with different network protocol technologies by performing the required protocol conversions.

`

LAB # 6

CASE STUDY ON ROUTING

AIM

1. To study the Routing protocol and Routed protocol
2. Routing basis
3. Static Routing

ROUTING PROTOCOL AND ROUTED PROTOCOL

You must understand the difference between a routing protocol and a routed protocol.

1. A routing protocol is used by routers to dynamically find all the networks in the internetwork and to ensure that all routers have the same routing table. Basically, a routing protocol determines the path of a packet through an internetwork.
   Examples of routing protocols are RIP, RIPv2, EIGRP, and OSPF.

2. Once all routers know about all networks, a routed protocol can be used to send user data (packets) through the established enterprise. Routed protocols are assigned to an interface and determine the method of packet delivery.
   Examples of routed protocols are IP and IPv6.

ROUTING BASIS

The term routing is used for taking a packet from one device and sending it through the network to another device on a different network. Routers do not really care about hosts; they only care about networks and the best path to each network.

The logical network address of the destination host is used to get packets to a network through a routed network, and then the hardware address of the host is used to deliver the packet from a router to the correct destination host.

If a network is not directly connected to the router, then the router must use one of two ways to learn how to get to the remote network:

- Static routing
- Dynamic routing

`

STATIC ROUTING

Static routing is a type of network routing technique. Static routing is not a routing protocol; instead, it is the manual configuration and selection of a network route, usually managed by the network administrator. It is employed in scenarios where the network parameters and environment are expected to remain constant.

Static routing is only optimal in a few situations. Network degradation, latency and congestion are inevitable consequences of the non-flexible nature of static routing because there is no adjustment when the primary route is unavailable.

There are some pros and cons of static routing

Pros

- There is no overhead on the router CPU, which means that you could possibly buy a cheaper router than you would use if you were using dynamic routing.
- There is no bandwidth usage between routers, which means you could possibly save money on WAN links.
- It adds security because the administrator can choose to allow routing access to certain networks only.

Cons

- The administrator must really understand the internetwork and how each router is connected in order to configure routes correctly.
- If a network is added to the internetwork, the administrator has to add a route to it on all routers—by hand.
- It's not feasible in large networks because maintaining it would be a full-time job in itself.

Dynamic Routing

Dynamic routing is a networking technique that provides optimal data routing. Unlike static routing, dynamic routing enables routers to select paths according to real-time logical network layout changes. In dynamic routing, the routing protocol operating on the router is responsible for the creation, maintenance and updating of the dynamic routing table. In static routing, all these jobs are manually done by the system administrator.

`

Dynamic routing uses multiple algorithms and protocols. The most popular are Routing Information Protocol (RIP) and Open Shortest Path First (OSPF).

There are some pros and cons of dynamic routing

Pros

- Dynamic routing is easy to configure on large networks.
- More intuitive at selecting the best route, detecting route changes and discovering remote networks.

Cons

- Because routers share updates, they consume more bandwidth than in static routing; the routers CPUs and RAM may also face additional loads as a result of routing protocols.
- Dynamic routing is less secure than static routing.

`

Lab 7

In this lab we will learn about:

1. Introduction to Wireshark
2. How to create a Troubleshooting Profile in Wireshark:

Introduction to Wireshark

Wireshark is a network packet analyzer which captures network packets and displays that packet data as detailed as possible. Wireshark is free open source software program available at www.wireshark.org

Intended Purposes:

When run on a host connected to a wired or wireless network, Wireshark captures and decodes the network frames. People use it to learn network protocol internals. Network administrators use it to troubleshoot network problems. Network security engineers use it to examine security problems.

Wireshark's Features

1- Available for UNIX and Windows.

2- Capture live packet data.

3- Display packets with very detailed protocol information.

4- Filter packets on many criteria.

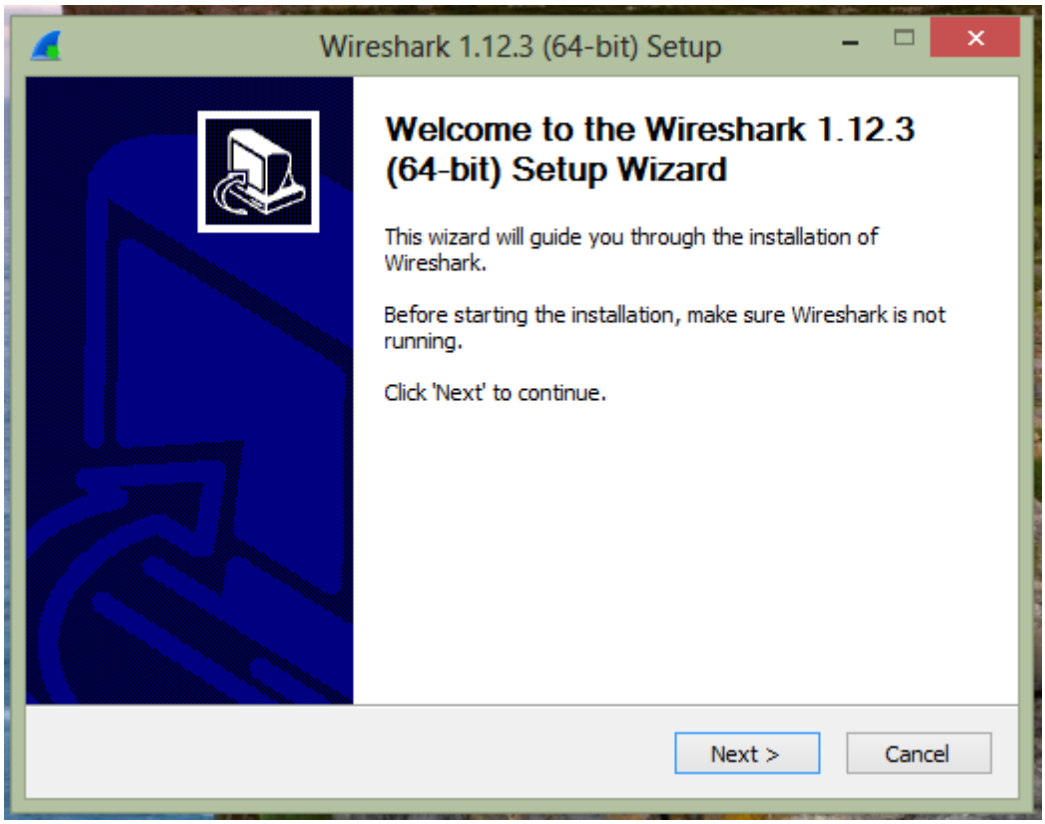
5- Save packet data captured.

Wireshark's Installation

Wireshark can be downloaded from website www.wireshark.org

Get Wireshark installer from: www.wireshark.org/download.html and execute it

`



Installation Components

• Wireshark - The network protocol analyzer.

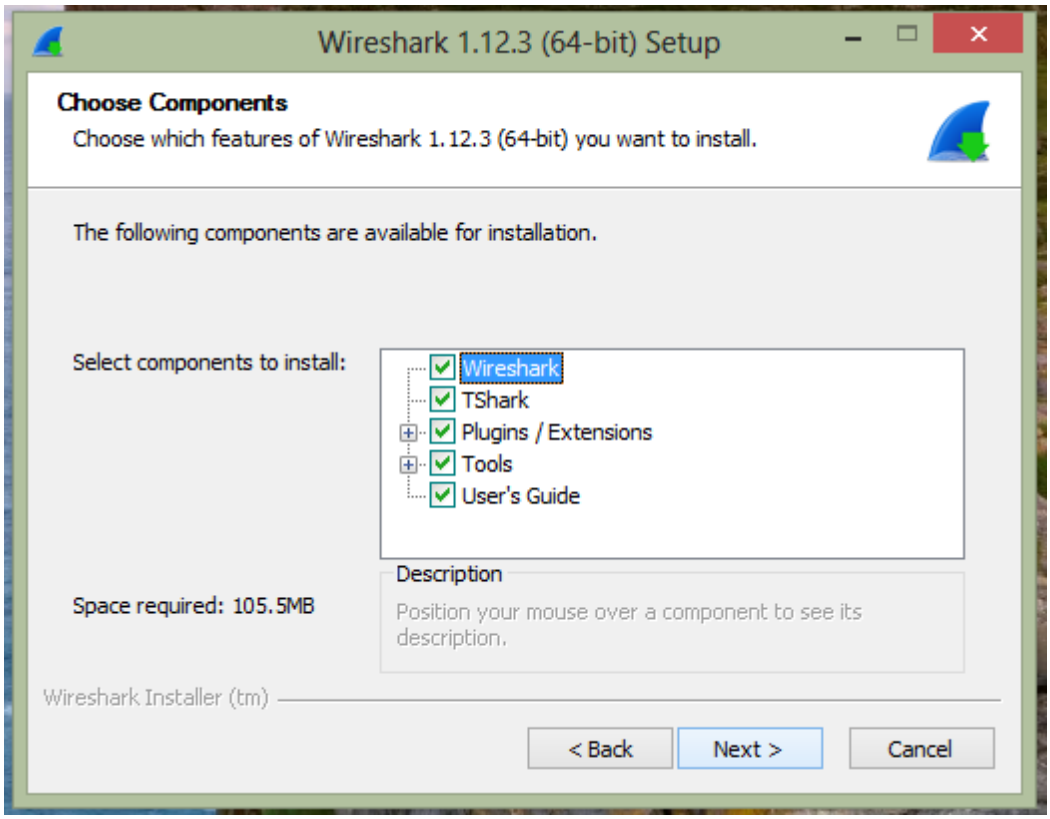
• TShark - A command-line network protocol analyzer.

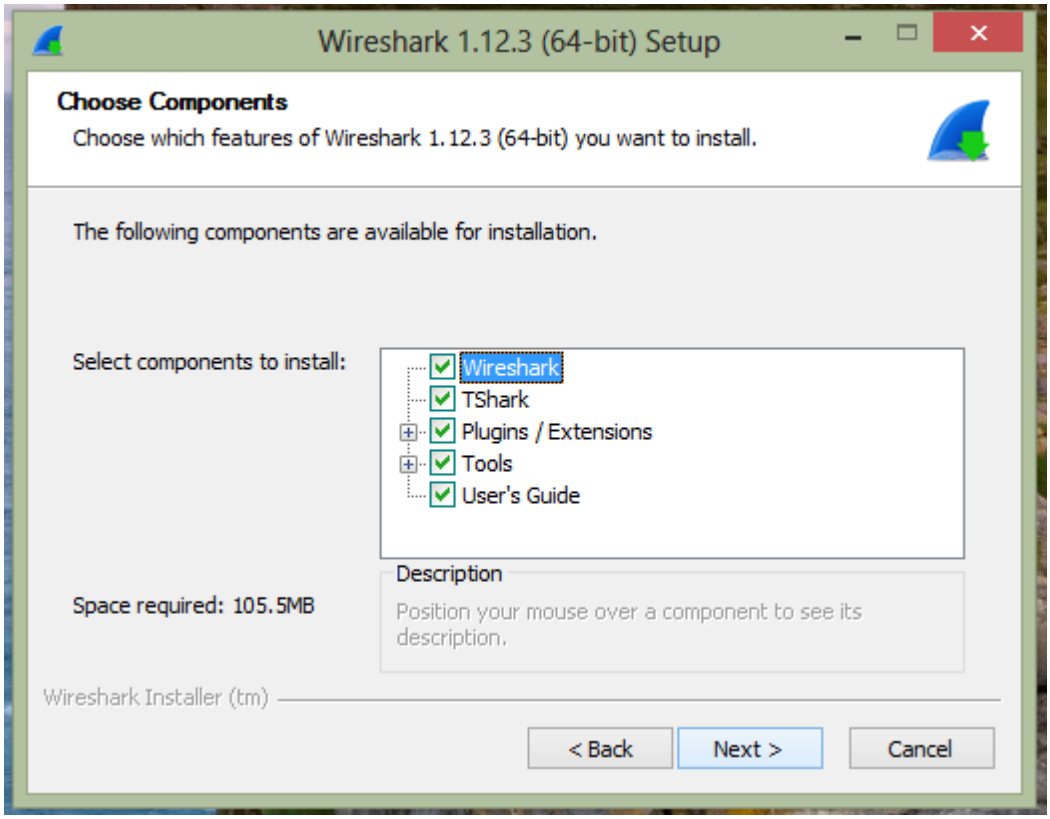
• Plugins & Extensions - Extras for the Wireshark and TShark dissection engines.

• Tools - Additional command line tools to work with capture files

• User's Guide

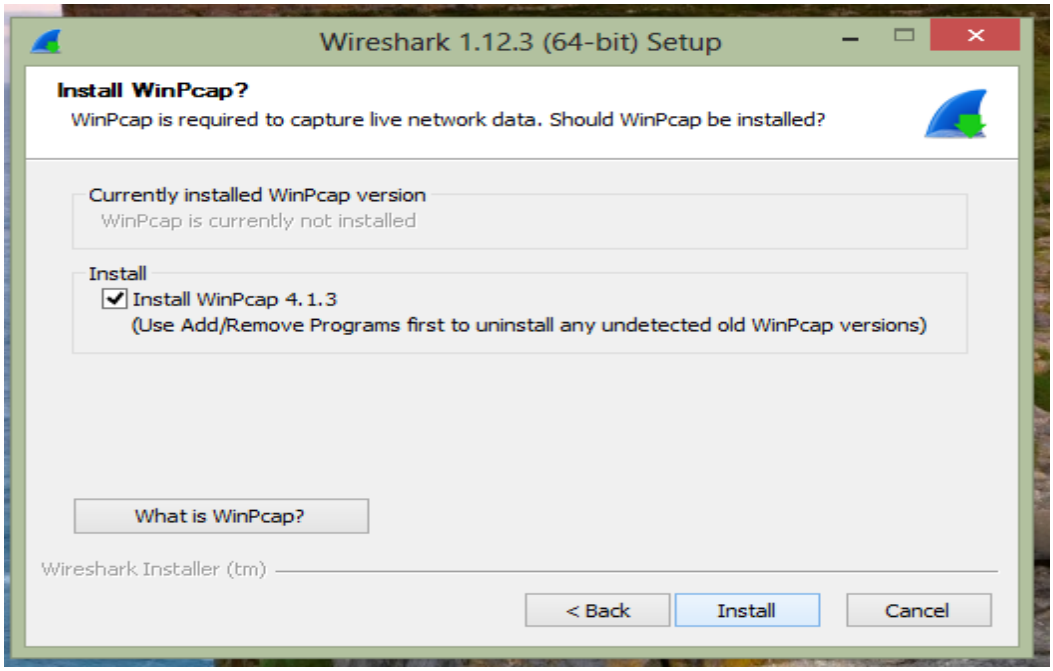
`

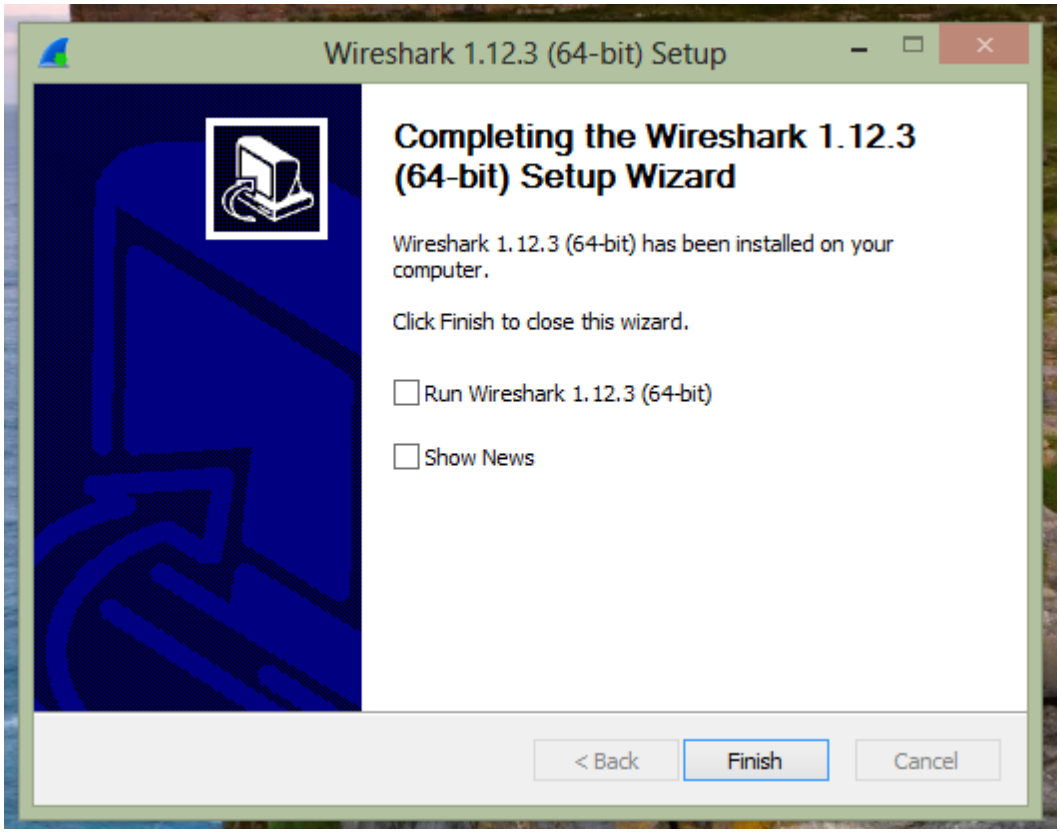

Wireshark 1.12.3 (64-bit) Setup
Choose Components
Choose which features of Wireshark 1.12.3 (64-bit) you want to install.
The following components are available for installation.
Select components to install:
Wireshark
TShark
Plugins / Extensions
Tools
User's Guide
Description
Position your mouse over a component to see its description.
Space required: 105.5MB
Wireshark Installer (tm)
< Back
Next >
Cancel


Wireshark 1.12.3 (64-bit) Setup
Choose Components
Choose which features of Wireshark 1.12.3 (64-bit) you want to install.
The following components are available for installation.
Select components to install:
Wireshark
TShark
Plugins / Extensions
Tools
User's Guide
Description
Position your mouse over a component to see its description.
Space required: 105.5MB
Wireshark Installer (tm)
< Back
Next >
Cancel

`

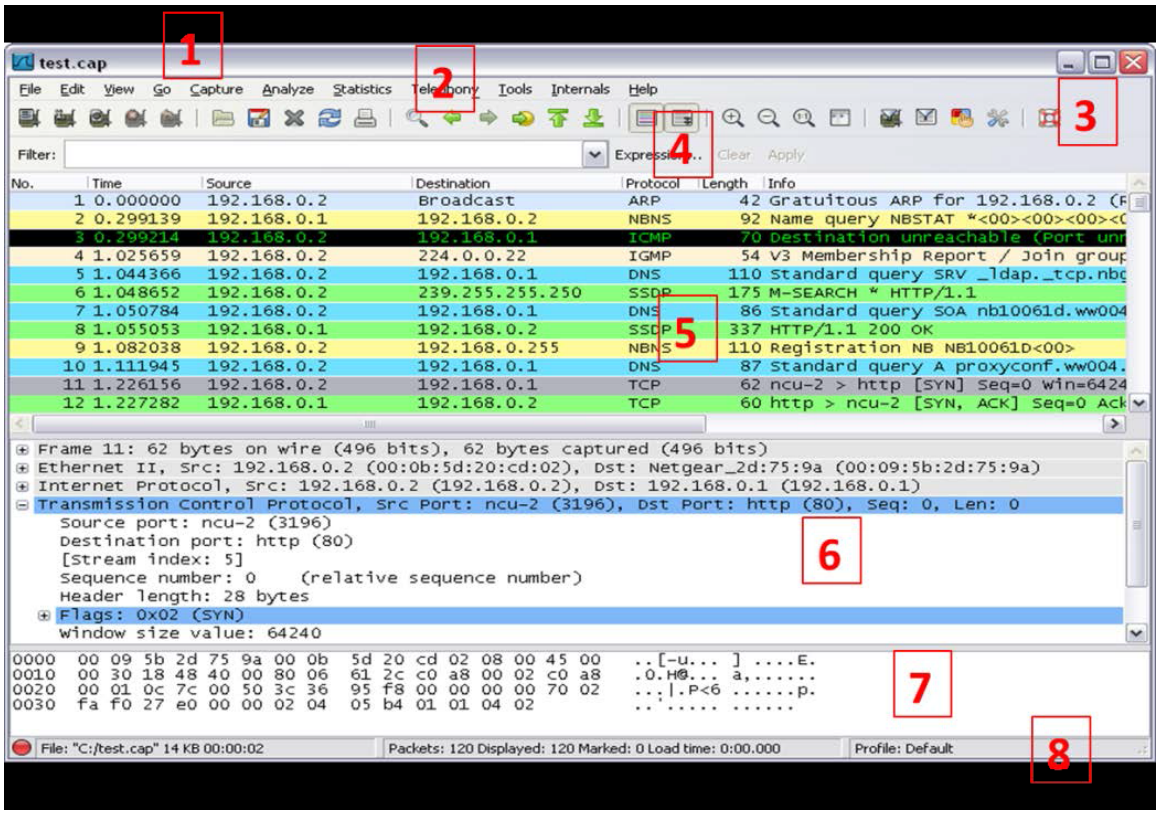
Installing WinPcap: With WinPcap installed you would be able to capture live network traffic.



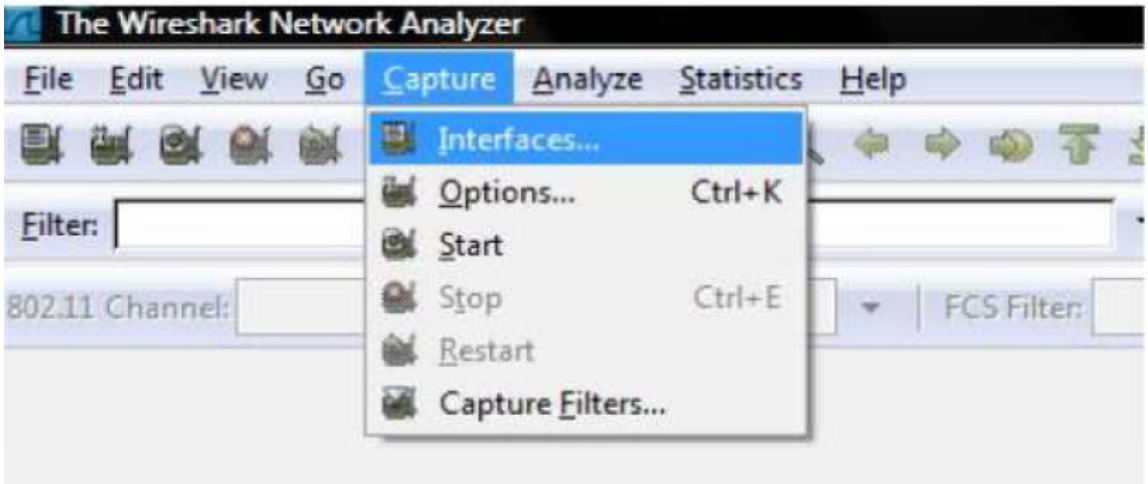


`

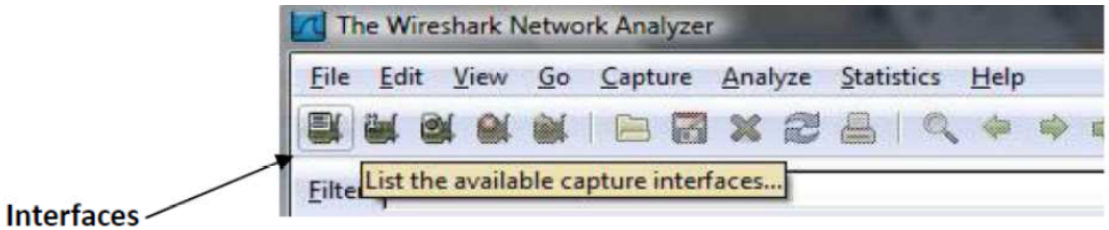
The main window is shown next.

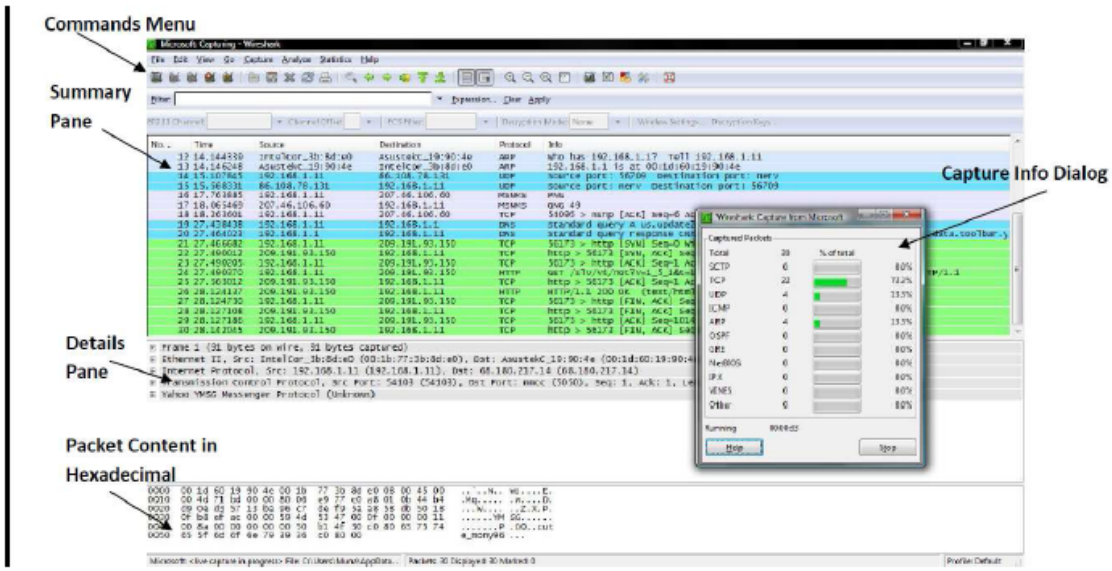


The different interfaces available that WinPcap driver sees in the machine are shown and you can either click start or click options for more options regarding capturing packets before starting the capture

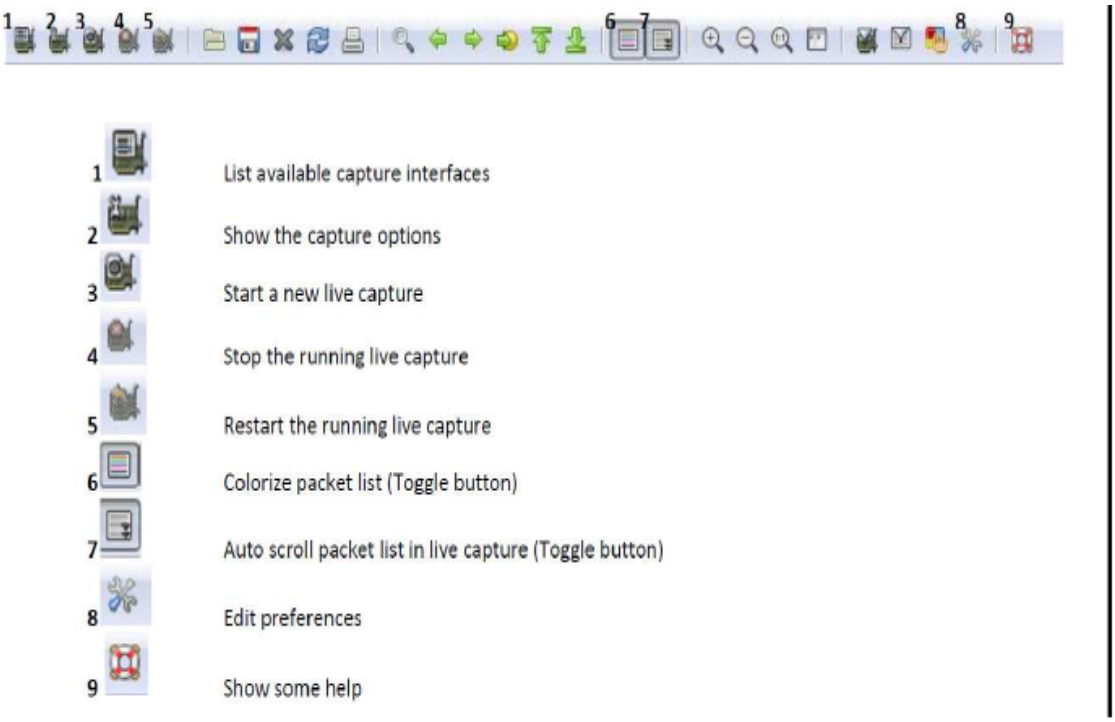


`


The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Help
Filter
List the available capture interfaces...
Interfaces


Commands Menu
Summary Pane
Capture Info Dialog
Details Pane
Packet Content in Hexadecimal

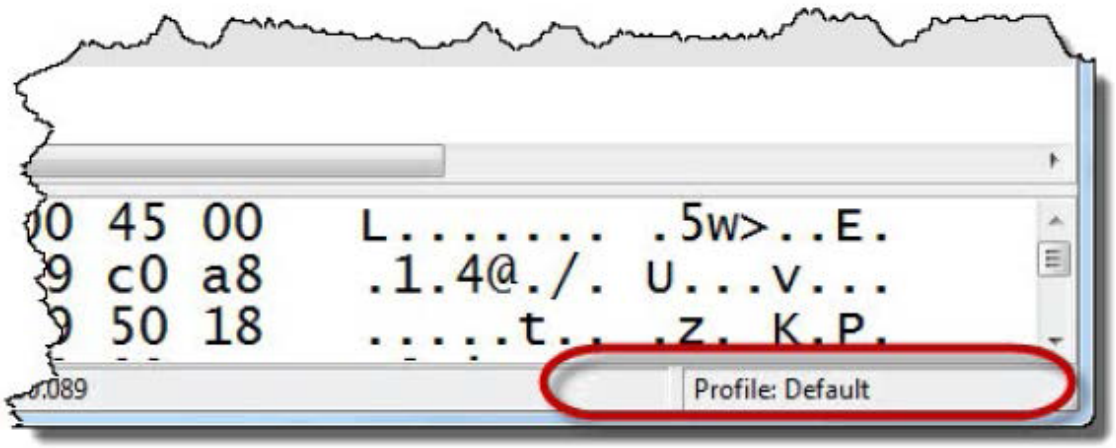
`



Task 1:

How to create a Troubleshooting Profile in Wireshark:

Until you create a new profile, you are working in Wireshark's Default profile. The profile you are working in is shown in the right side column of the Status Bar. This is shown next.



`

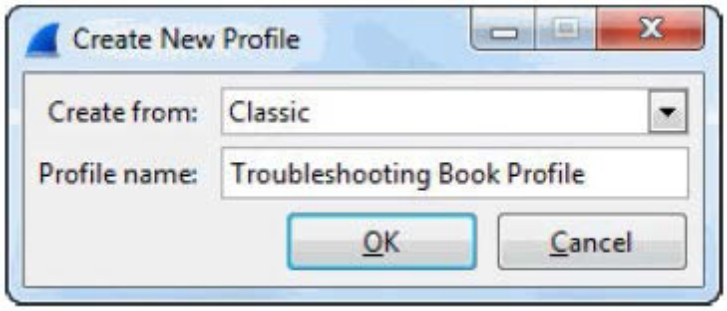
You can create profiles to customize Wireshark with buttons, colors, and more. You can create separate profiles for different needs. For example, you may want to make a VoIP profile, a WLAN profile, and a general troubleshooting profile. You can quickly switch between profiles depending on your needs.

Step1: Right-click the **Profile** column on the Status Bar.

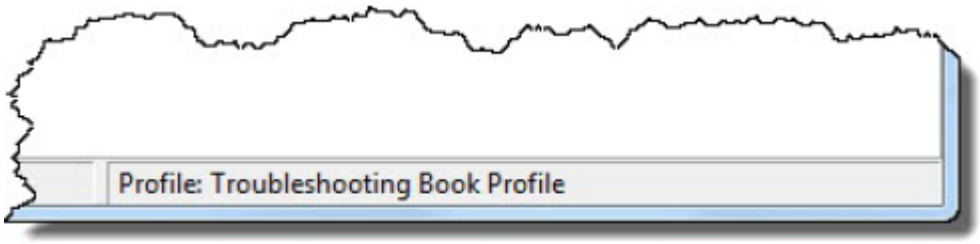
Step2: In the Configuration Profile window, select **New**.

Step3: Click the **arrow** in the **Create from** area, expand the **Global section** and select **Classic**. This profile uses the most vibrant colors.

Step4: Enter **Troubleshooting Book Profile** in the Profile Name area. Click **OK**.



As soon as you create your new profile, the Wireshark Status Bar indicates that you are working in the Troubleshooting Book Profile, as shown next



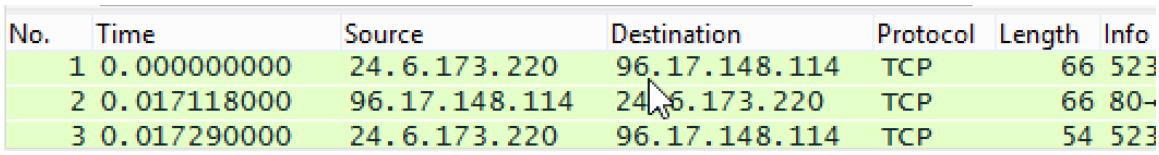
You will be able to add capabilities and customization to this new profile. Wireshark also allows download/import a predefined profile for immediate use.

`

Task-2

How to enhance the Packet List Pane Columns in Wireshark

By default, the Packet List pane contains: No. (number), Time, Source, Destination, Protocol, Length, and Info columns. This is shown next.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	24.6.173.220	96.17.148.114	TCP	66	523
2	0.017118000	96.17.148.114	24.6.173.220	TCP	66	80-
3	0.017290000	24.6.173.220	96.17.148.114	TCP	54	523

You can add columns to display additional information about packets to speed up your analysis process.

`

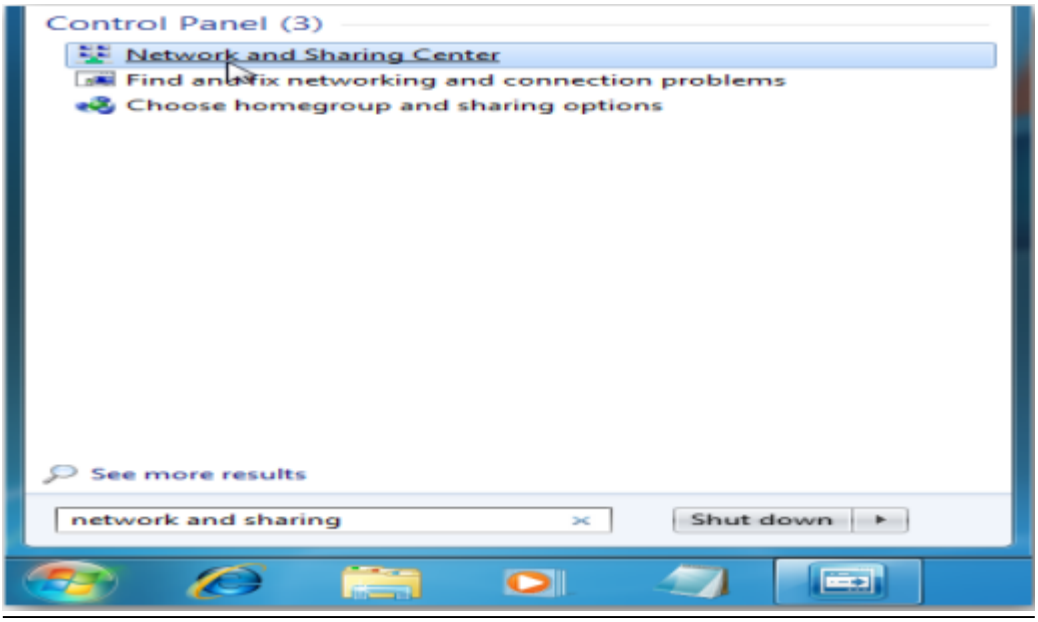
Lab 8

To assign IP address to the PC connected to the internet

Learning Objective: At the end of the lab you will be able to know how to assign IP address to a PC connected to the Internet.

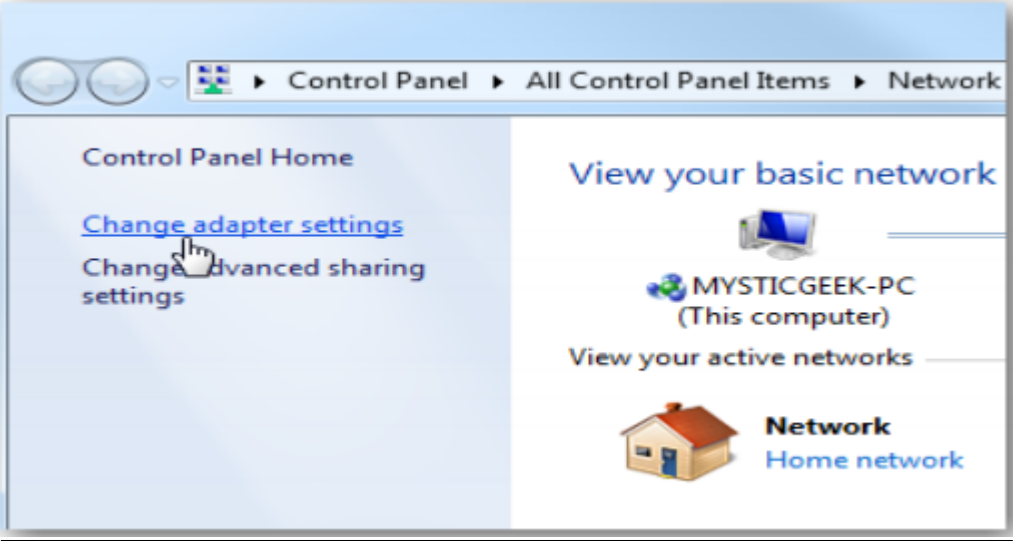
Assigning IP address in Windows 7 or Windows 10

Step-1: To assign or change the computer's IP address in Windows, type network and sharing into the Search box in the Start Menu and select Network and Sharing Center when it comes up. If you're in Windows 7 or 10 it'll be in the start menu.

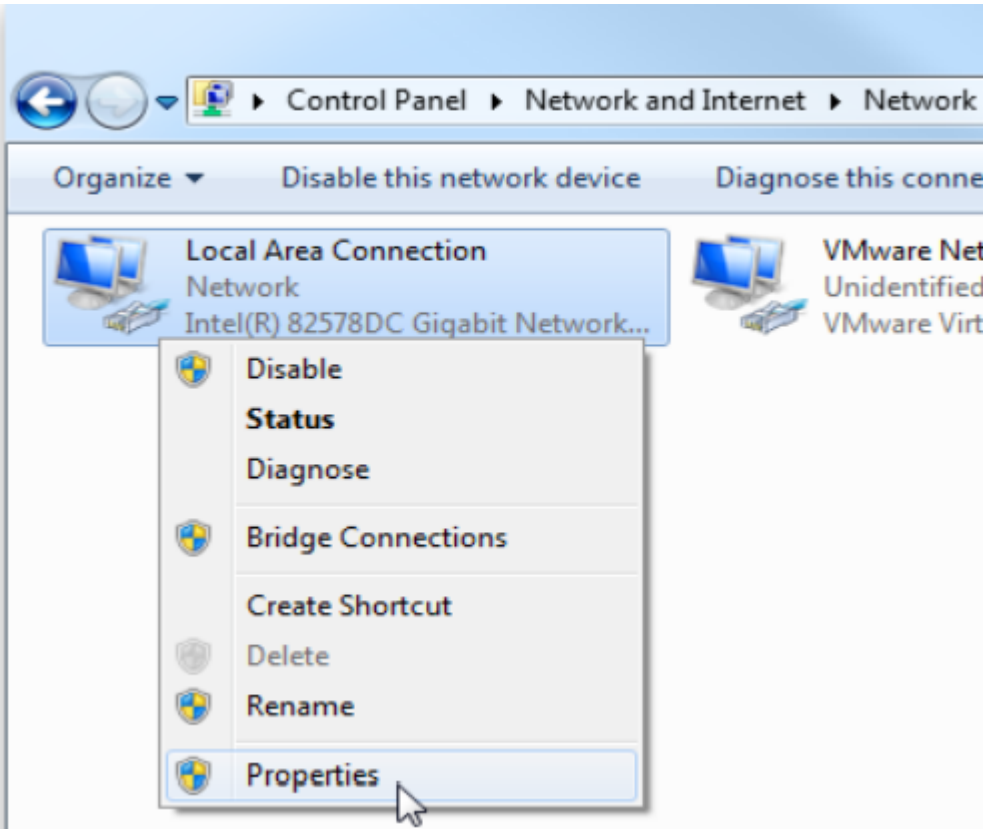


Step-2: Then when the Network and Sharing Center opens, click on Change adapter settings. This will be the same on Windows 7 or 10

`

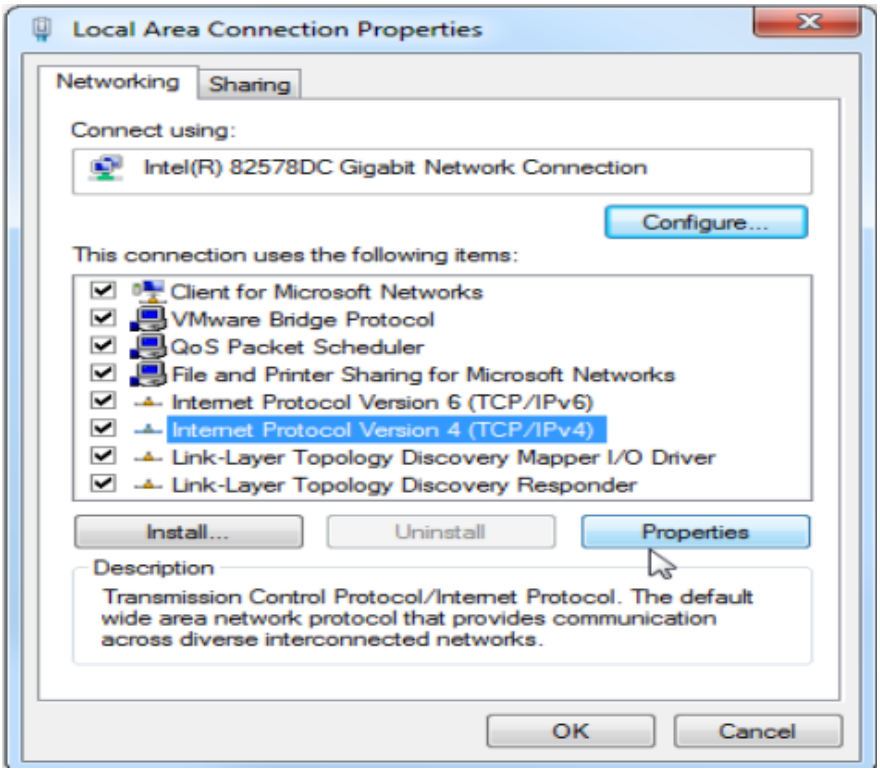


Step-3: Right-click on your local adapter and select Properties.



`

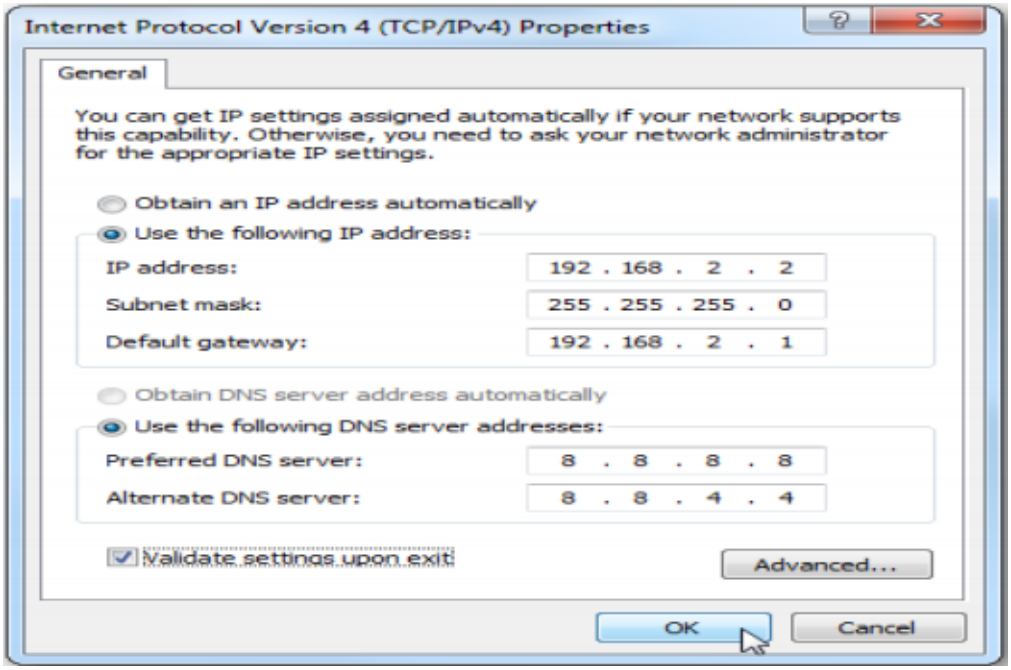
Step-4: In the Local Area Connection Properties window highlight Internet Protocol Version 4 (TCP/IPv4) then click the Properties button.



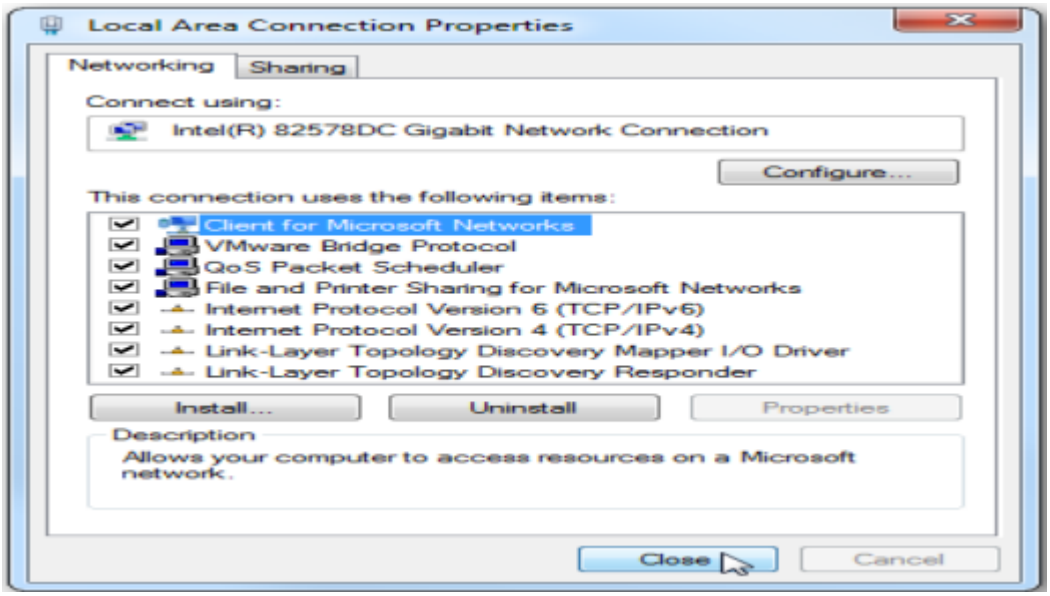
Now select the radio button Use the following IP address and enter in the correct IP, Subnet mask, and Default gateway that corresponds with your network setup. Then enter your Preferred and Alternate DNS server addresses. Here we're on a home network and using a simple Class C network configuration and Google DNS.

Step-5: Check Validate settings upon exit so Windows can find any problems with the addresses you entered. When you're finished click OK

`

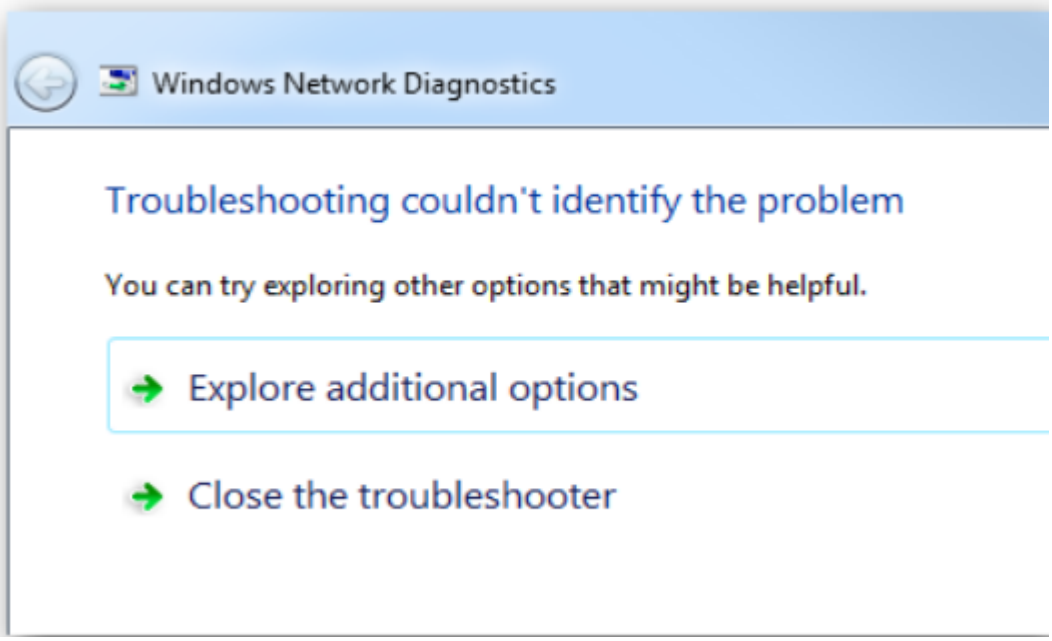


Step-6: Now close out of the Local Area Connections Properties window.

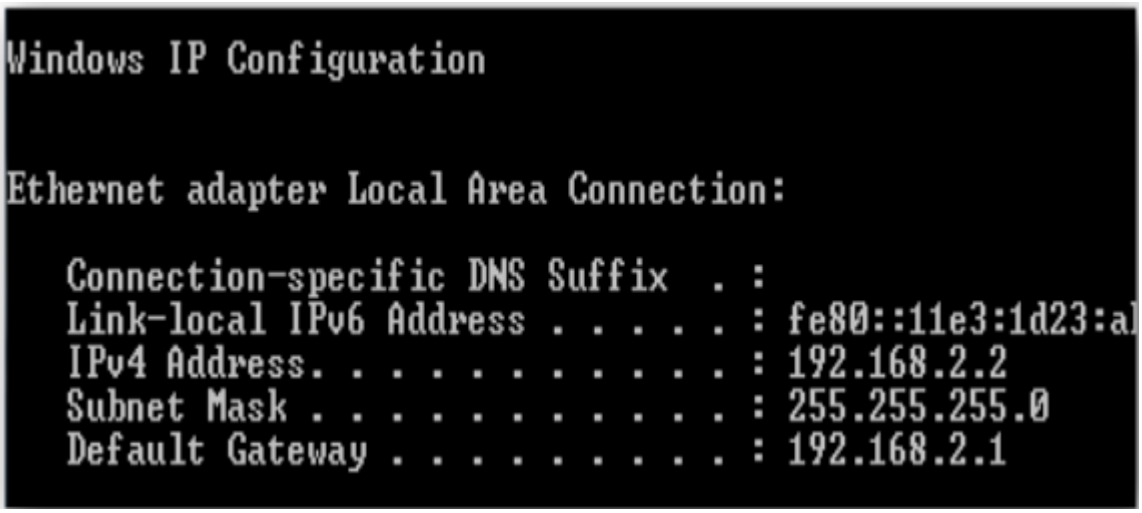


`

Windows will run network diagnostics and verify the connection is good. Here we had no problems with it, but if you did, you could run the network troubleshooting wizard.



Step-7: Now you can open the command prompt and do an ipconfig to see the network adapter settings have been successfully changed.



`

Lab 9

Objective:

- Introduce Address Resolution Protocol (ARP) and the arp –a workstation command.
- Explore the arp command help feature using the -? Option.

Background / Preparation

ARP is used as a tool for confirming that a computer is successfully resolving network Layer 3 addresses to Media Access Control (MAC) Layer 2 addresses.

ARP maintains a table in the computer of IP and MAC address combinations. In other words, it keeps track of which MAC address is associated with an IP address. If ARP does not know the MAC address of a local device, it issues a broadcast using the IP address. This broadcast searches for the MAC address that corresponds to the IP address. If the IP address is active on the LAN, it will send a reply from which ARP will extract the MAC address. ARP will then add the address combination to the local ARP table of the requesting computer.

MAC addresses and therefore ARP are only used within the LAN. When a computer prepares a packet for transmission, it checks the destination IP address to see if it is part of the local network. It does this by checking to see if the network portion of the IP address is the same as the local network. If it is, the ARP process is consulted to get the MAC address of the destination device using the IP address. The MAC address is then applied to the data packet and used for delivery.

If the destination IP address is not local, the computer will need the MAC address of the default gateway. The default gateway is the router interface that the local network is connected to in order to provide connectivity with other networks. The gateway MAC address is used because the packet will be delivered there and the router will then forward it to the network it is intended for.

If the computer does not receive any packets from an IP address after a few minutes, it will drop the MAC/IP entry from the ARP table assuming the device has logged off. Later attempts to access that IP address will cause ARP to do another broadcast and update the table.

The Structure of Arp Header:

Below listed fields are there in an ARP message.

1. **Hardware Type:** This is to specify the type of hardware used by the local network to transmit the Address Resolution Protocols message. Once common hardware under this category would be the ‘Ethernet’ with a value equal to 1.
2. **Protocol Type:** The protocol type is a 16-bit field used to specify the type of protocol.
3. **Hardware size:** This is the length in bytes for the MAC address; generally, we see the ethernet has a MAC address of 6 bytes long.
4. **Protocol Size**: It represents the length of the IPV4 logical address, IPV4 address is generally 4 bytes long.
5. **OpCode:** . Operation Code indicates that the packet is an ARP Request (1) or an ARP Response (2).It specifies the nature of the ARP message.An ARP Request has an assigned value of 1, whereas the ARP reply holds the value of 2.
6. **Sender MAC address:** Layer 2 (MAC) address for the device sending the message.
7. **Sender IP address:** Protocol address in IPV4 for the device sending the message.

`

8. **Target MAC address:** Layer 2 (MAC) address of the intended receiver. This field does not hold any value during the request phase and works only during the reply phase.
9. **Target IP address**: This address the protocol address for the intended receiver.

Step 1

Establish a network connection If the connection to the Internet is dial-up, connect to the ISP to ensure that the computer has an IP address. In a TCP/IP LAN with a Dynamic Host Configuration Protocol (DHCP) server it should not be necessary to do this step.

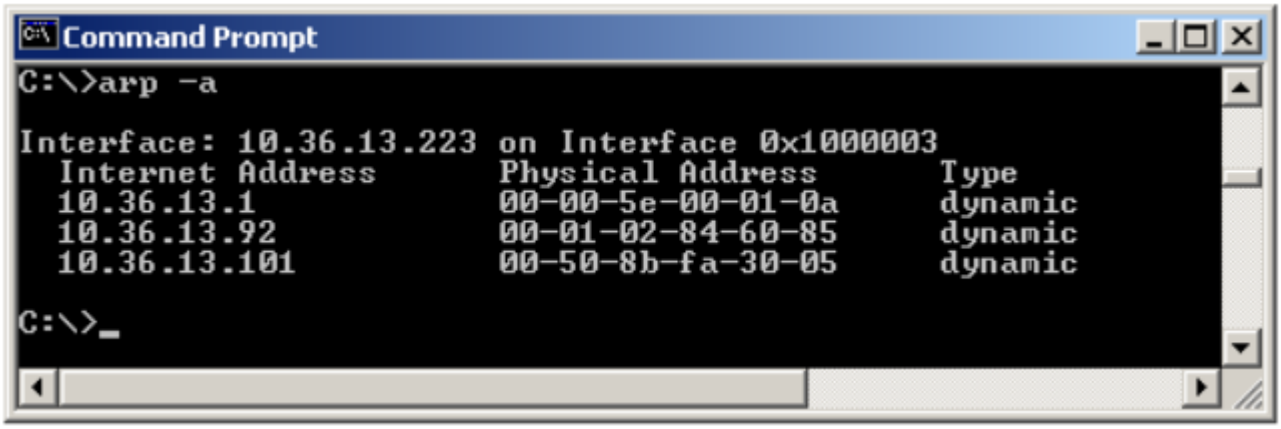
Step 2

Access a command prompt

One of the quickest ways to launch the **Command Prompt**, in any modern version of Windows, is to use the Run window. Then, type **cmd** or **cmd**.exe and press **Enter** or click/tap OK

Step 3

a. Display the ARP table a. In the window type arp -a and press Enter. Do not be surprised if there are no entries. The message displayed will probably be, 'No ARP Entries Found'. Windows computers remove any addresses that are unused after a couple minutes.
b. Try pinging a couple local addresses and a website URL. Then re-run the command. The figure below shows a possible result of the arp -a command. The MAC address for the website will be listed because it is not local, but that will cause the default gateway to be listed. In the example below 10.36.13.1 is the default gateway while the 10.36.13.92 and 10.36.13.101 are other network computers. Notice that for each IP address there is a Physical Address, or MAC, and type, indicating how the address was learned.



Step 4

Ping several URLs

a. Ping the following URLs and note the IP address of each. Also select one additional URL to ping and record it below:

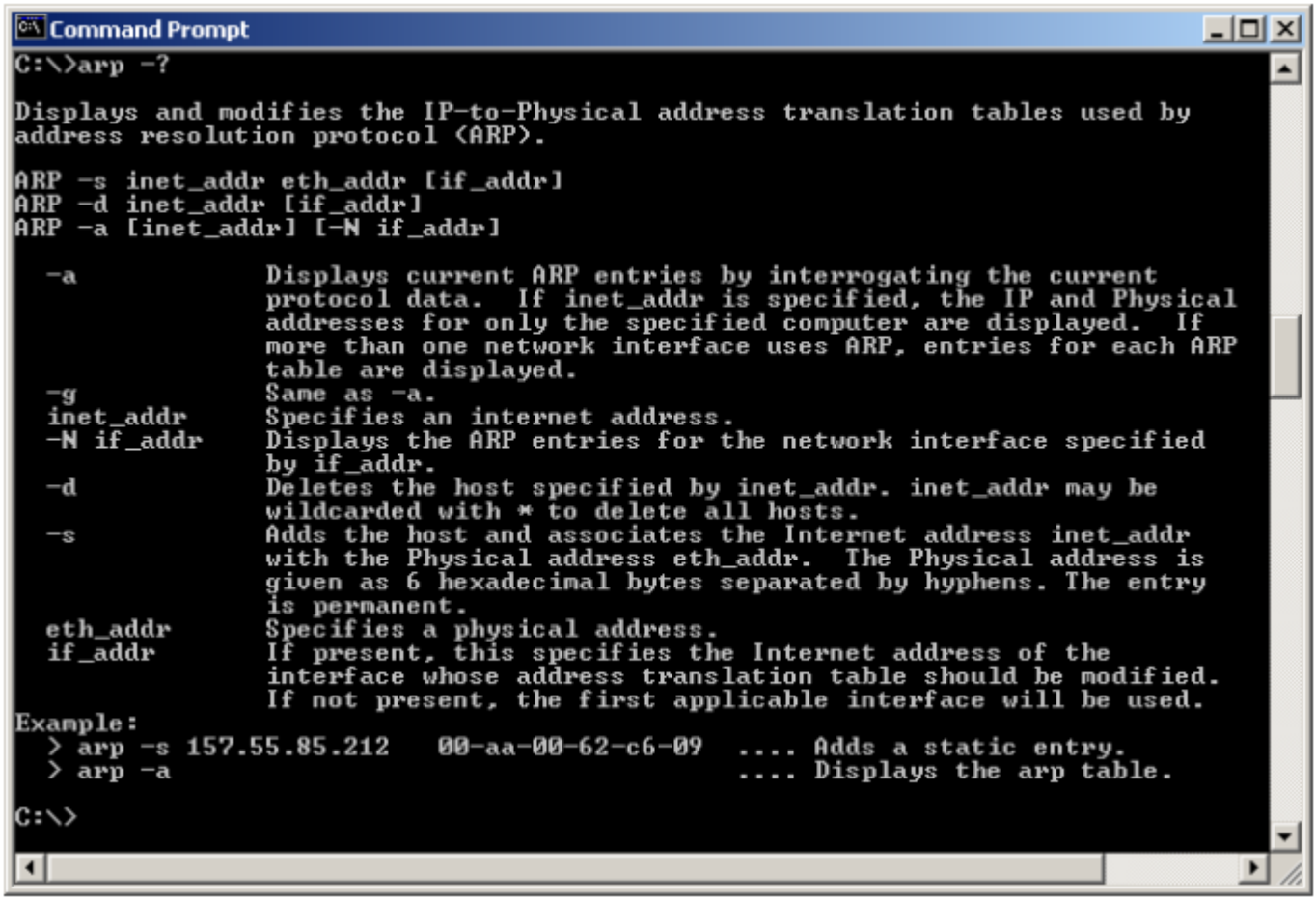
`

www.cisco.com: _____
www.vu.edu.pk:_____

a. Now run the arp –a command again and record the MAC addresses for each of the above next to their IP addresses.

Use the ARP Help Feature

Try the command arp -? To see the help feature and look over the options.



The purpose of this step is not so much the ARP command options but to demonstrate using the? To access help, if available.

`

Lab 10

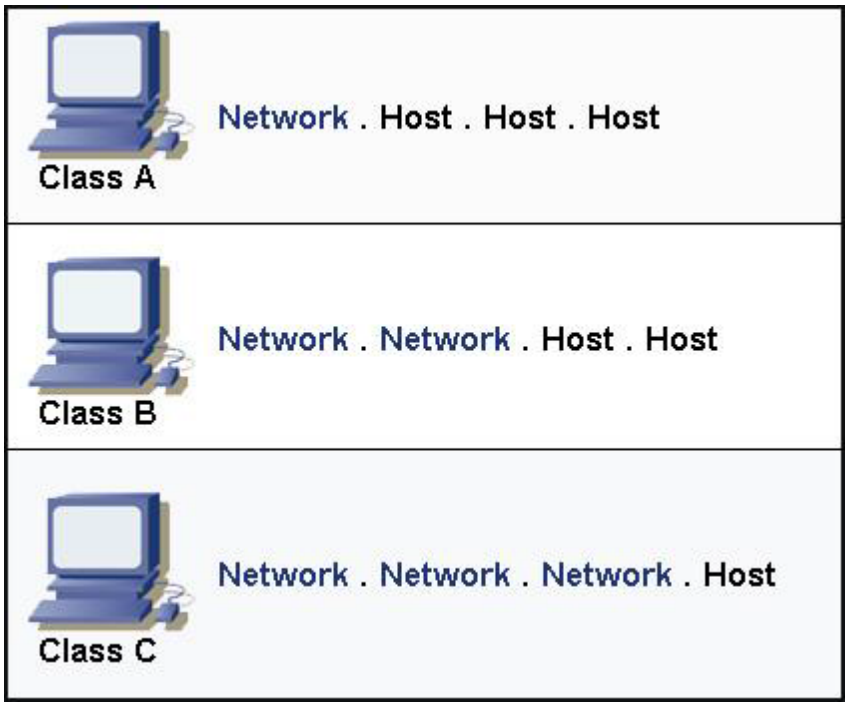
IPv4 Address Subnetting

In this lab we learn about:

1. Classful Addressing
2. Determining Address Class
1. IP Address Class A, B and C Network and Host Capacities
3. IP Addresses with Special Meanings

2. Classful Addressing

The original IP addressing scheme is set up so that the dividing line occurs only in one of a few locations: on octet boundaries. There are five classes of available IP ranges: Class A, Class B, Class C, Class D and Class E, while only A, B, and C are commonly used. Each class allows for a range of valid IP addresses. They allow the Internet to provide addressing for a small number of very large networks, a moderate number of medium-sized organizations, and a large number of smaller companies.

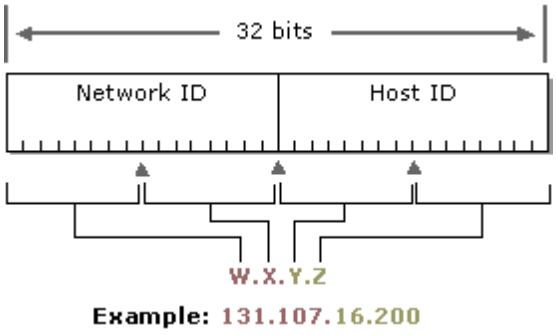


`

IP Address Class	Network ID Bits	Host ID Bits	Intended Use
Class A	8	24	Used for addressing very large organizations with hundreds of thousands or millions of hosts to connect to the Internet.
Class B	16	16	Used for addressingmedium-to-large organizations with hundreds to thousands of hosts to connect to the Internet.
Class C	24	8	Used for addressing smaller organizations with no more than about 250 hosts to connect to the Internet.
Class D	n/a	n/a	IP multicasting.
Class E	n/a	n/a	Reserved for "experimental use".

Network Identifier (Network ID): A certain number of bits, starting from the left-most bit, are used to identify the network where the host or other network interface is located. This is also sometimes called the network prefix or even just the prefix.

Host Identifier (Host ID): The remainder of the bits is used to identify the host on the network.



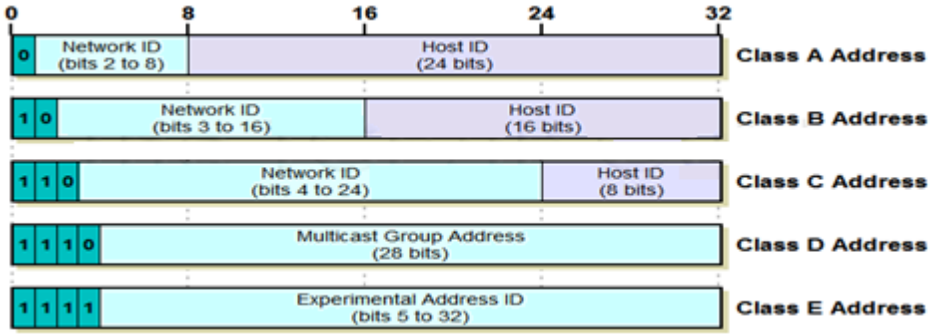
As IP address split into network ID and host ID components, these addresses are assigned special meanings. For example, if the network ID is used with all ones in the host ID portion, this indicates a ***broadcast to the entire network***. Similarly, if the network ID is used by itself with all zeroes in the host portion indicates ***the network ID***.

If you move a device from one network to a different one the network ID must change to that of the new network. Therefore, the IP address must change as well.

`

3. Determining Address Class from the First Octet Bit Pattern

IP Address Class	Lowest Binary Value of First Octet	Highest Binary Value of First Octet	Range of First Octet Decimal Values	Theoretical IP Address Range
Class A	**0**000 0001	**0**111 1110	1 to 126	**1**.0.0.0 to**126**.255.255.255
Class B	**10**00 0000	**10**11 1111	128 to 191	**128.0**.0.0 to**191.255**.255.255
Class C	**110**0 0000	**110**1 1111	192 to 223	**192.0.0**.0 to**223.255.255**.255
Class D	**1110** 0000	**1110** 1111	224 to 239	224.0.0.0 to 239.255.255.255
Class E	**1111** 0000	**1111** 1111	240 to 255	240.0.0.0 to 255.255.255.255



4. IP Address Class A, B and C Network and Host Capacities

IP Address Class	Network ID Bits/ Host IDBits	First Octet of IP Address	Number of Possible Network IDs	# Of Host IDs Per Network ID
Class A	8 / 24	**0**xxx xxxx	$2^8 = 256$	$2^{24}-2 = 16{,}277{,}214$
Class B	16 / 16	**10**xx xxxx	$2^{16} = 65{,}536$	$2^{16}-2 = 65{,}534$
Class C	24 / 8	**110**x xxxx	$2^{24} = 16{,}277{,}216$	$2^8-2 = 254$

5. Special Address Patterns Special IP addresses are constructed by replacing the normal host ID in an IP address with one of two special patterns. The two patterns are:

All Zeroes: When the host bits are replaced by a set of all zeroes, the special meaning is the equivalent of the Network ID that represent all hosts in that network especially in the routing table.

All Ones: When the host bits are replaced by a set of all ones, this has the special meaning of the broadcast ID of the network, this address used to send a common message for all hosts exists in this network.

`

- There are several other sets of IP addresses set aside for various special uses, which are not available for normal address assignment.

Reserved Addresses Several blocks of addresses were designated just as "reserved" with no specific indication given of what they were reserved for. They may have been set aside for future experimentation, or for internal use in managing the Internet, or for other purposes. These address extract along 0.0.0.0 class and 255.0.0.0 class.

Loopback Addresses

The purpose of the loopback range is testing of the TCP/IP protocol implementation on a host.



Private/Unregistered/Non-Routable Addresses

Class	Private Networks	Subnet Mask	Address Range
A	10.0.0.0	255.0.0.0	10.0.0.0 - 10.255.255.255
B	172.16.0.0 - 172.31.0.0	255.240.0.0	172.16.0.0 - 172.31.255.255
C	192.168.0.0	255.255.0.0	192.168.0.0 - 192.168.255.255

Lab 11

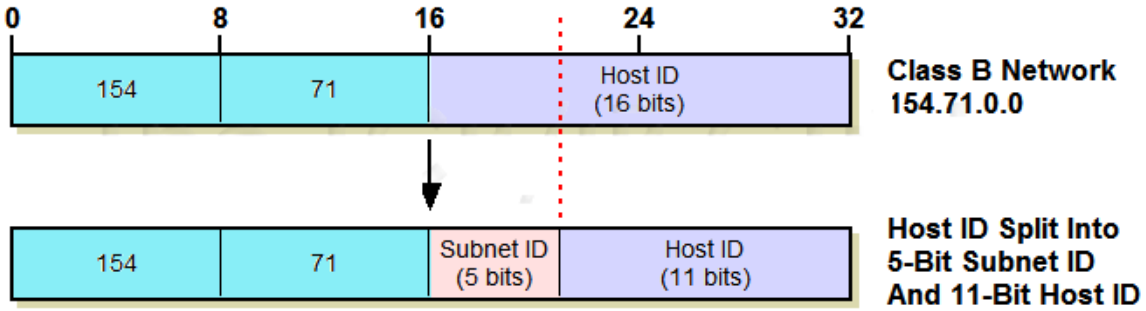
IPv4 Address Subnetting

In this lab we will learn about:

1. Subnetting of Classful IP Addressing
2. Classless Addressing
3. IP Subnet Addressing ("Subnetting") Concepts
4. IP Subnet Masks, Notation and Subnet Calculations
5. IP Default Subnet Masks for Address Classes A, B and C
6. Deciding How Many Subnet Bits to Use
7. Trading Off Bit Allocations to meet Subnetting Requirements

1. Subnetting of Classful IP Addressing

In the subnet addressing system, the two-tier network/host division of the IP address is made into a three-tier system by taking some number of bits from a class A, B or C host ID and using them for a *subnet identifier or number*. The network ID is unchanged. The *subnet ID* is used for routing within the different subnetworks that form a complete network, providing extra flexibility for administrators. For example, consider a class C address that normally uses the first 24 bits for the network ID and remaining 8 bits for the host ID. The host ID can be split into, say, 3 bits for a subnet ID and 5 for the host ID.



2. Classless Addressing

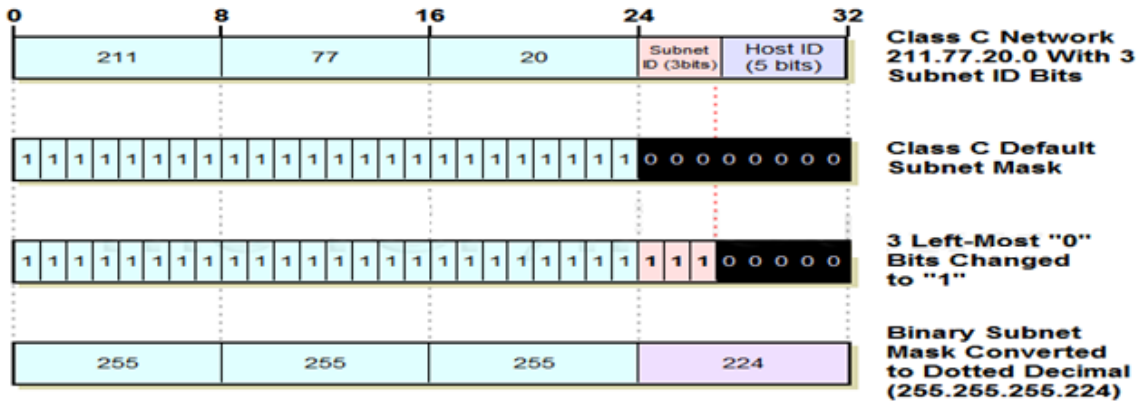
In the classless system, the division between the network ID and host ID can occur at an arbitrary point, not just on octet boundaries like in the “Classful” scheme. In the original “Classful” scheme the division between network ID and host ID is implied. However, if either Subnetting or classless addressing is used, then the subnet mask or “slash number” are required to fully qualify the address.

3. IP Subnet Addressing ("Subnetting") Concepts

The original “Classful” IP addressing scheme conceptually divides a large internetwork into a simple two-level hierarchy: many *networks* of different sizes, each of which contains a number of *hosts*. The system works well for smaller organizations that may connect all their machines in a single network. However, it lacks flexibility for large organizations that often have many subnetworks, or *subnets*. To better meet the administrative and technical requirements of larger organizations, the “Classful” IP addressing system was enhanced through a technique known as *subnet addressing*, or *Subnetting*.

A three-level hierarchy is thus created: networks, which contain subnets, each of which then has a number of hosts.

4. IP Subnet Masks, Notation and Subnet Calculations

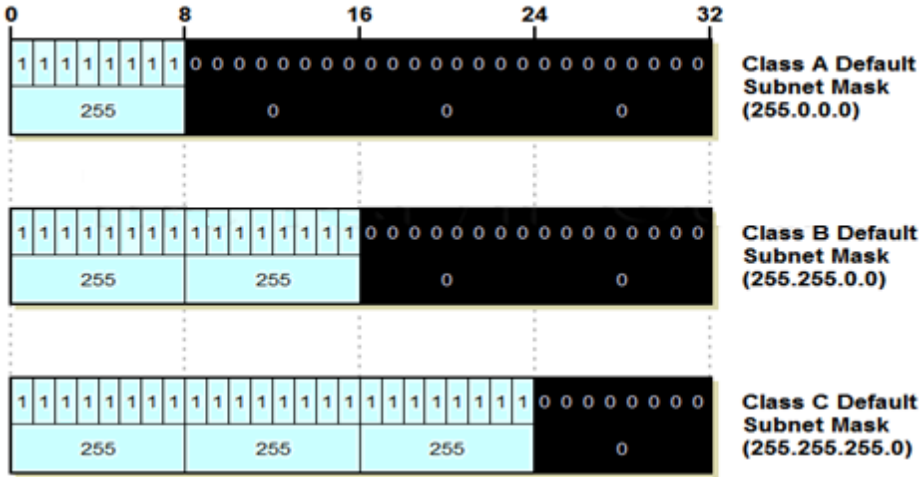


We use the subnet mask by applying the Boolean *AND* function between it and the IP address to obtain the Network ID and subnet ID of that IP address.

Instead of specifying IP address of 154.71.150.42 with subnet mask of 255.255.248.0, we can just say 154.71.150.42/**21**. This is sometimes called *slash notation* or Classlesss Inter-Domain Routing *(CIDR) notation*. It is more commonly used in variable-length masking (VLSM) environments, and as the second name implies, is also the standard for specifying classless addresses under the CIDR addressing scheme.

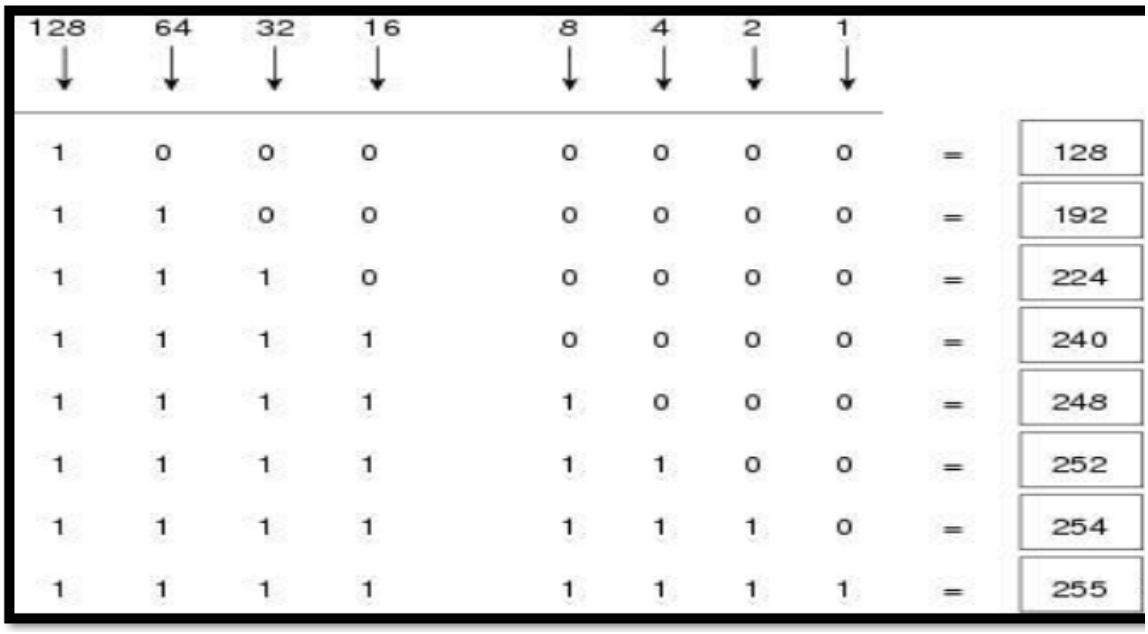
5. IP Default Subnet Masks for Address Classes A, B and C

IP Address Class	Network ID bits / Host ID bits	Default Subnet Mask			
		First Octet	**Second Octet**	**Third Octet**	**Fourth Octet**
Class A	**8 / 24**	11111111(255)	00000000(0)	00000000(0)	00000000(0)
Class B	**16 / 16**	11111111(255)	11111111(255)	00000000(0)	00000000(0)
Class C	**24 / 8**	11111111(255)	11111111(255)	11111111(255)	00000000(0)



6. Deciding How Many Subnet Bits to Use

The key decision in Subnetting is how many bits to take from the host ID portion of the IP address to put into the subnet ID. The number of subnets possible on our network is two to the power of the number of bits we use to express the subnet ID, and the number of hosts possible per subnet is two to the power of the number of bits left in the host ID (less two, one for network id and one for broadcast).



128	64	32	16	8	4	2	1		
1	0	0	0	0	0	0	0	=	128
1	1	0	0	0	0	0	0	=	192
1	1	1	0	0	0	0	0	=	224
1	1	1	1	0	0	0	0	=	240
1	1	1	1	1	0	0	0	=	248
1	1	1	1	1	1	0	0	=	252
1	1	1	1	1	1	1	0	=	254
1	1	1	1	1	1	1	1	=	255

7. Trading Off Bit Allocations to Meet Subnetting Requirements

The key design decision in Subnetting is how to divide the “Classful” host ID into subnet ID and host ID bits. We must make this choice based on our requirements for the number of subnets that exist in the network, and also on the maximum number of hosts that need to be assigned to each subnet in the network.

Class A

# of Subnet ID Bit	# of Host ID Bits	Subnets Per Network	# of Hosts Per Subnet	Subnet Mask(Binary / Dotted Decimal)	Subnet Mask CIDR
0	**24**	1	16,277,214	11111111.00000000.0000000.00000000 (255.0.0.0)	/8
1	**23**	2	8,388,606	11111111.10000000.0000000.00000000 (255.**128**.0.0)	/9
2	**22**	4	4,194,302	11111111.11000000.0000000.00000000 (255.**192**.0.0)	/10
3	**21**	8	2,097,150	11111111.11100000.00000000.00000000 (255.**224**.0.0)	/11
4	**20**	16	1,048,574	11111111.11110000.00000000.00000000 (255.**240**.0.0)	/12
5	**19**	32	524,286	11111111.11111000.00000000.00000000 (255.**248**.0.0)	/13
6	**18**	64	262,142	11111111.11111100.00000000.00000000 (255.**252**.0.0)	/14
7	**17**	128	131,070	11111111.11111110.00000000.00000000 (255.**254**.0.0)	/15
8	**16**	256	65,534	11111111.11111111.00000000.00000000(255.**255**.0.0)	/16
9	**15**	512	32,766	11111111.11111111.10000000.00000000 (255.**255.128**.0)	/17
10	**14**	1,024	16,382	11111111.11111111.11000000.00000000 (255.**255.192**.0)	/18
11	**13**	2,048	8,190	11111111.11111111.11100000.00000000 (255.**255.224**.0)	/19
12	**12**	4,096	4,094	11111111.11111111.11110000.00000000 (255.**255.240**.0)	/20
13	**11**	8,192	2,046	11111111.11111111.11111000.00000000(255.**255.248**.0)	/21
14	**10**	16,384	1,022	11111111.11111111.11111100.00000000 (255.**255.252**.0)	/22
15	**9**	32,768	510	11111111.11111111.11111110.00000000 (255.**255.254**.0)	/23
16	**8**	65,536	254	11111111.11111111.11111111.00000000 (255.**255.255**.0)	/24
17	7	131,072	126	11111111.11111111.11111111.10000000(255.**255.255.128**)	/25
18	**6**	262,144	62	11111111.11111111.11111111.11000000(255.**255.255.192**)	/26
19	**5**	524,288	30	11111111.11111111.11111111.11100000(255.**255.255.224**)	/27
20	**4**	1,048,576	14	11111111.11111111.11111111.11110000(255.**255.255.240**)	/28
21	**3**	2,097,152	6	11111111.11111111.11111111.11111000(255.**255.255.248**)	/29
22	**2**	4,194,304	2	11111111.11111111.11111111.11111100(255.**255.255.252**)	/30

Class B

# of Subnet ID Bit	# of Host ID Bits	# of Subnets Per Network	# of Hosts Per Subnet	Subnet Mask(Binary / Dotted Decimal)	Subnet Mask CIDR
0	**16**	1	65,534	11111111.11111111.00000000.00000000 (255.255.0.0)	/16
1	**15**	2	32,766	11111111.11111111.10000000.00000000(255.255.**128**.0)	/17
2	**14**	4	16,382	11111111.11111111.**11**000000.00000000 (255.255.**192**.0)	/18
3	**13**	8	8,190	11111111.11111111.**111**00000.00000000 (255.255.**224**.0)	/19
4	**12**	16	4,094	11111111.11111111.**1111**0000.00000000 (255.255.**240**.0)	/20
5	**11**	32	2,046	11111111.11111111.**11111**000.00000000(255.255.**248**.0)	/21
6	**10**	64	1,022	11111111.11111111.**111111**00.00000000 (255.255.**252**.0)	/22
7	**9**	128	510	11111111.11111111.**1111111**0.00000000(255.255.**254**.0)	/23
8	**8**	256	254	11111111.11111111.**11111111**.00000000 (255.255.**255**.0)	/24
9	**7**	512	126	11111111.11111111.**11111111.1**0000000 (255.255.**255.128**)	/25
10	**6**	1,024	62	11111111.11111111.**11111111.11**000000 (255.255.**255.192**)	/26
11	**5**	2,048	30	11111111.11111111.**11111111.111**00000 (255.255.**255.224**)	/27
12	**4**	4,096	14	11111111.11111111.**11111111.1111**0000 (255.255.**255.240**)	/28
13	**3**	8,192	6	11111111.11111111.**11111111.11111**000 (255.255.**255.248**)	/29
14	**2**	16,384	2	11111111.11111111.**11111111.111111**00 (255.255.**255.252**)	/30

Class C

# of Subnet ID Bit	# of Host ID Bits	# of Subnets Per Network	# of Hosts Per Subnet	Subnet Mask(Binary / Dotted Decimal)	Subnet Mask CIDR
0	**8**	1	254	11111111.11111111.11111111.00000000(255.255.255.0)	/24
1	**7**	2	126	11111111.11111111.11111111.10000000(255.255.255.**128**)	/25
2	**6**	4	62	11111111.11111111.11111111.**11**000000 (255.255.255.**192**)	/26
3	**5**	8	30	11111111.11111111.11111111.**111**00000 (255.255.255.**224**)	/27
4	**4**	16	14	11111111.11111111.11111111.**1111**0000 (255.255.255.**240**)	/28
5	**3**	32	6	11111111.11111111.11111111.**11111**000 (255.255.255.**248**)	/29
6	**2**	64	2	11111111.11111111.11111111.**111111**00 (255.255.255.**252**)	/30

Lab 12

IPv4 Address Subnetting-Part2

In this lab we learn about:

1. Requirements Analysis
2. Partitioning Network Address Host Bits
3. Class B Subnetting Design Example
4. Class C Custom Subnet Mask Calculation Example
5. Class B Custom Subnet Mask Calculation Example
6. Determining Subnet Identifiers and Subnet Addresses
7. VLSM & Route Summarization
8. Class B Subnet ID and Address Determination Example
9. CIDR ("Slash") Notation

1: Requirements Analysis

Analyzing the requirements of the network for subnetting isn't difficult, because there are only a few issues that we need to consider. Since requirements analysis is usually done by asking questions, here's a list of the most important questions in analyzing subnetting requirements:

- What class is our IP address block?
- How many physical subnets are on the network? (A “physical subnet” generally refers to a broadcast domain on a LAN; a set of hosts on a physical network bounded by routers.)
- Are we adding more physical networks in the near future? If so then how many?
- How many hosts do we have in the largest subnet today?
- How many hosts do we have in the largest subnet in the near future?

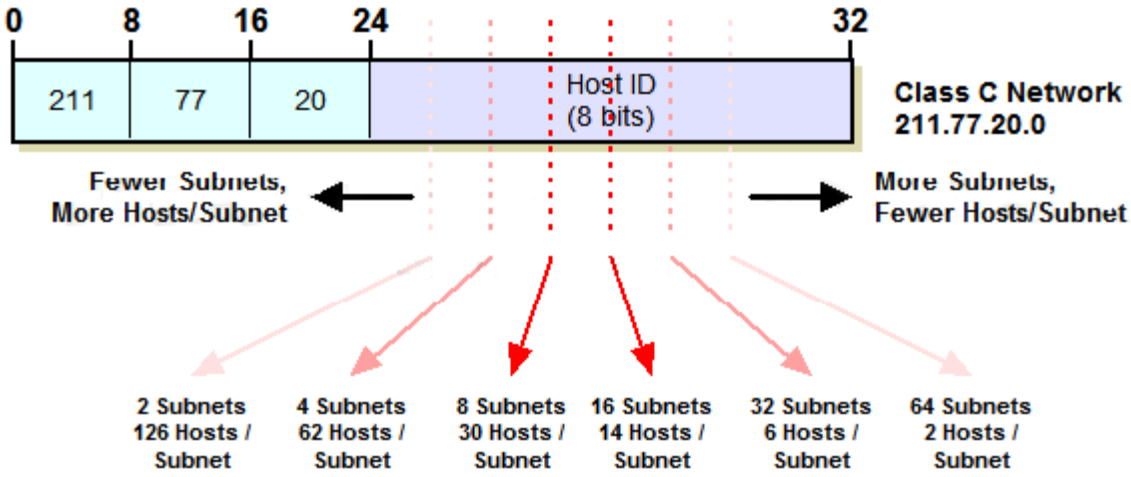
We need to analyze the requirements above not only for the present network, but for the near future as well.

2: Partitioning Network Address Host Bits

After we complete our brief requirements analysis, we should know the two critical parameters that we must have in order to subnet our network: the number of subnets required for the network, and the maximum number of hosts per subnetwork. In using these two figures to design our Subnetted network, we will decide while subnetting: how to divide the 8, 16 or 24 bits in the "classful" host ID into subnet ID and host ID. We need to decide how many bits to borrow from the host ID to use for the subnet ID.

Fundamental trade-off in choosing this number is as follows:

There are six possible ways this decision can be made for a Class C network, as the following figure illustrates:

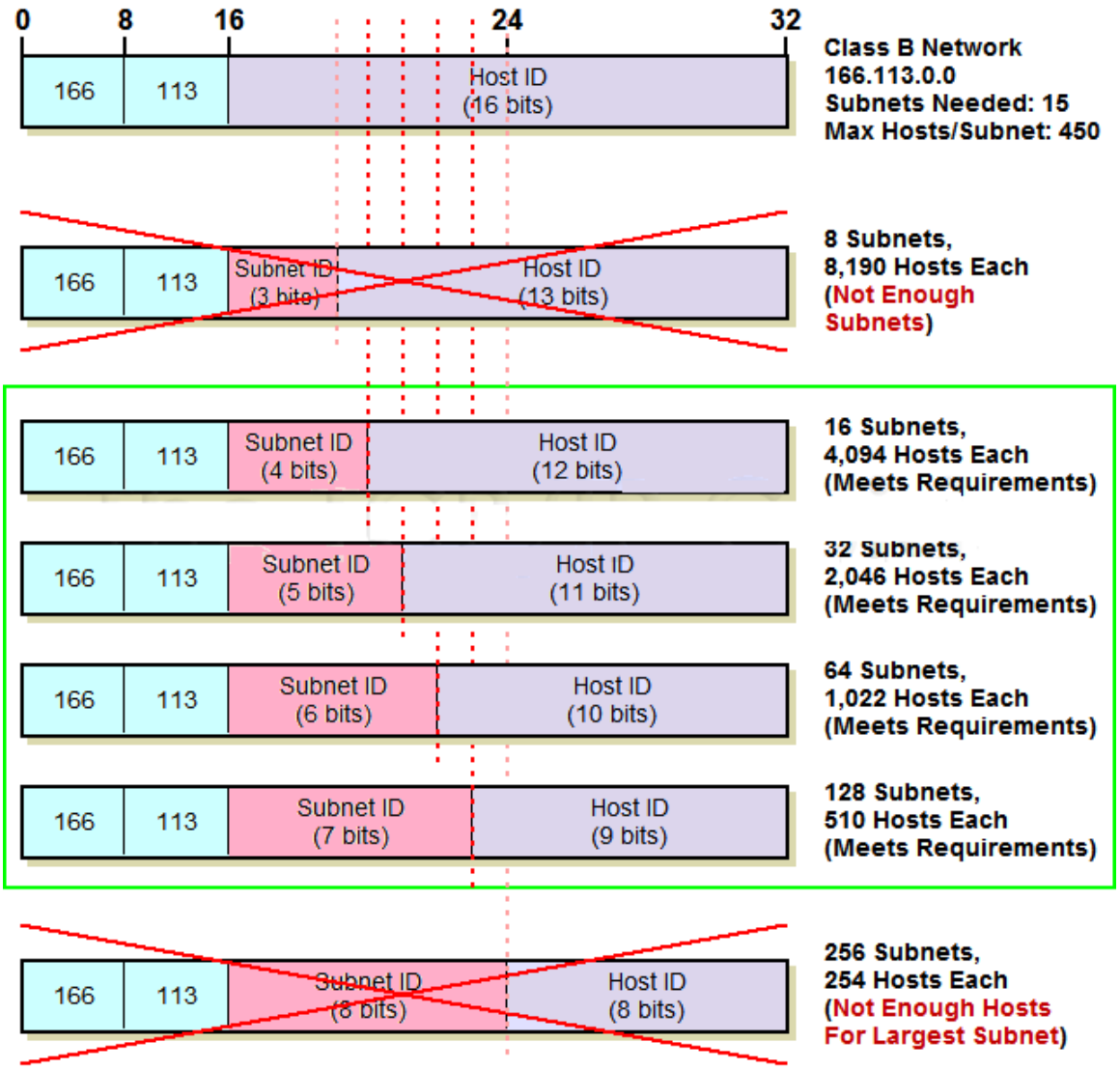


The relationship between the bits and the number of subnets and hosts is as follows:

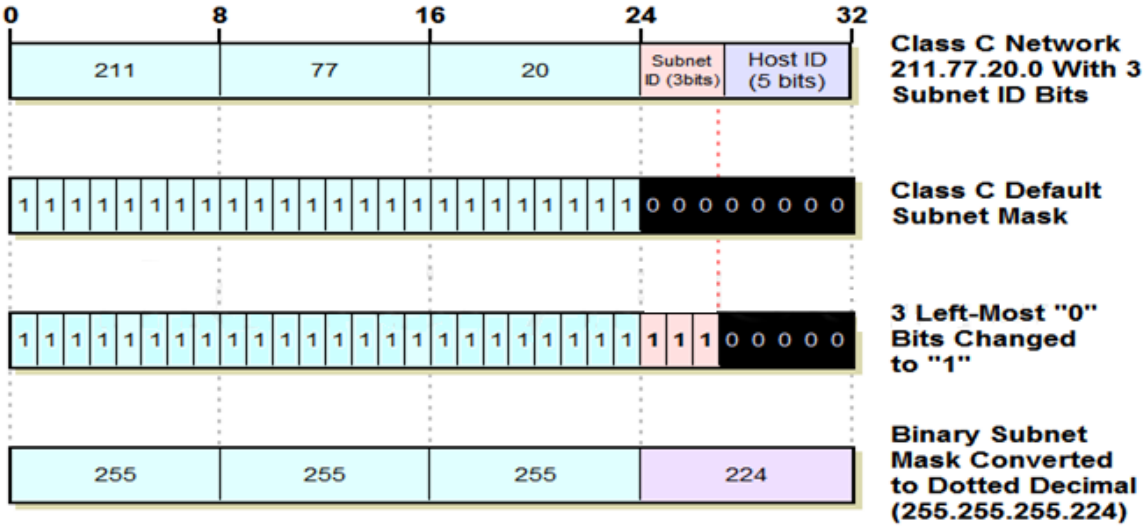
- The number of subnets allowed in the network is two to the power of the number of subnet ID bits.
- The number of hosts allowed per subnet is two to the power of the number of host ID bits, minus two.

We subtract two from the number of hosts in each subnet to exclude the "special meaning" cases where the host ID is all zeroes or all ones. First we must calculate the number of subnets and hosts when we use the subnet ID bits and leave the rest for the host ID.

3. Class B Subnetting Design Example



4. Class C Custom Subnet Mask Calculation Example



1. **Determine Default Subnet Mask:** Each of Classes A, B and C has a default subnet mask, which is the subnet mask for the network prior to subnetting. It has a 1 for each network ID bit and a 0 for each host ID bit. For Class C, the subnet mask is 255.255.255.0.

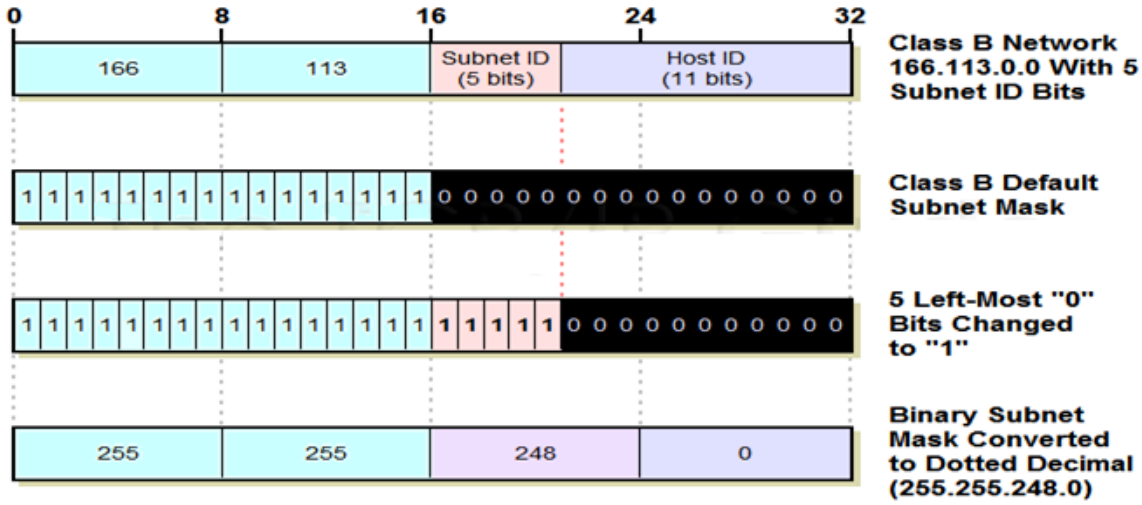
2. **Change Left-Most Zeroes to Ones for Subnet Bits:** We have decided to use 3 bits for the subnet ID. The subnet mask has to have a 1 for each of the network ID or subnet ID bits. The network ID bits are already 1 from the default subnet mask, so, we change the 3 **left-most** 0 bits in the default subnet mask from a 0 to 1

3. **Convert Subnet Mask To Dotted Decimal Notation:** We take each of the octets in the subnet mask and convert it to decimal. The result is our custom subnet mask in the form we usually see it: 255.255.255.224.

4. **Express Subnet Mask In "Slash Notation":** Alternately, we can express the subnet mask in "slash notation". This is just a slash followed by the number of ones in the subnet mask. 255.255.255.224 is equivalent to "/27".

5. Class B Custom Subnet Mask Calculation Example

Now, let's do the same example with our Class B network (166.113.0.0) with 5 bits for the subnet ID (with a bit less narration this time):



1. **Determine Default Subnet Mask:** For Class B, the subnet mask is 255.255.0.0. In binary, this is:11111111 11111111 00000000 00000000

2. **Change Left-Most Zeroes to Ones for Subnet Bits:** We have decided to use 5 bits for the subnet ID, so, we change the 5 left-most 0 bits from 0 to 1.

3. **Convert Subnet Mask to Dotted Decimal Notation:** We take each of the octets in the subnet mask and convert it to decimal, to give us a custom subnet mask of 255.255.**248**.0

4. **Express Subnet Mask in "Slash Notation":** We can express the subnet mask 255.255.248.0 as "/21", since it is 21 ones followed by 11 zeroes. In other words, its prefix length is 21.

6: Determining Subnet Identifiers and Subnet Addresses:

The network ID assigned to our network applies to the entire network. This includes all subnets and all hosts in all subnets. Each subnet, however, needs to be identified with a unique subnet identifier called **subnet ID**, so it can be differentiated from the other subnets in the network. This is of course the purpose of the subnet ID bits that we took from the host ID bits in subnetting. After we have identified each subnet we need to determine the address of each subnet, so we can use this in assigning hosts specific IP addresses.

The key to understanding how to determine subnet IDs and subnet addresses is to always work in binary form, and then convert to decimal later.

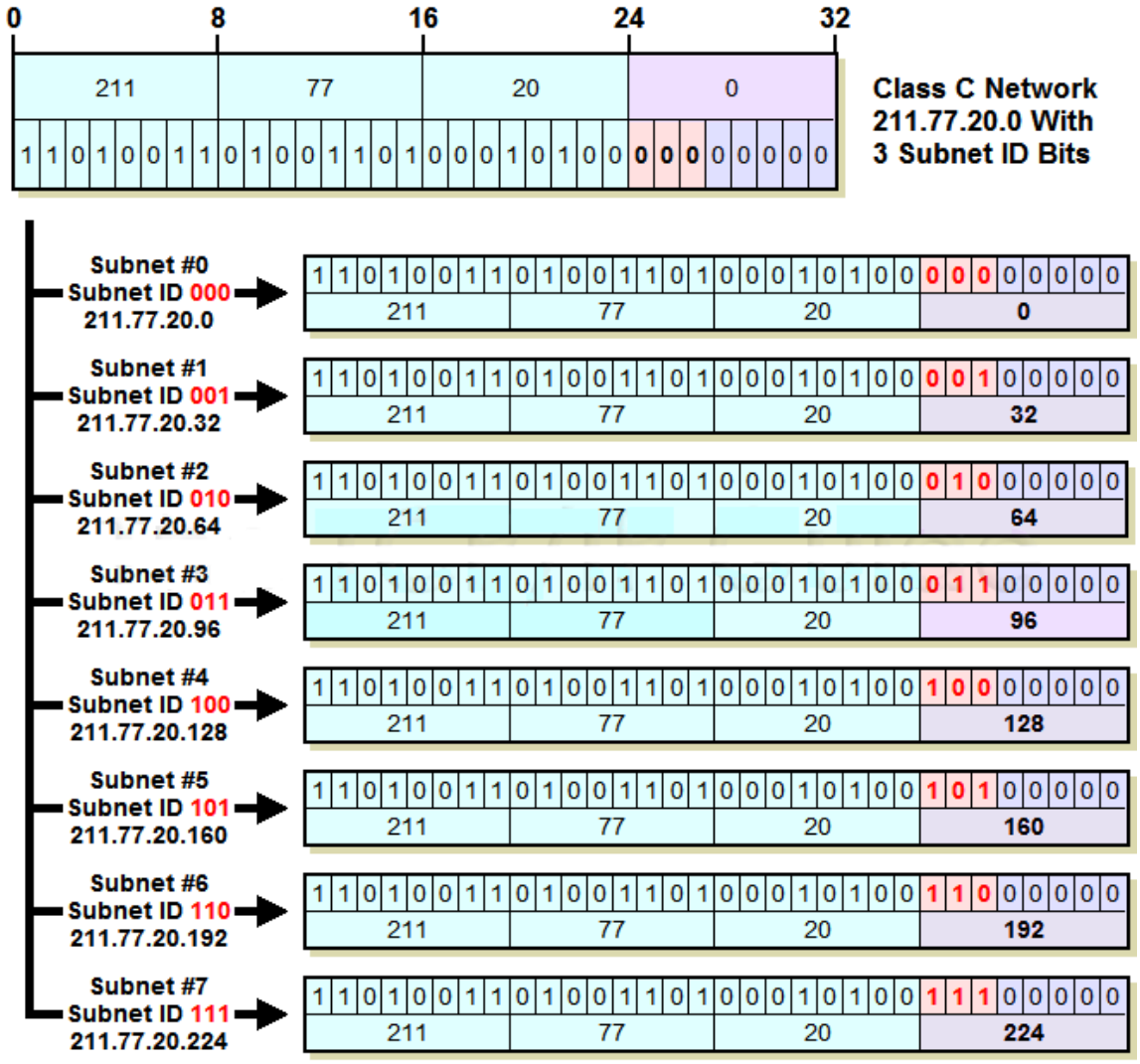
7. VLSM & Route Summarization:

We determine the subnet IDs and addresses as follows:

1. **Subnet ID:** This is just the subnet number, and can be expressed in either binary or decimal form.

2. **Subnet Address:** This is the address formed by taking the address of the network as a whole, and substituting the (binary) subnet ID in for the subnet ID bits. We need to do this in binary, but only for the octets where there are subnet ID bits; the ones where there are only network ID bits or only host ID bits are left alone.

Class C Subnet ID and Address Determination Example

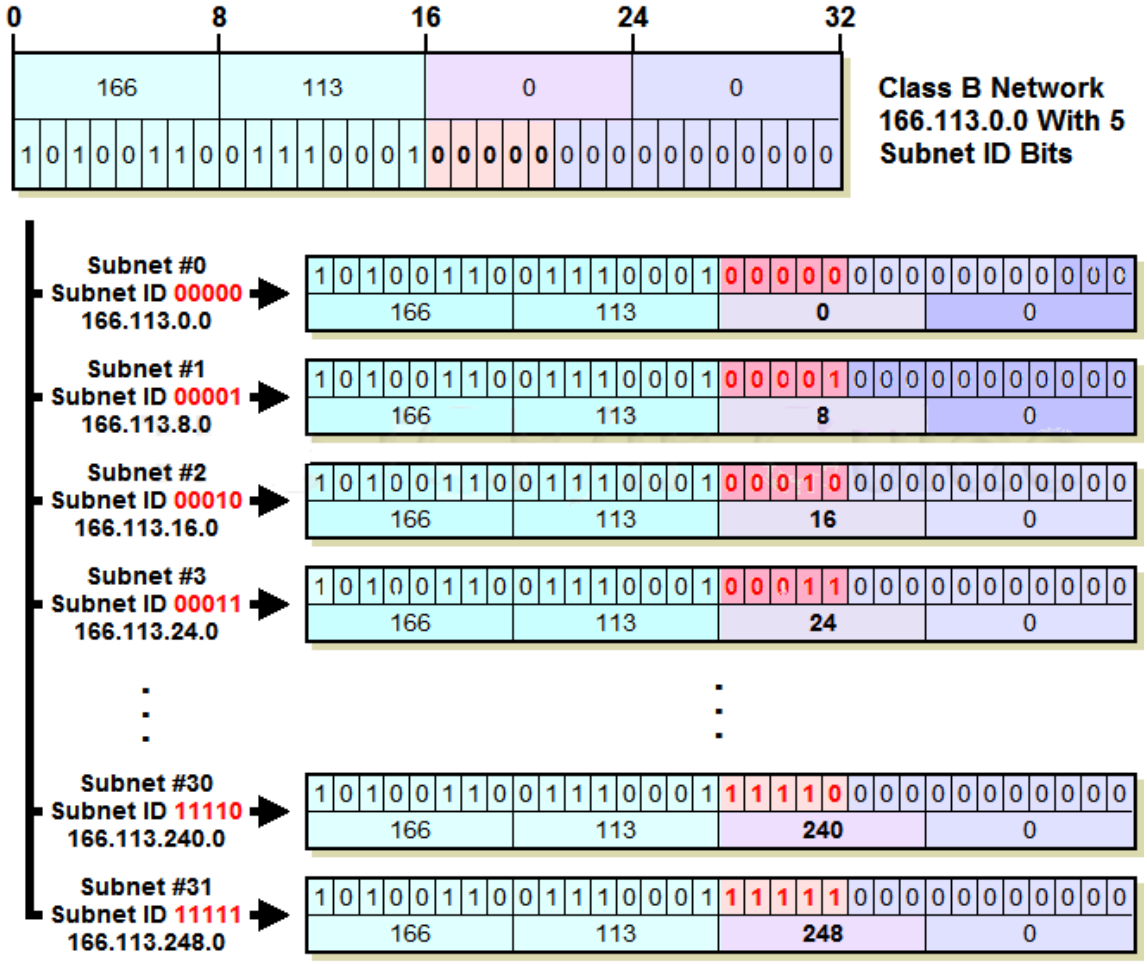


This diagram shows each of the 8 possible subnets created when we use 3 bits for the subnet ID in a Class C network. The binary subnet ID is simply substituted for the subnet bits, and the resulting 32-bit number converted to dotted decimal form. The address of any subnet can be found by adding 32 to the last octet of the previous subnet. This pattern occurs for all subnetting choices; the increment depends on how many bits we are using for the subnet ID.

Here, the increment is 32, which is 25; 5 is the number of host ID bits left after we took 3 subnet ID bits.

8. Class B Subnet ID and Address Determination Example

Class B network 166.113.0.0. We are using 5 bits for the subnet ID, leaving 11 hosts ID bits.



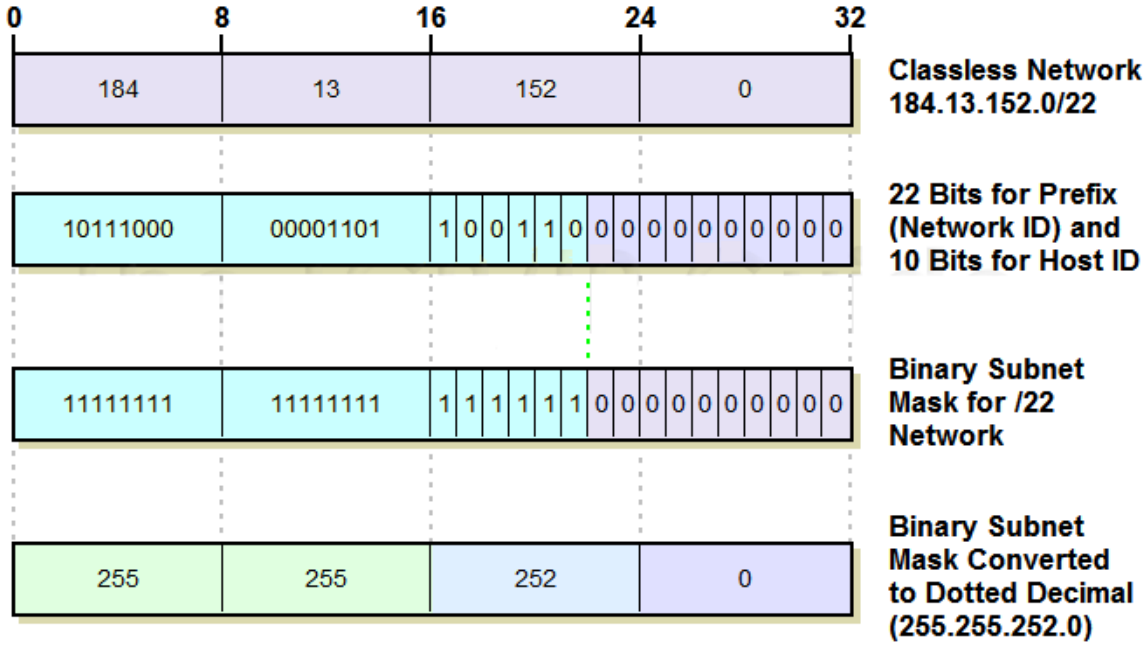
This diagram shows how both subnet addresses and host addresses are determined in a two-step process. The subnet addresses are found by substituting subnet ID values (shown in red) for the subnet ID bits of the network. Then, for any given subnet address, we can determine a host address by substituting a host number (shown in blue) for the host ID bits within that subnet. So, for example, host #2 in subnet #6 has "110" for the subnet ID and "00010" for the host ID, resulting in a final octet value of "11000010" or 194.

9. CIDR ("Slash") Notation

Classless addressing scheme sometimes called Classless Inter-Domain Routing (CIDR).

Just as subnetting required the use of a subnet mask to show which bits belong to the network ID or subnet ID and which to the host ID, CIDR uses a subnet mask to show where the line is drawn between host ID and network ID. However, for simplicity, under CIDR we don't usually work with 32-bit binary subnet masks. Instead, we use slash notation, more properly called CIDR notation. In this method, we show the size of the network, sometimes called the prefix length, by following an IP address by an integer that tells us how many bits are used for the network ID (prefix).

For example, consider the network specification 184.13.152.0/22. The “22” means this network has 22 bits for the network ID and 10 bits for the host ID. This is equivalent to specifying a network with an address of 184.13.152.0 and a subnet mask of 255.255.252.0. This sample network provides a total of 1,022 hosts (210 minus 2). The table in the following topic shows all the different possible network sizes that can be configured under CIDR.



Important note about subnetting:

The network ID is the same for all hosts in all subnets, and all subnets in the network.

The subnet ID is the same for all hosts in each subnet, but unique to each subnet in the network.

The host ID is unique within each subnet. Each subnet has the same set of host IDs.

Network ID Bits	Host ID Bits	Hosts Per Network	CIDR	Equivalent Subnet Mask
1	31	2,147,483,646	/1	128.0.0.0
2	30	1,073,741,822	/2	192.0.0.0
3	29	536,870,910	/3	224.0.0.0
4	28	268,435,454	/4	240.0.0.0
5	27	134,217,726	/5	248.0.0.0
6	26	67,108,862	/6	252.0.0.0
7	25	33,554,430	/7	254.0.0.0
8	24	16,777,214	/8	255.0.0.0
9	23	8,388,606	/9	255.128.0.0
10	22	4,194,302	/10	255.192.0.0
11	21	2,097,150	/11	255.224.0.0
12	20	1,048,574	/12	255.240.0.0
13	19	524,286	/13	255.248.0.0
14	18	262,142	/14	255.252.0.0
15	17	131,070	/15	255.254.0.0
16	16	65,534	/16	255.255.0.0
17	15	32,766	/17	255.255.128.0
18	14	16,382	/18	255.255.192.0
19	13	8,190	/19	255.255.224.0
20	12	4,094	/20	255.255.240.0
21	11	2,046	/21	255.255.248.0
22	10	1,022	/22	255.255.252.0
23	9	510	/23	255.255.254.0
24	8	254	/24	255.255.255.0
25	7	126	/25	255.255.255.128
26	6	62	/26	255.255.255.192
27	5	30	/27	255.255.255.224
28	4	14	/28	255.255.255.240
29	3	6	/29	255.255.255.248
30	2	2	/30	255.255.255.252

Lab 13

Practice question about IPv4 subnetting:

1. How many subnets and hosts per subnet can you get from the network 172.22.0.0/28?

   Answer: 4096 subnets and 14 hosts

2. Write the broadcast address for the network 172.25.253.0/24:

   Answer: 172.25.253.255

3. What subnet mask would you use for the 172.29.0.0 network, such that you can get 120 subnets and 280 hosts per subnet?

   Answer: 255.255.254.0

4. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 192.168.16.0/24:

   Answer: Subnets: 1, Hosts per subnet: 254

5. Write the broadcast address for the network 172.28.38.0 255.255.254.0:

   Answer: 172.28.39.255

6. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 172.23.0.0 255.255.255.240:

   Answer: Subnets: 4096, Hosts per subnet: 14

7. Write the last valid host on the network 10.207.14.0/23:

   Answer: 10.207.15.254

8. What is the Subnet Mask corresponding to a shorthand of /21?

Answer: 255.255.248.0

9. Write the valid host range for the network that the IP address 10.2.181.162 255.255.255.252 is a part of:

   Answer: First Host: 10.2.181.161, Last Host: 10.2.181.162

10. Write the first valid host on the network 192.168.177.0/24:
    Answer: 192.168.177.1

11. Write the broadcast address for the network 172.16.176.0/23
    Answer: 172.16.177.255

12. What is the shorthand corresponding to a subnet mask of 255.224.0.0?
    Answer: /11

13. Write the first valid host on the network 10.66.79.224 255.255.255.252:
    Answer: 10.66.79.225

14. What is the Subnet Mask corresponding to a shorthand of /15?
    Answer: 255.254.0.0

15. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 172.27.0.0/24:
    Answer: Subnets: 256, Hosts per subnet: 254

16. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 192.168.101.0/28:
    Answer: Subnets: 16, Hosts per subnet: 14

17. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 172.18.0.0/28:
    Answer: Subnets: 4096, Hosts per subnet: 14

18. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 172.20.0.0/22:
    Answer: Subnets: 64, Hosts per subnet: 1022

19. Write the maximum number of valid subnets and usable hosts per subnet that you can get from the network 192.168.152.0 255.255.255.248:
    Answer: Subnets: 32, Hosts per subnet: 6

20. What are the subnet mask, first available host address of subnet one (NOT subnet zero), and the maximum number of hosts per subnet for the network below?

    Network ID: 184.97.0.0

    Subnets Required: 117

    Answer:

    Subnet Mask: 255.255.254.0

    1st Available Host Address of Subnet 1: 184.97.2.1

    Max # of hosts/subnet: 510

21. What are the subnet mask, first available host address of subnet one (NOT subnet zero), and the maximum number of hosts per subnet for the network below?

    You need to allow for maximum number of hosts. Also, you can use subnet zero and the all-ones subnet.

Network ID: 219.59.71.0

Subnets Required: 6

Answer:

Subnet Mask: 255.255.255.224

1st Available Host Address of Subnet 1: 219.59.71.33

Max # of hosts/subnet: 30

22. What are the network address, broadcast address, and the subnet mask for a host with the IP Address below?

    IP Address: 128.161.72.218 / 30

    Answer:

    Network Address: 128.161.72.216

    Broadcast Address: 128.161.72.219

Subnet Mask: 255.255.255.252

23. What are the network address, broadcast address, and the subnet mask for a host with the IP Address below?
    IP Address: 2**0**6. 232. 120. 201/ 24

    Answer:

    Network Address: 206.232.120.0

    Broadcast Address: 206.232.120.255

    Subnet Mask: 255.255.255.0

24. What are the network address, broadcast address, and the subnet mask for a host with the IP Address below?
    IP Address: 150. 20. 204. 138/ 25

    Answer:

    Network Address: 150.20.204.128

    Broadcast Address: 150.20.204.255

    Subnet Mask: 255.255.255.128

25. What are the network address, broadcast address, and the subnet mask for a host with the IP Address below?
    IP Address: 202. 7. 57. 105/ 20

    Answer:

    Network Address: 202.7.48.0

    Broadcast Address: 202.7.63.255

    Subnet Mask: 255.255.240.0

26. You have been allocated a class B network address of 135.1.0.0 and and need to create 4 subnets each with around 200 hosts what is the easiest mask to use to satisfy the criteria?

Solution:

Easiest is to sub net on a byte boundary which would mean a subnet mask of 255.255.255.0

This would allocate 8 bits for the subnet and 8 bits for the host.

We need to accommodate around 200 hosts which requires 8 bits which we have.

We need 4 subnets which require 4 bits and we have 8 bits. So we have more than enough.

27. Write the IP address 222.1.1.20 mask 255.255.255.192 in CIDR notation

Decimal 192 =11000000 binary which means that 2 bits of this octet are used for the subnet. Now add the 24 bits 255.255.255 and we have 26 bits. So we write:

Solution:

222.1.1.20/262

28. Write the IP address 135.1.1.25 mask 255.255. 248.0 in CIDR notation

Decimal 248 =11111000 binary which means that 5 bits of this octet are used for the subnet. Now add the 16 bits 255.255. and we have 21 bits. So we write:

Solution:

135..1.1.25/21

29. You have been allocated a class C network address of 211.1.1.0 and are using the default subnet mask of 255.255.255.0 how may hosts can you have?

Solution:

A class C address has 8 bits of the host which will give 28 -2 =254 hosts

30. Subnet the Class C IP Address 195.1.1.0 So that you have 10 subnets each with a maximum 12 hosts on each subnet. List the Address on host 1 on subnet 0,1,2,3,10

Solution:

Current mask= 255.255.255.0

Bits needs for 10 subnets =4 =24 =16 possible subnets

Bits needs for 12 hosts = 4 = 24 = 16-2=14 possible hosts.

So our mask in binary =11110000= 240 decimal

Final Mask =255.255.255.240

Hosts on Subnets 0,1,2,3,10

- Subnet 0 host 1 IP address = 195.1.1.1 **0000 0001**
- Subnet 1 host 1 IP address = 195.1.1.17 **0001 0001**
- Subnet 2 host 1 IP address = 195.1.1.33 **0010 0001**
- Subnet 3 host 1 IP address = 195.1.1.49 **0011 0001**
- Subnet 10 host 1 IP address = 195.1.1.161 **1010 0001**

31. Subnet the Class C IP Address 205.11.2.0 so that you have 30 subnets.

What is the subnet mask for the maximum number of hosts?

How many hosts can each subnet have?

What is the IP address of host 3 on subnet 2 ?

Solution:

Current mask= 255.255.255.0

Bits needs for 30 subnets =5 =25 =32 possible subnets

Bits left for hosts = 3 = 23 = 8-2=6 possible hosts.

So our mask in binary =11111000= 248 decimal

Final Mask =255.255.255.248

Address of host 3 on subnet 2 is

subnet 2 =00010000 host 3 =000000011

Add the two together =00010011=19

therefore IP address of host 3 on subnet 2 =205.11.2.19

32. Subnet the Class C IP Address 195.1.1.0 So that you have at least 2 subnets each subnet must have room for 48 hosts .

    What are the two possible subnet masks?

Solution:

Current mask= 255.255.255.0

Bits needs for 48 hosts = 6 = 26 = 64-2=62 possible hosts.

Bits needs for 2 subnets =1 =21 =2 possible subnets

Total of 7 bits needed so therefore we can use either 1 bit or 2 bits for the subnet. So we could have

1 bit subnet 7 bits hosts or 2 bits subnet 6 bit host

masks are 10000000 and 11000000 =128 decimal and 192 decimal.

Final possible masks are:

255.255.255.128 and 255.255.255.192

Real world Scenario:

Consider the table below for real word example

Department	No of Hosts
Engineering	27
Customer Care	18
IT	8
Finance	12
HR	4

The requirement is for 5 subnets with the largest number of hosts being 27.

In order to create the subnets, bits will be borrowed from the fourth octet by extending the subnet mask from 24 to 27. The three extra borrowed bits are required as 2^3 will allow for eight subnets. By borrowing three bits however there are now only 5 bits remaining for hosts. That will be sufficient as 2^5 = 32 and our biggest requirement is for 27.

The network plan will be

Department	No of Hosts	Subnet	Spare IP
Engineering	27	192.168.0.0/27	3
Customer Care	18	192.168.0.32/27	12
IT	8	192.168.0.64/27	22
Finance	12	192.168.0.96/27	18
HR	4	192.168.0.128/27	26

The network requirement have been satisfied in a theoretical sense, by providing 5 subnets and sufficient IP host addresses to meet the requirement. However, it is not a good real world design. HR and IT subnets are very wasteful. By splitting the address range into eight equal subnets, three subnets went unused, it was less than optimal. Is there another way to subnet?

Design the Network using VLSM

Using the same network scenario redesign the IP Plan using subnetting with Variable Length Subnet Masking (VLSM).

With this task it is necessary to consider what masks we can apply for each subnet that provides a more efficient use of the address space.

Hint: Start with the largest departments first, then size the subnet mask to fit by using the minimum number of host bits. Remember, the binary bits to decimal mapping.

Solution

Department	No of Hosts	Subnet /Mask
Engineering	27	192.168.0.0/27 (30 Hosts)

Customer Care	18	192.168.0.32/27 (30 Hosts)
IT	8	192.168.0.64/28 (14 Hosts)
Finance	12	192.168.0.80/28 (14 Hosts)
HR	4	192.168.0.96/29 (6 Hosts)

By addressing the number of hosts required per department, a more efficient IP address plan can be constructed, which minimizes IP wasted addresses albeit that is not a concern with private addresses. However, practice on conserving address space is necessary for when real valuable IP addresses are used.

Scenario:

You work for a large communications corporation named GlobeComm which has been assigned a Class A network address. Currently, the company has 1,000 subnets in offices around the world. You want to add 100 new subnets over the next three years, and you want to allow for the largest possible number of host addresses per subnet.

Which subnet mask would you choose?

Solution:

Answer:255.255.224.0

Requirements: 1,000 subnets + 100 subnets = 1,100 subnets, and as many host addresses as possible

2^10-2 = 1022 we Know from memory that 2^10= 1024, and will yield ONLY 1022 usable subnets with 10 bits of subnetting.

Because our requirements ask for over 1024 subnets, we must borrow one more host bit for our subnetting:

2^11-2 = (1024 * 2) - 2 = 2046 unique usable subnets

11111111 00000000 00000000 00000000 = 255.0.0.0 Default class A subnet mask

11111111 11111111 11100000 00000000 = 255.255.224.0 Default Class A subnet mask with our 11 bits of subnetting

Scenario:

NuTex Corporation has been assigned the Class B network address 165.87.0.0. NuTex needs to divide the network into eight subnets. What subnet mask should be applied to the network to provide the most hosts per subnet?

Solution:

Answer: 255.255.240.0
2^3-2 = 6 subnets
2^4-2 = 14 subnets
(2^16 - 2^4) -2 = 2^12 - 2 = 4094 hosts possible per subnet

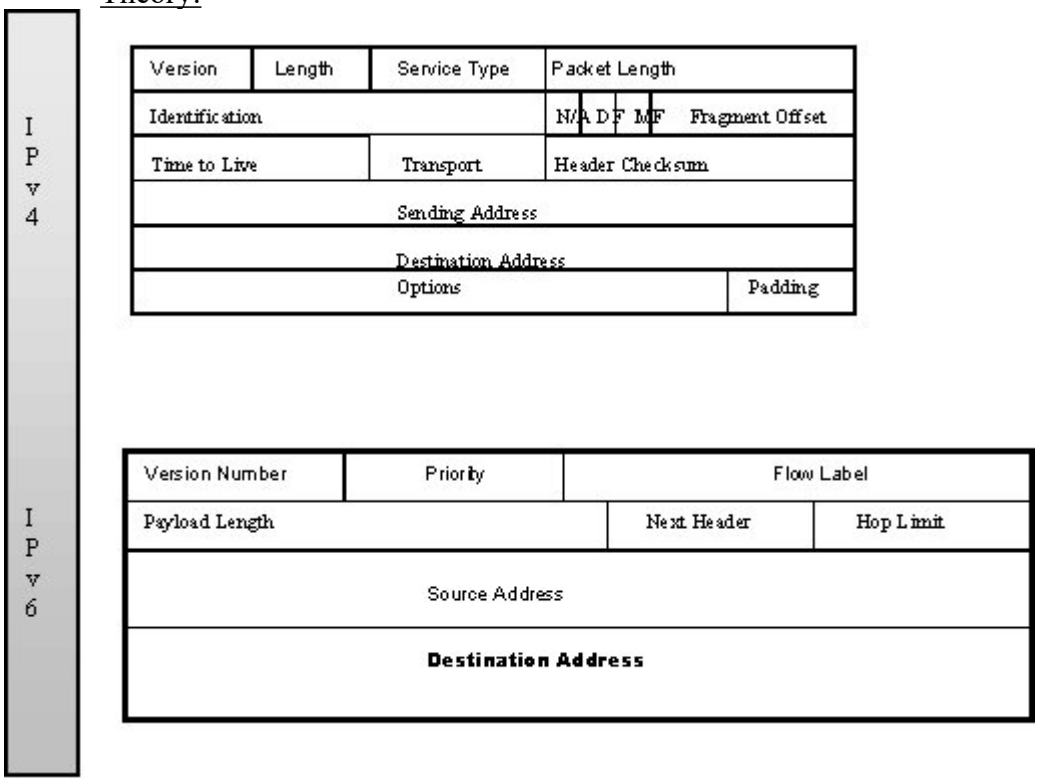
Because 3 bits of subnetting will only yield 6 usable subnets (7 if ip subnet-zero is enabled), you will need to borrow another bit. To meet your minimum requirement of 8 subnets, and the maximum number of hosts per subnet, this Subnet mask will be correct: 255.255.240.0

Lab 14

Title: **Study of IPv4 and IPv6.**
Objectives:

1. Identify the different IP addressing versions notation.
2. Understand the difference between IPv4 and IPv6.

Theory:



IPv4:-

- Fourth revision of development of Internet Protocol (IP)
- Most widely used protocol at present
- Connection-less protocol used on packet-switched networks
- A system of addresses
    - To identify devices
    - 32 bits
        - 2^{32} combinations
        - Nearly 4.3 billion
- Not enough and more addresses are needed

Problems of IPv4:-

- **Insufficiency**

- Only four bytes
- Maximum nodes ~ 4.3 billion
- Much less than the human population
- Not enough for growing number of users
- Will be exhausted in near future

- **Increasing routing information**

- Rapid growth of routing tables in backbones

- **Increasing the number of Network Address Translation (NAT)**
  Breaking the Internet architecture
- **Security issues**

- Number of ways to encrypt IPv4 traffic

  Example: IPsec

- No real standard encryption method

Network Address Translation (NAT):-

- Allows to maintain public IP addresses and private IP addresses separately
- Also used in server load balancing
- This operates inside the router
- Maintains an address pool
- First the destination is checked
- The address is translated
- The network address information in the datagram is modified

IPv6:-

- Next development version of Internet protocol
- Address size is 128 bits
- IPv6 address representation

  - xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx
  - Each x is a 4bit hexadecimal digit
  - IPv6 addresses range from

    0000:0000:0000:0000:0000:0000:0000:0000 to
    ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff.

- **Two Other Formats**

- **Omit leading zeros** : Specify IPv6 addresses by omitting leading zeros E.g. 1050:0000:0000:0000:0005:0600:300c:326b can be written as 1050:0:0:0:5:600:300c:326b
- **Double colon** :Specify IPv6 addresses by using double colons (::) in place of a series of zeros E.g. : ff06:0:0:0:0:0:0:c3 may be written as ff06::c3
- Double colons can be used only once in one IP address

- **Bigger Address Space**

- 128 bits for addressing
- No NATs are needed

- **Allows full IP connectivity**

- For Computers, Mobile devices
- Mobility
- Facility for mobile devices
- Allows roaming between different networks

- Built-in security

    - Includes IPsec

        - Authentication
        - Encryption
        - Compression

- **IPv6 Address Notation**

- Three possible notation types
- Depends on content of address you need to represent

1. **Standard Notation**:

    Represent the address as eight 16-bit hexadecimal words separated by ``:" (colons) E.g. :
    FEDC:BA98:0332:0000:CF8A:000C:2154:7313 or
    FEDC:BA98:332:0:CF8A:C:2154:7313.

1. **Compressed notation**
   Multiple fields of zeros

    -

        -

- Represent a single contiguous group of zero fields within an IPv6 address
- Uses a double colon``: ‘’

E.g.:1762:0:0:0:0:B03:1:AF18 FF01:0:0:0:CA:0:0:20:0:0:0:0:0:0:1:0:0:0:0:0:0:0 can be represented as 1762::B03:1:AF18 FF01::CA:0:0:2 ::1 ::

1. **Mixed notation**

    1.

        i. For IPv4 addresses encapsulated in IPv6 addresses
        ii. Represented using the original IPv4 ``." notation

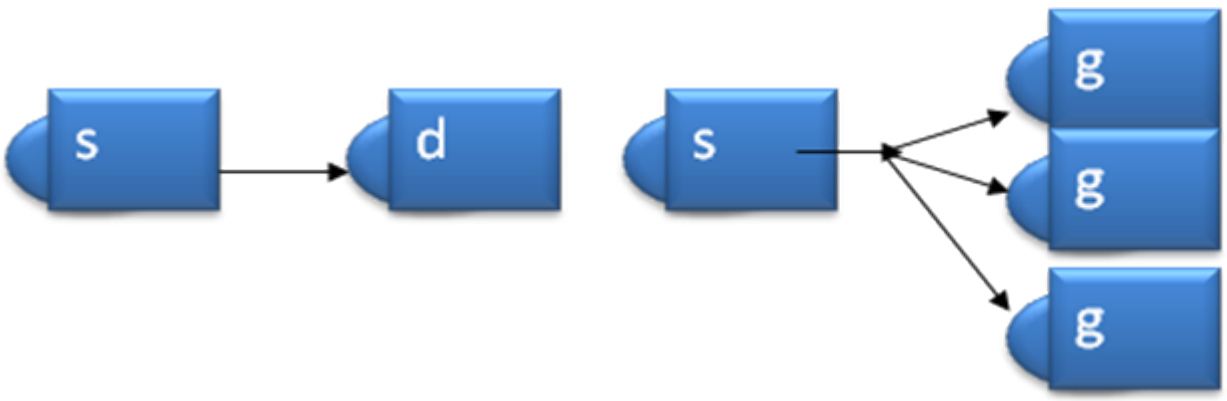
            1. 0:0:0:0:0:0:127.32.67.15
            2. 0:0:0:0:0:FFFF:127.32.67.15

        iii. Possible to use the compressed notation

            1. ::127.32.67.15
            2. ::FFFF:127.32.67.15

Types of addresses:

- Unicast
    - Communicate specified 1 computer
- Multicast
    - Communicate group of computers
- Anycast
    - Send group address that can receive multiple computers, but received by 1 computer



Unicast **Multicast** **Anycast**

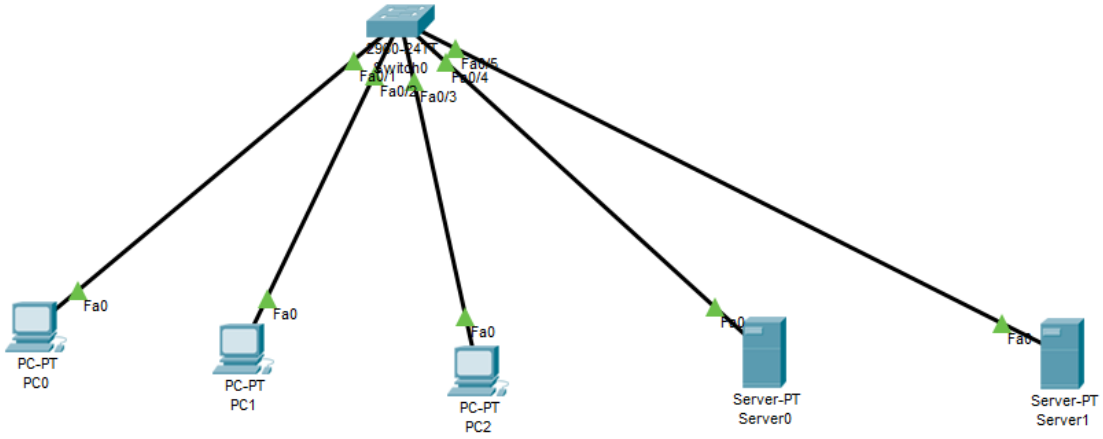
Difference between IPv4 and IPv6

IPv4	**IPv6**
IPv4 addresses are 32 bit length.	IPv6 addresses are 128 bit length.
IPv4 addresses are binary numbers represented in decimals.	IPv6 addresses are binary numbers represented in hexadecimals.
IPsec support is only optional.	Inbuilt IPsec support.
Fragmentation is done by sender and forwarding routers.	Fragmentation is done only by sender.
No packet flow identification.	Packet flow identification is available within the IPv6 header using the Flow Label field.
Options fields are available in IPv4 header.	No option fields, but IPv6 Extension headers are available.
Broadcast messages are available.	Broadcast messages are not available. Instead a link-local scope "All nodes" multicast IPv6 address (FF02::1) is used for broadcast similar functionality.
Manual configuration (Static) of IPv4 addresses or DHCP (Dynamic configuration) is required to configure IPv4 addresses.	Auto-configuration of addresses is available.

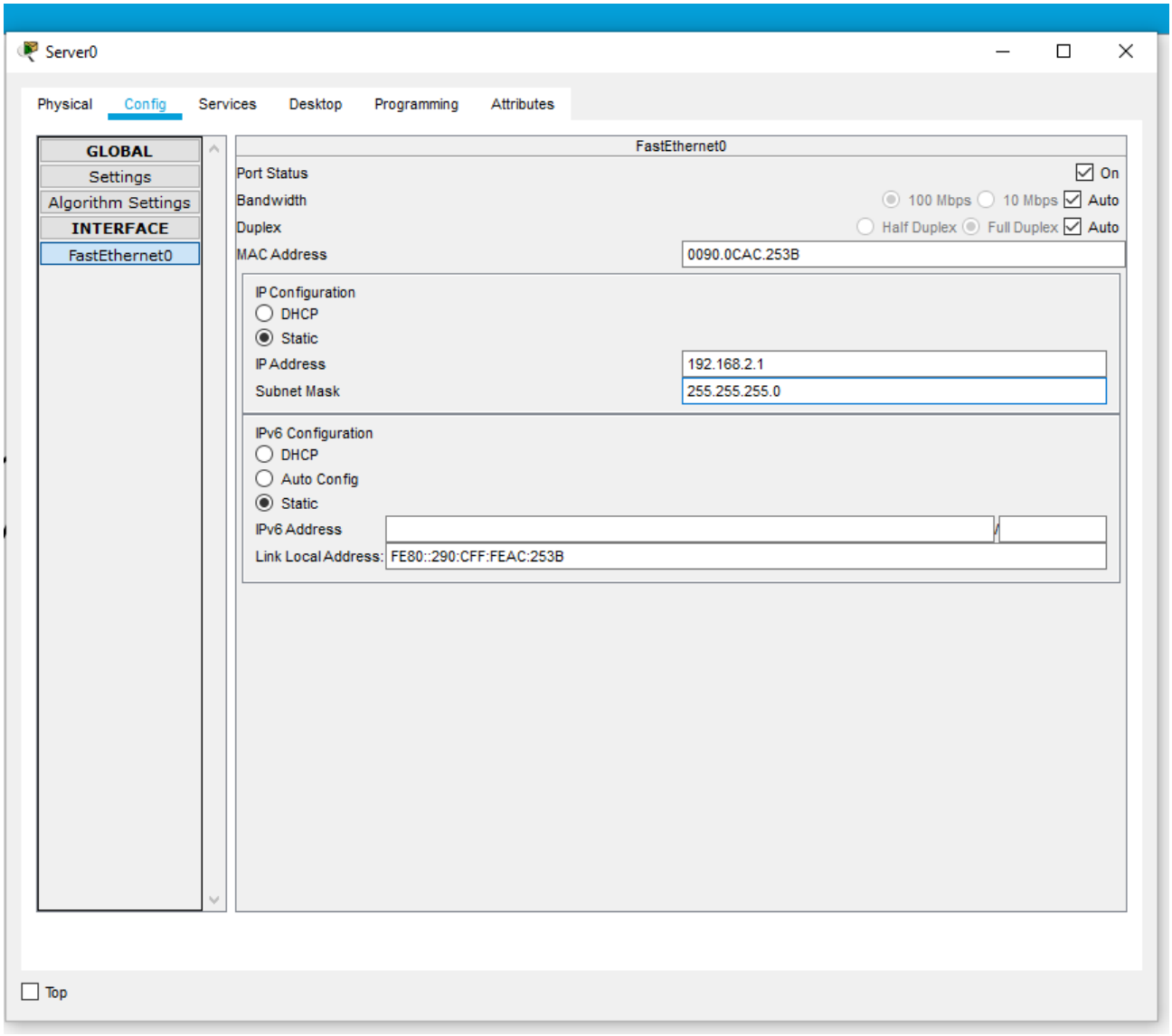
Lab 15

Configuring DHCP and Web server

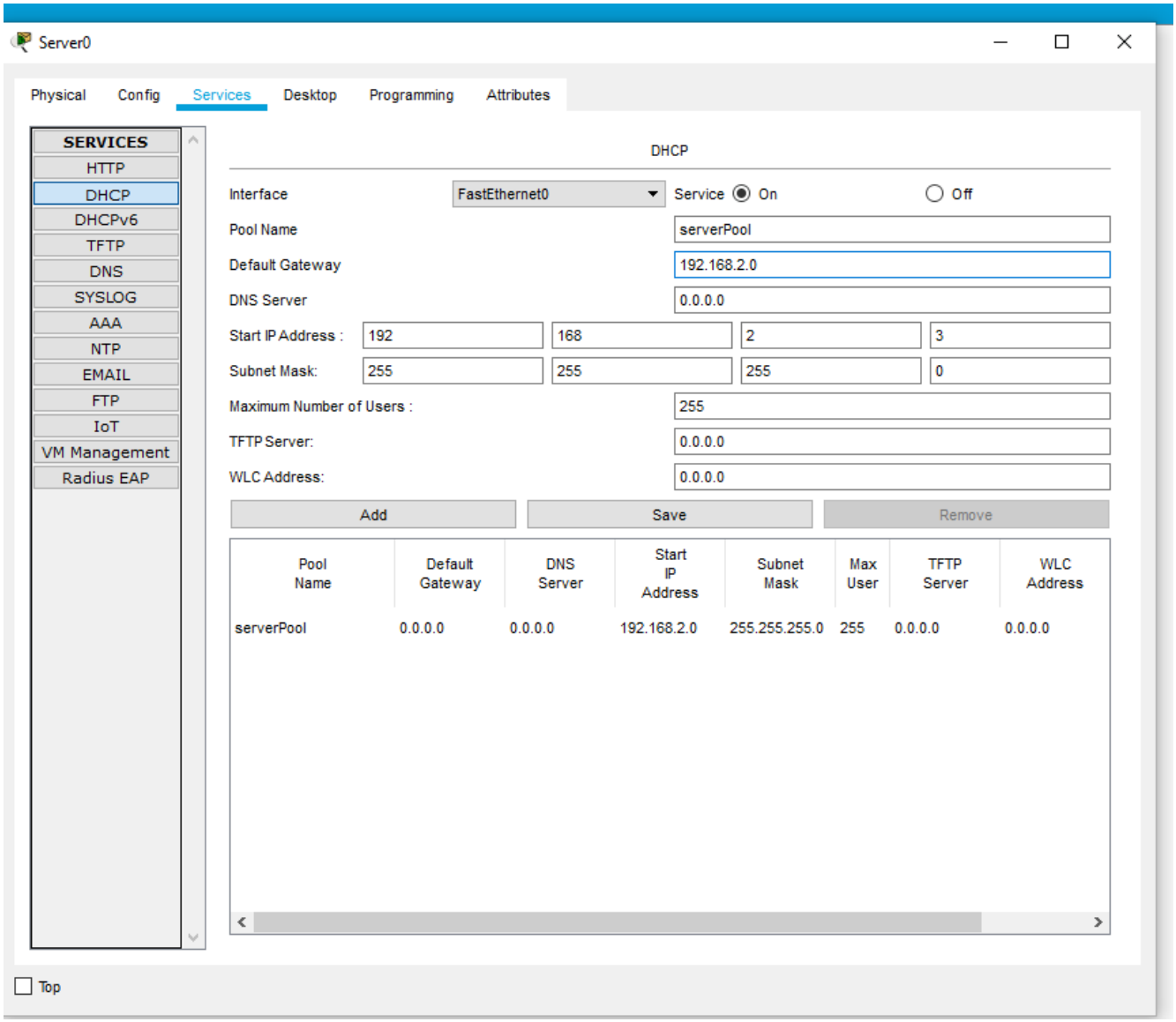
Add 3 hosts and two servers in packet tracer and attach them with a switch as given below:



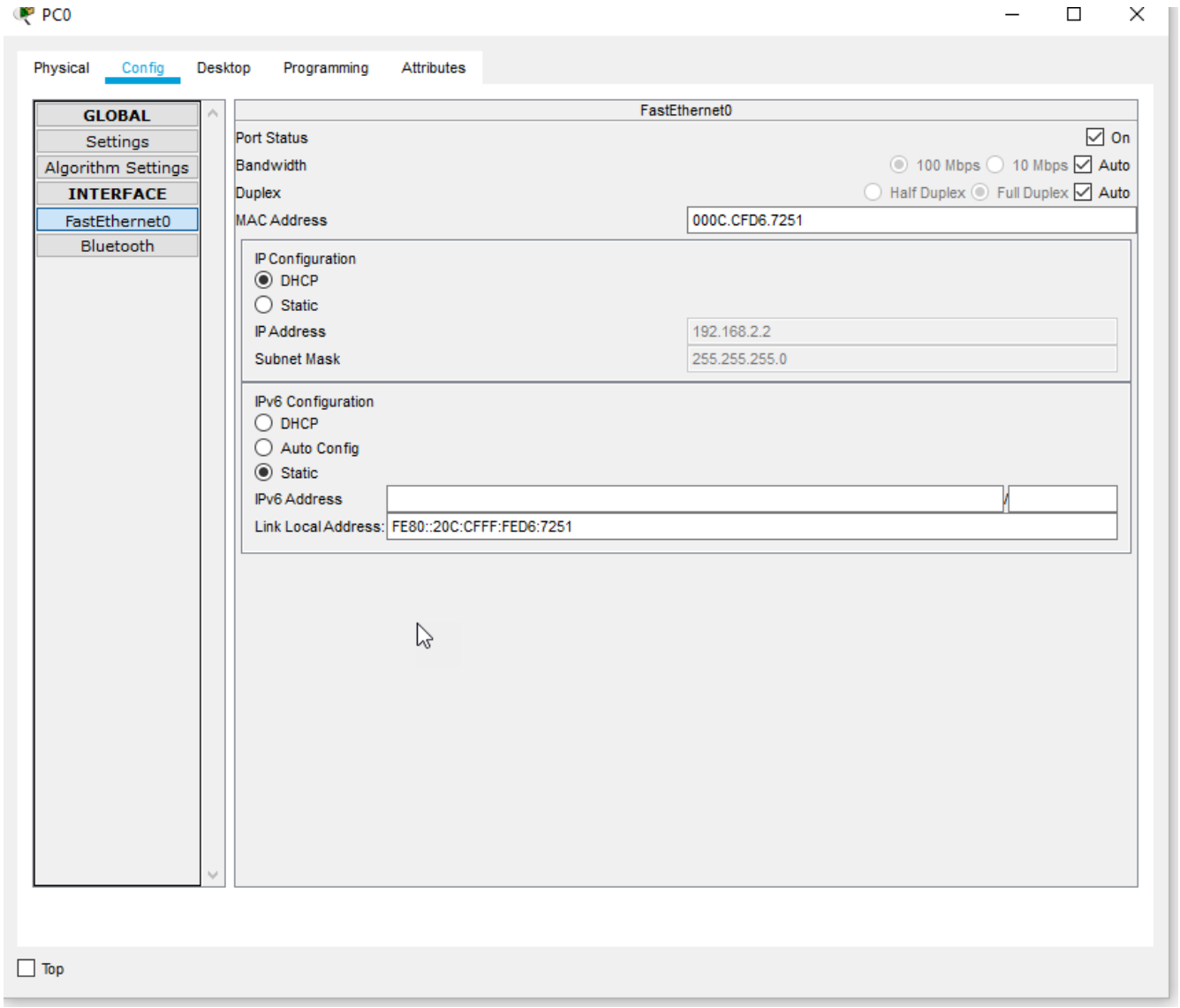
Now configure IP addresses on each host and servers as given below:
PC0= 192.168.2.3
PC1= 192.168.2.4
PC2= 192.168.2.5
Sever0= 192.168.2.1
Sever1= 192.168.2.2
DHCP pool start :192.168.2.3
Now configure IP address on sever0



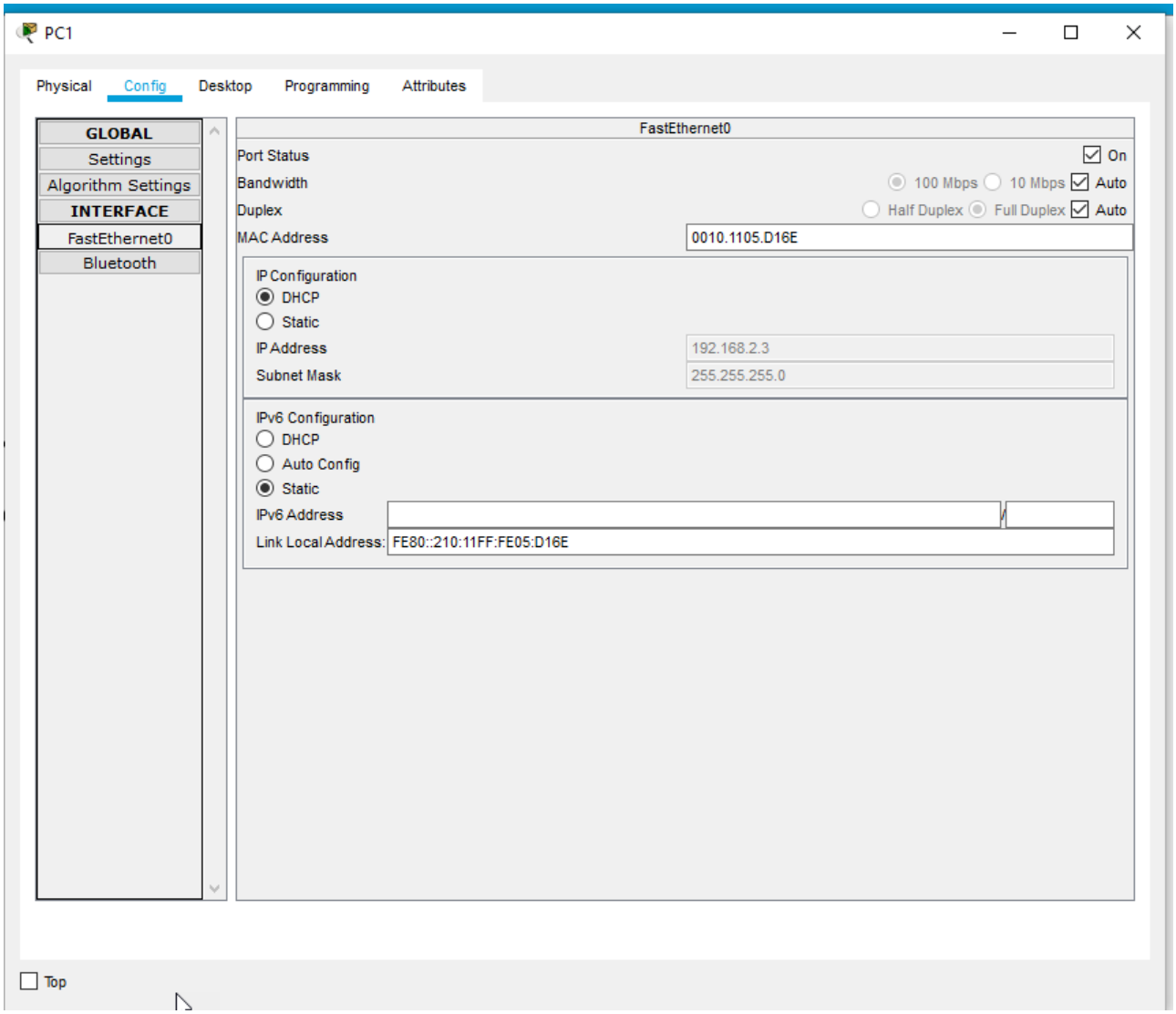
Now configure DHCP services on this server0 and add start IP address:



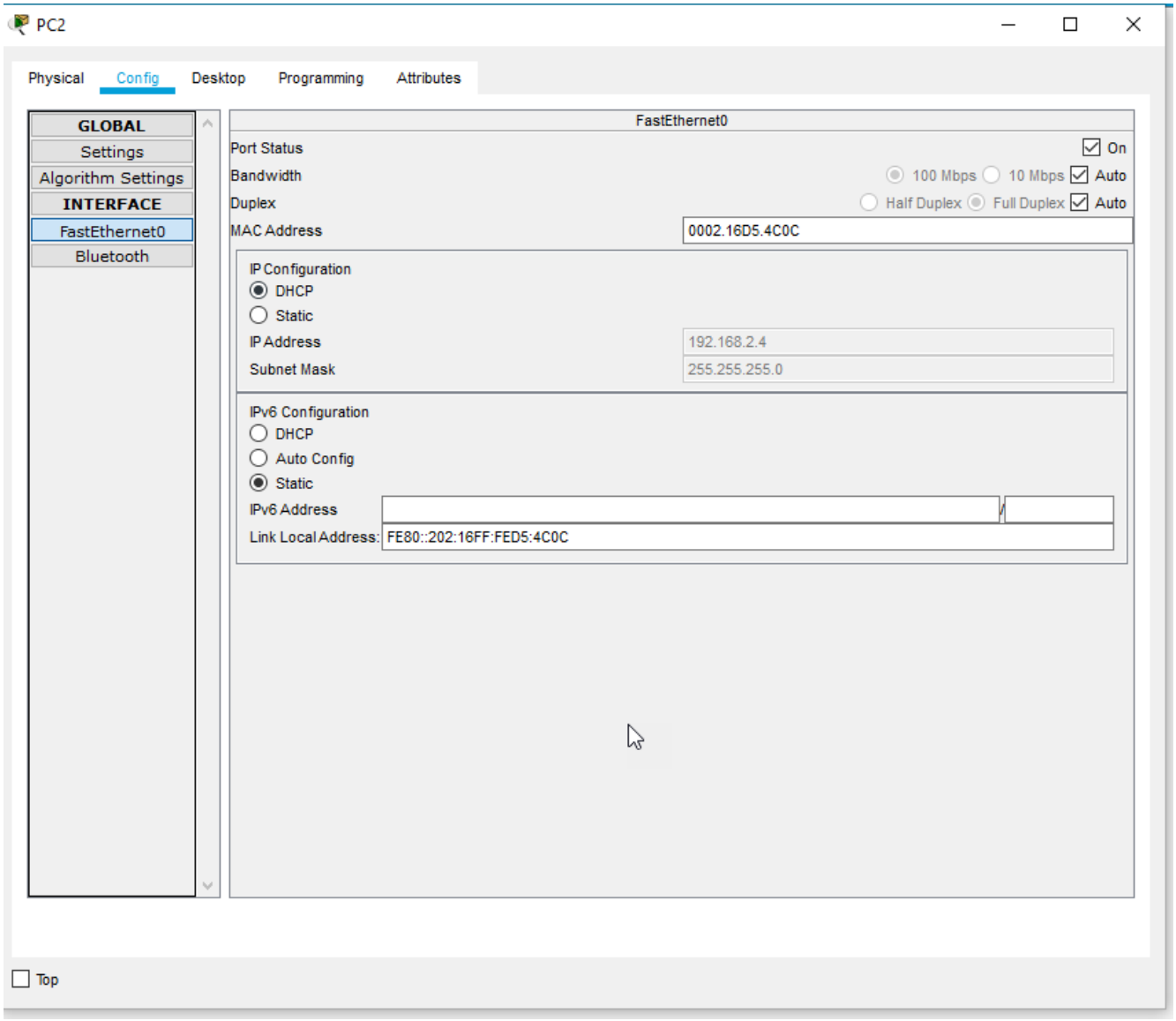
Now configure IP address on each host using DHCP server.You will check DHCP radio box to auto assign IP address.



Now configure IP address on pc1 using DCHP server:



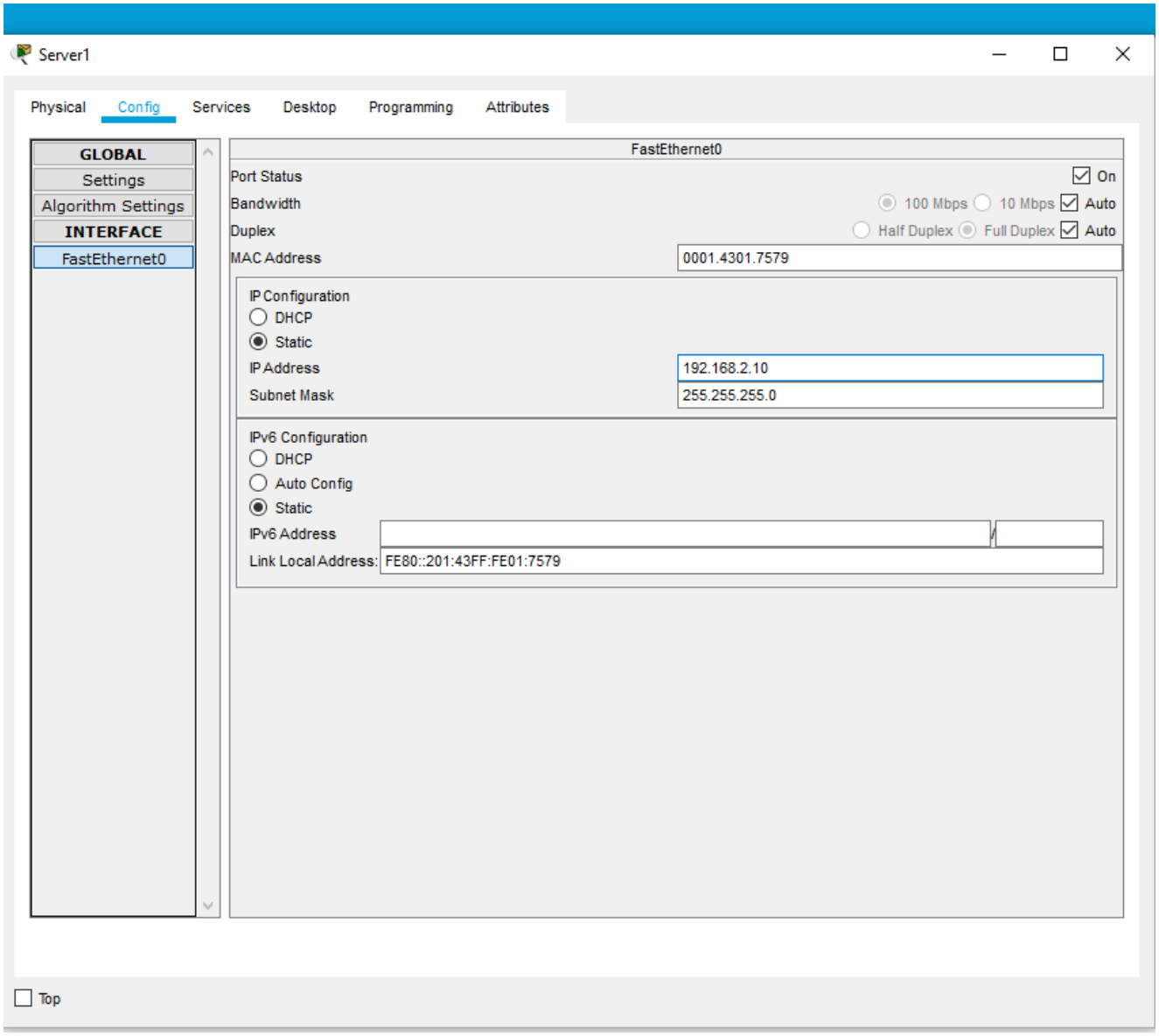
Now configure IP address on PC2 using DHCP server:



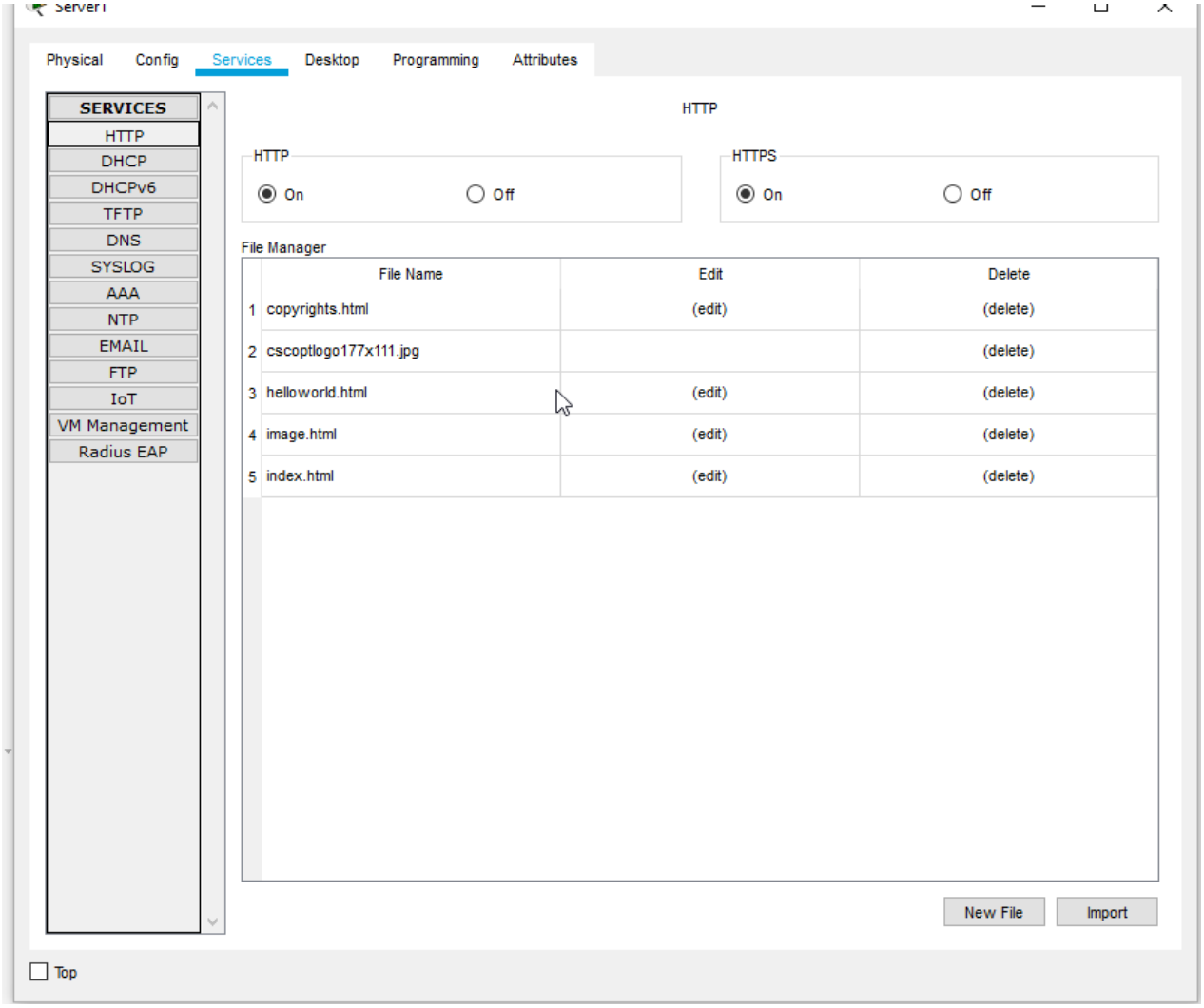
TASK-2

Now we will configure Web sever

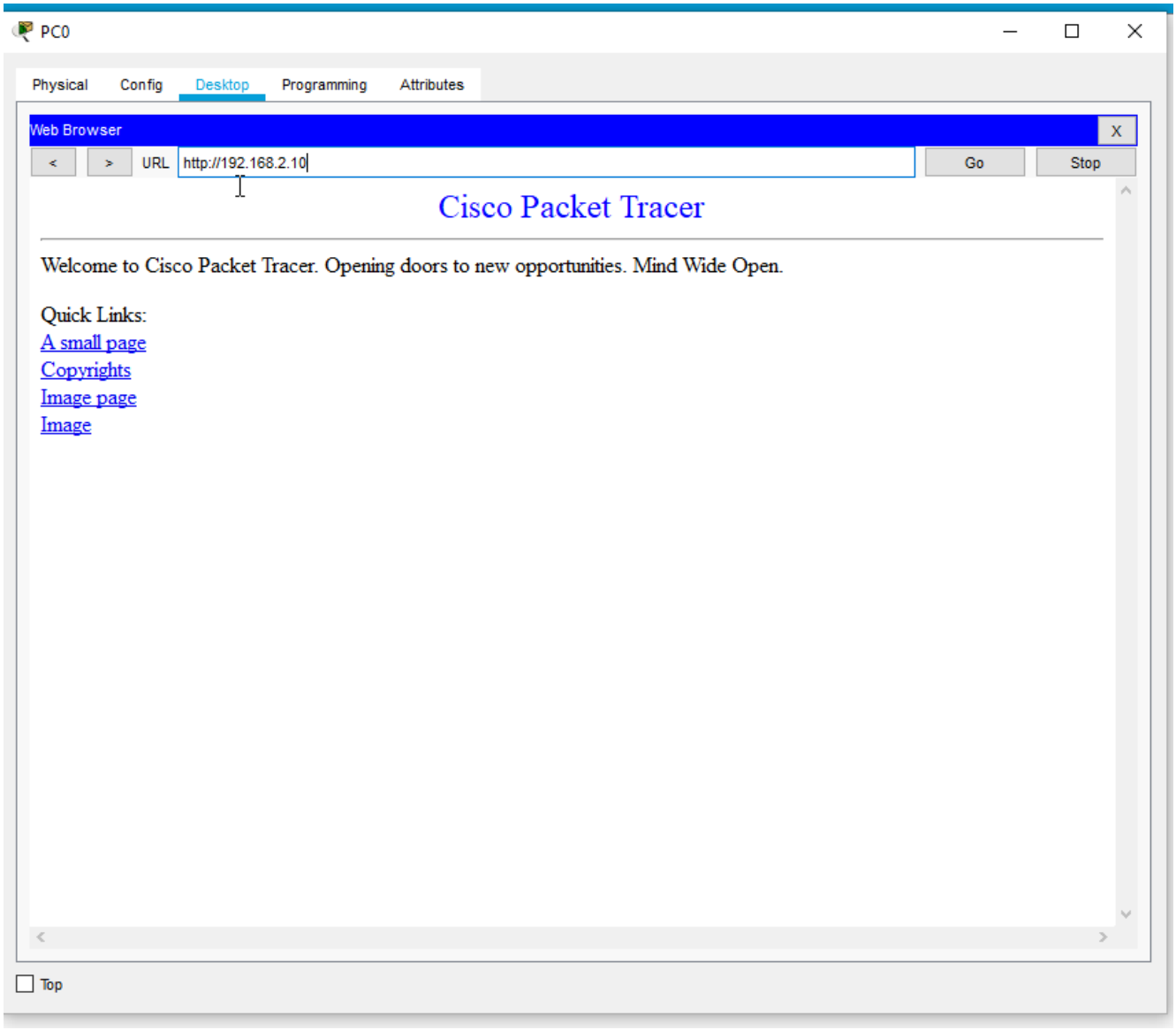
First configure IP address on this server1 as given below



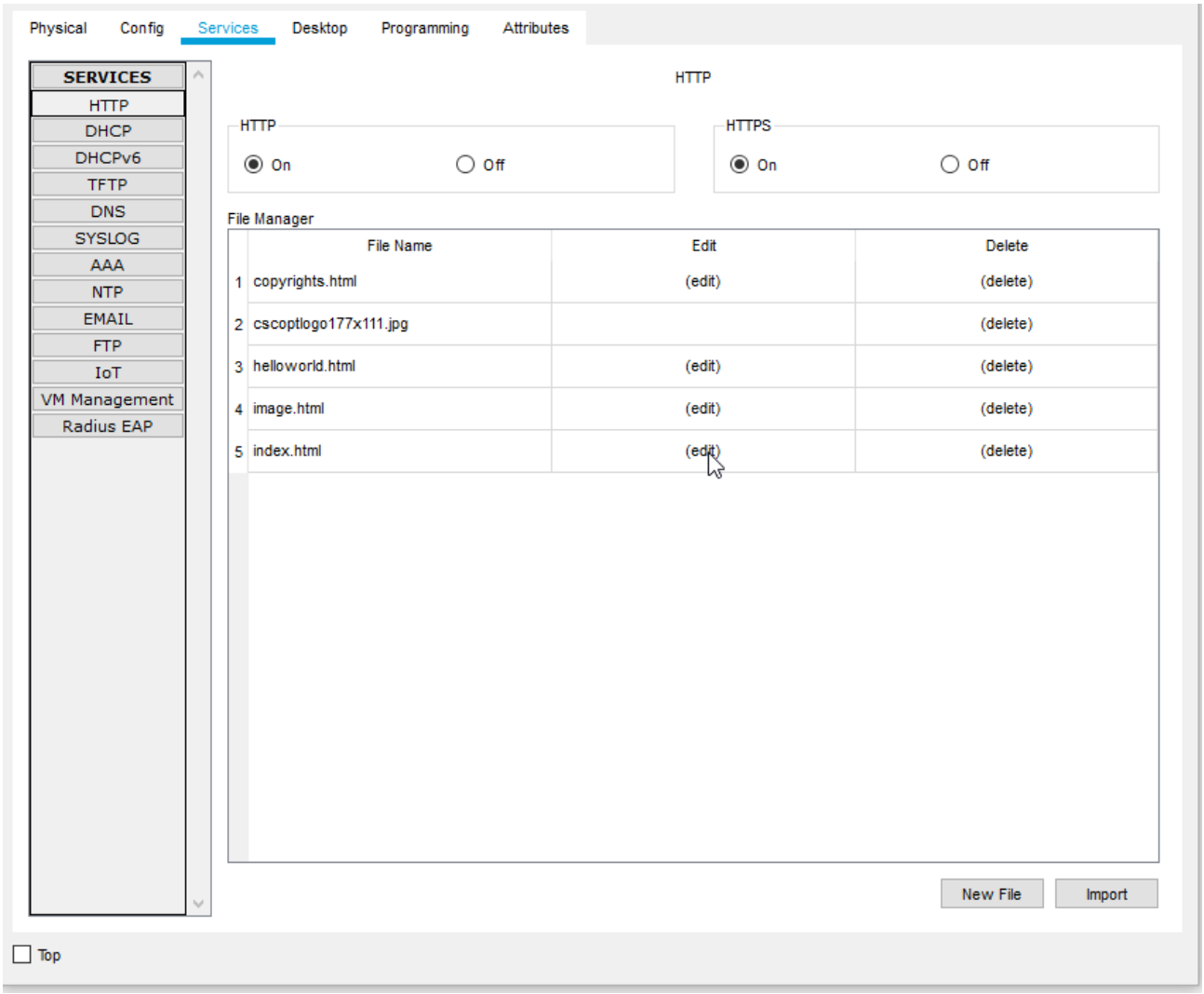
Now configure web services on this server1



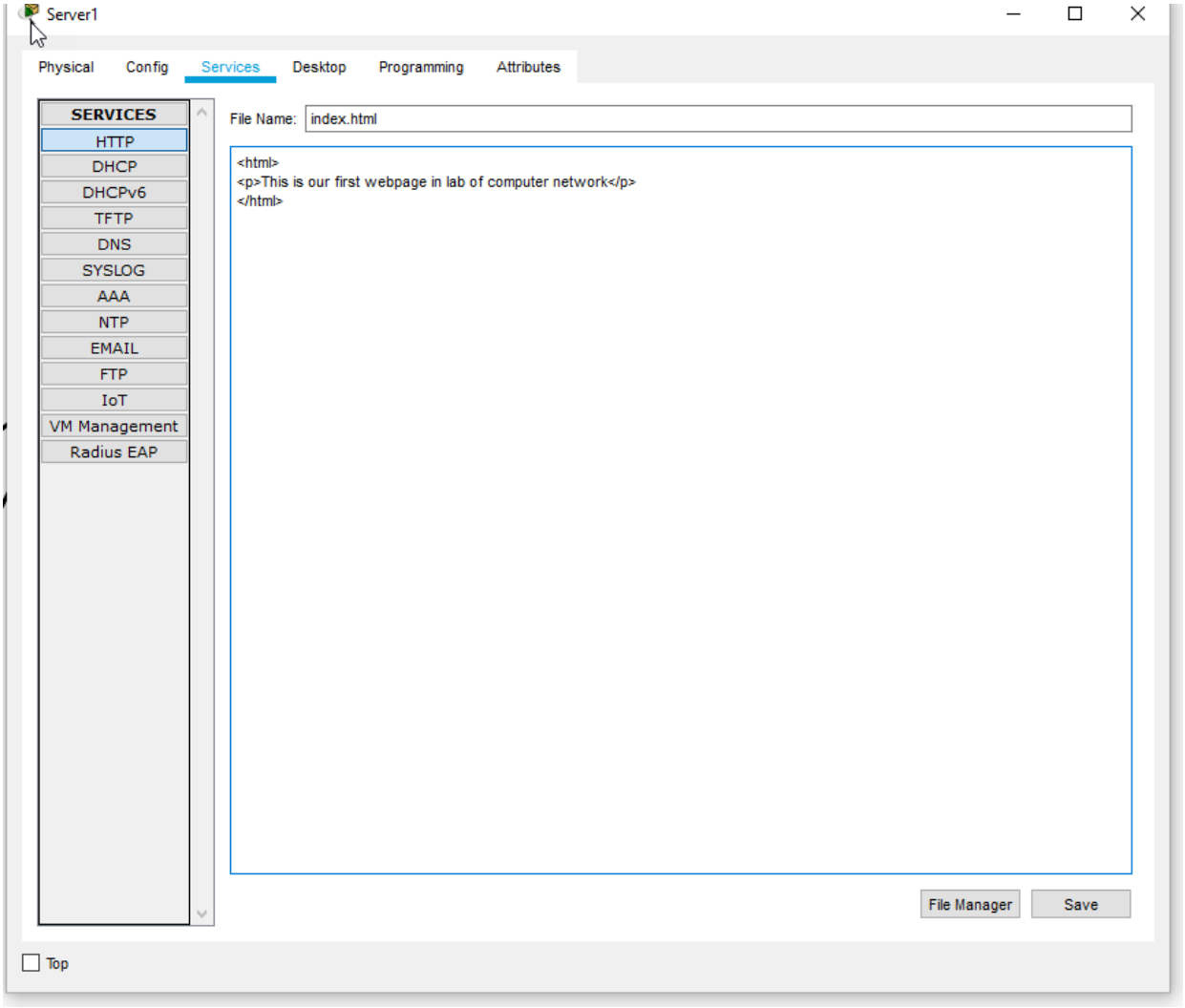
Now we will access this web sever using PC0 web browser:



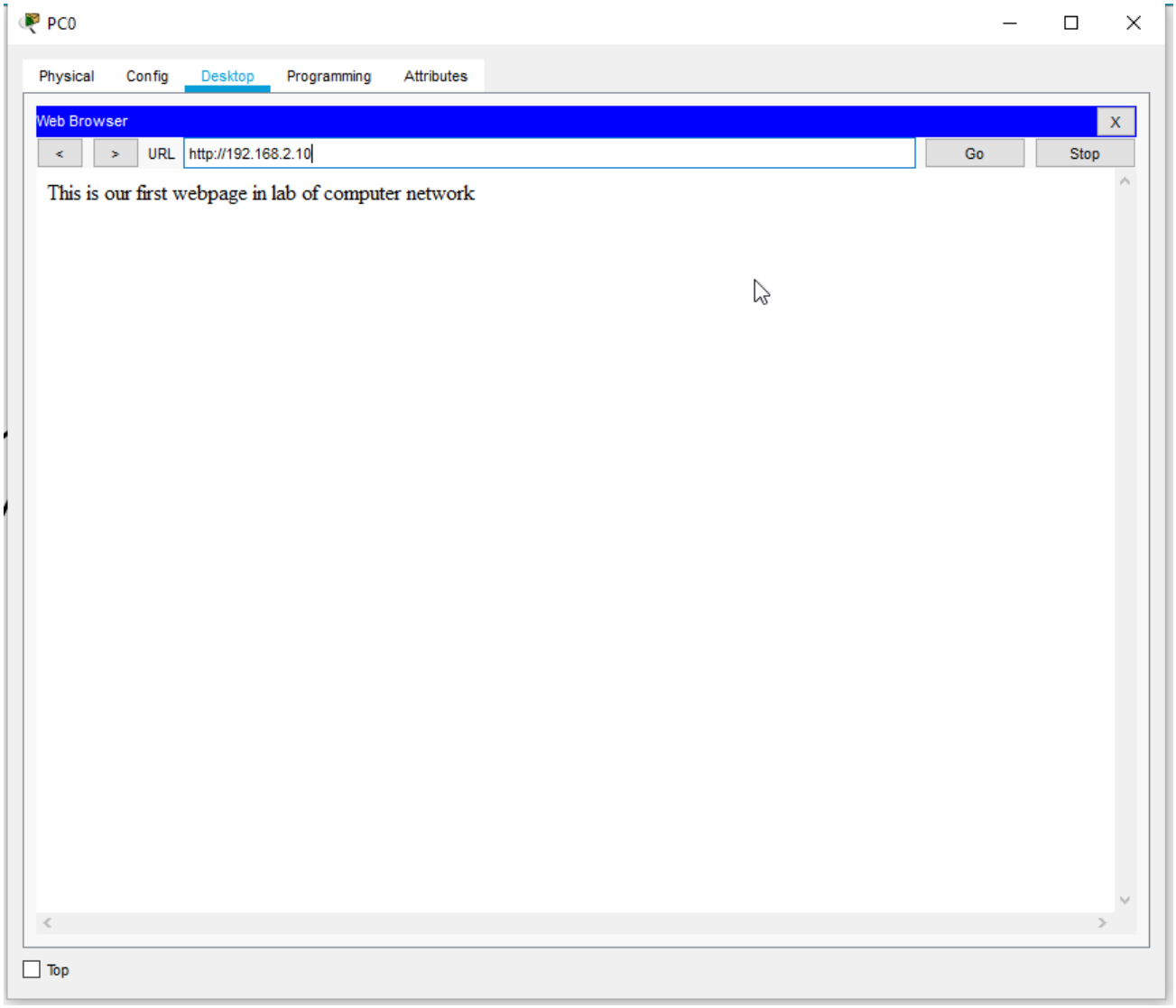
Web server is successful configured now.
Now you can create your own web page and can access from any pc:



Now we will edit index.html file



Now save this page and access this page from pc0

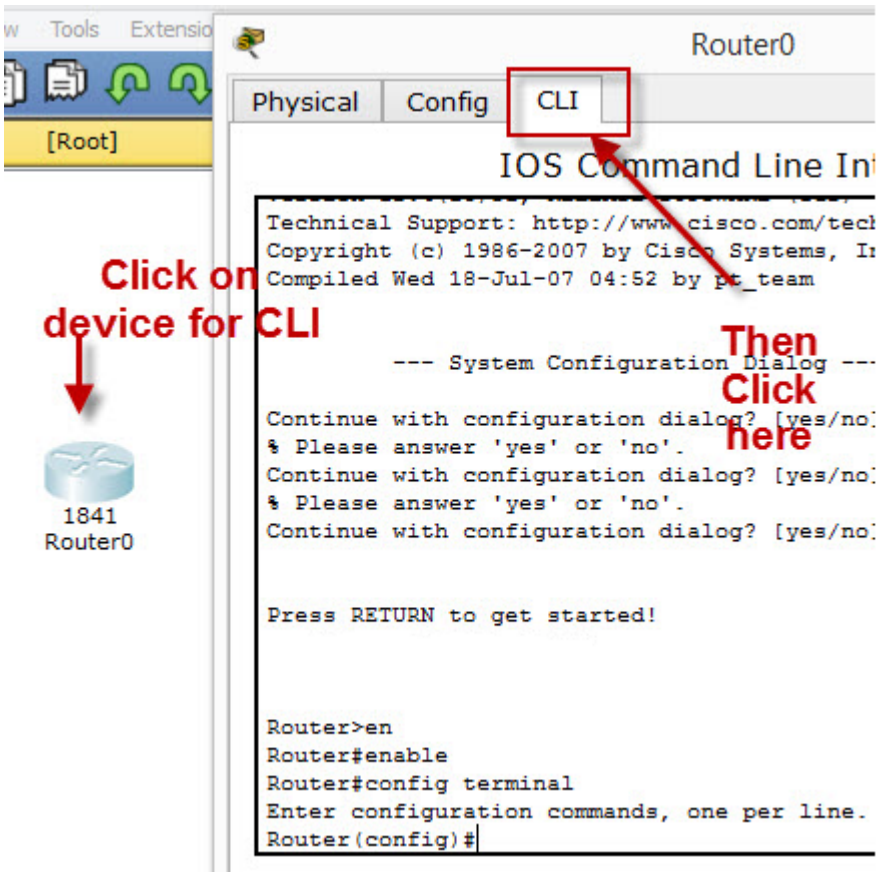

PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.2.10
Go
Stop
This is our first webpage in lab of computer network
Top

Lab 16

Packet Tracer Cisco CLI Commands list

Cisco Router Configuration Step By Step

To configure any device in packet tracer you are required to open or access its CLI. You can do it by clicking any device and then navigating to CLI tab. Once you are at CLI you can perform all Cisco Commands here.



Cisco IOS supports numerous command modes which can be practice with packet tracer, followings are the main command modes of cisco CLI with specific commands to navigate from one mode to another.

Mode	Symbol	How to access this mode	Command for leaving this mode
User EXEC Mode	Router >	Default mode after booting. Press enter for accessing this.	Use **exit** command

Privileged EXEC mode	Router #	Use **enable** command from user exec mode for entering into this mode	exit
Global Configuration mode	Router(config)#	Use **configure terminal** command from privileged exec mode	Exit or Ctrl+Z for user EXEC mode
Interface Configuration	Router(config-if)#	Use **interface <interface name+number>** command from global configuration mode	Use **exit** command to return in global mode

IOS commands are not case sensitive it means that you can use them in uppercase, lowercase, or mixed case, but passwords are case sensitive. Therefore make sure you type it in correctly. In any mode, you can obtain a list of commands available on that mode by entering a question mark (?).

```
Braunch_office_router(config)#router ?
  bgp     Border Gateway Protocol (BGP)
  eigrp   Enhanced Interior Gateway Routing Protocol (EIGRP)
  ospf    Open Shortest Path First (OSPF)
  rip     Routing Information Protocol (RIP)
Braunch_office_router(config)#router
```

How to Change the Cisco Router name

You can change the cisco router name by using command **hostname** in global configuration mode.

```
Router(config)#hostname HR-Router
HR-Router(config)#
```

How to set the Enable password:

You can set the password for protecting enable mode by following command: (Following command will set the password to cisco)

```
Router(config)#enable secret Cisco
Router(config)#exit
Router>en
Password:
Router#
```

How to set the IP address to Cisco interface:

You can set the IP address to any Cisco device interface by using the following commands:

Router(config)#interface <interface name&number>

Router(config-if)#ip address <IP address> <subnet mask>

How to enable a port or interface

Router(config-if)#no shut

Example:

```
Router(config)#interface fastEthernet 4/0
Router(config-if)#ip address 192.168.77.88 255.255.255.0
Router(config-if)#no shut
```

How to check the IP address of all interfaces:

You can use the "**show ip interface brief**" command in Privileged EXEC mode for checking the IP address of all interface of Cisco device.

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        unassigned      YES NVRAM  administratively down down

FastEthernet1/0        unassigned      YES NVRAM  administratively down down

Serial2/0              unassigned      YES NVRAM  administratively down down

Serial3/0              unassigned      YES NVRAM  administratively down down

FastEthernet4/0        192.168.77.88   YES manual down                  down
```

How to save the configurations:

You can use the following command for router configuration to Nvram for use at next boot up.

Router#**copy running-config startup-config**